

COMPOSITION OF THE SYSTEM

System control unit

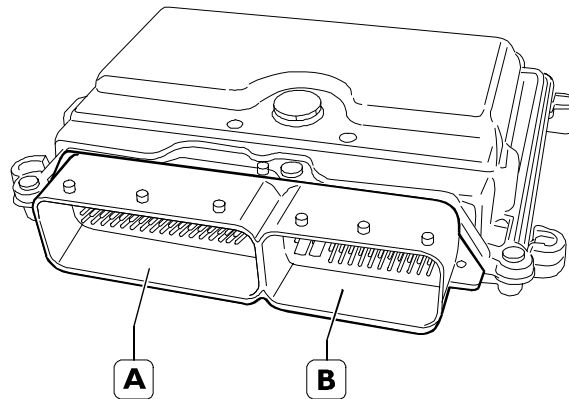
The electronic control unit receives the signals from the sensors/switches: the management and control of the system under the different operating conditions of the gearbox are based on the above signals.

It is interfaced with other electronic systems available on the vehicle, such as EDC and ABS, through CAN communication lines.

A - Connector on the gearbox side

B - Connector on the vehicle side

Figure 131/16



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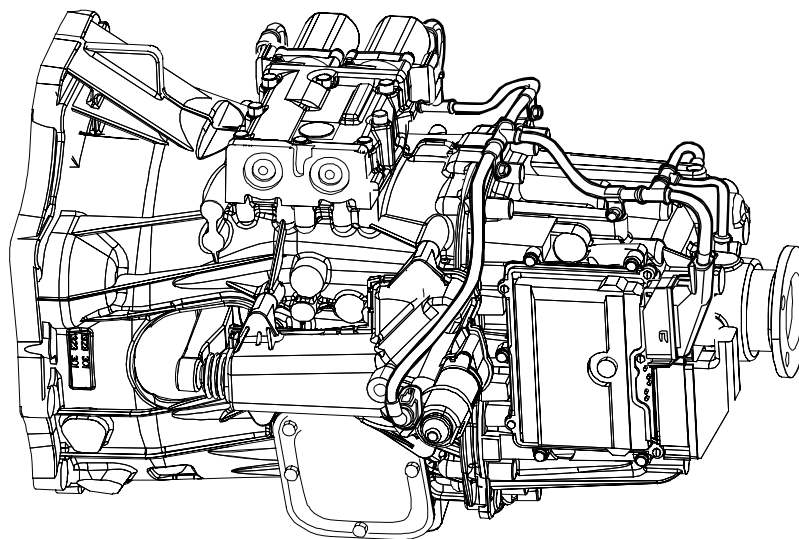
Through the connection with the EDC I6 system, the gearbox control unit is able to detect the position of the accelerator pedal and also the engine revs number.

Connection with the ABS8/ESP8 control unit is used to prevent the "UP" gear shift at bends and also control driving under poor grip conditions in case of mode "A" driving.

The new ABS system controls "smart" warning lights incorporated into the on-board panel. These warning lights come on to indicate braking system failure.

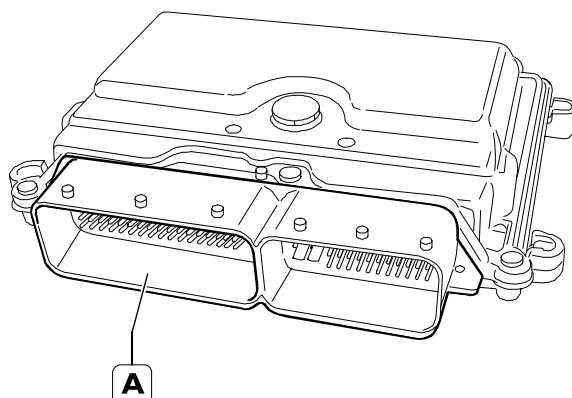
NOTE The "Brake" signal, upon start-up, comes directly from the EDC control unit. The reverse gear signal is direct.

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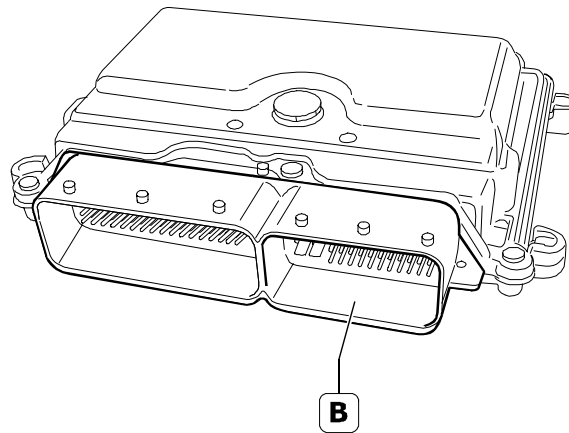
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Position of the electronic control unit

Connector control unit PIN-OUT – gearbox side (A)**Figure 131/18**

101869

Pin	Function
1	Electric motor positive (gearbox actuator selector pin 6)
2	Electric motor negative (gearbox actuator selector pin 1)
3	Electric motor positive (clutch actuator pin 3)
4	Electric motor negative (clutch actuator pin 4)
5	Electric motor positive (gear engaging gearshift actuator pin 6)
6	Electric motor negative (gear engaging gearshift actuator pin 1)
11	Sensor direction signal (clutch actuator pin 4)
12	Sensor speed signal (clutch actuator pin 2)
13	Clutch actuator pin 5 sensor voltage signal (5 V)
15	Sensor direction signal (gear engaging gearshift actuator pin 4)
16	Sensor voltage signal, 12 V (gear engaging gearshift actuator pin 3)
17	Sensor voltage signal, 12 V (gearbox actuator selector pin 3)
18	Sensor direction signal (gearbox actuator selector pin 4)
33	Earth (clutch actuator pin 1)
35	Sensor speed signal (gear engaging gearshift actuator pin 5)
36	Earth (gear engaging gearshift actuator pin 2)
37	Earth (gearbox actuator selector pin 2)
38	Sensor speed signal (gearbox actuator selector pin 5)

Connector control unit PIN-OUT – vehicle side (B)**Figure 131/19**

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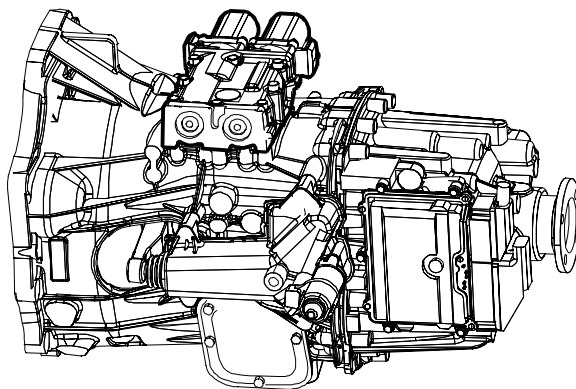
Pin	Function
1	Earth
2	Earth
3	Earth
4	Battery positive
5	Battery positive
6	Battery positive
7	Earth
8	Earth
9	KL 30
14	PTO actuation request signal (option)
17	CAN H VDB line (female junction)
18	CAN H VDB line (female junction)
28	CAN L VDB line (male junction)
29	CAN L VDB line (female junction)
33	Free
37	Free
38	Free
39	KL 30
40	CAN L --- ZF LINE
41	CAN H --- ZF LINE
43	KL 15
44	On-board panel signal (doors open)
45	On-board panel signal (emergency brake)
46	PTO state signal
47	PTO ON signal
49	Power supply from gearbox electronic control unit
51	Speed signal from on-board panel
52	Diagnosis K line

Gearbox actuator

The function of the gear actuator is to continuously exchange information with the electronic control unit for gear selection and engagement.

It is made up of two electric motors, control cylinders, and respective sensors.

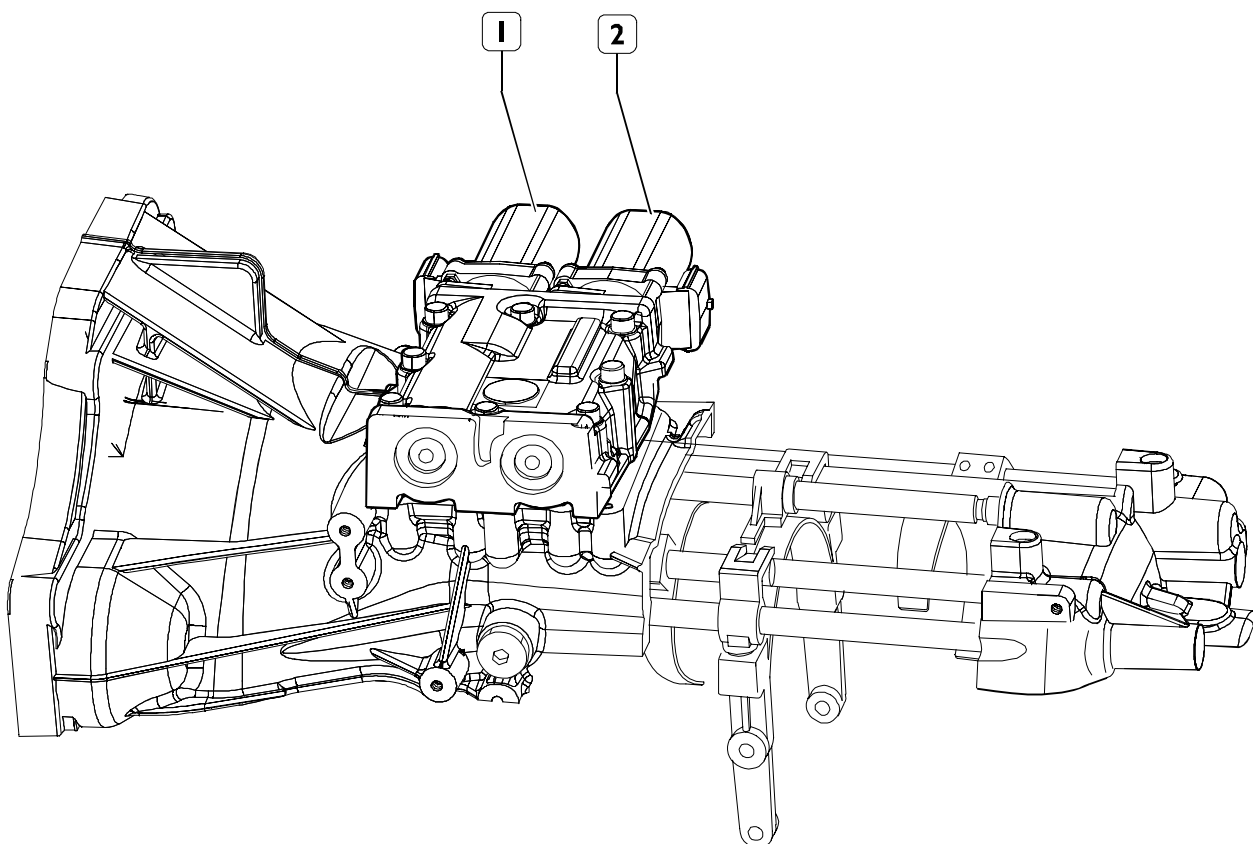
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90139

Position of actuator

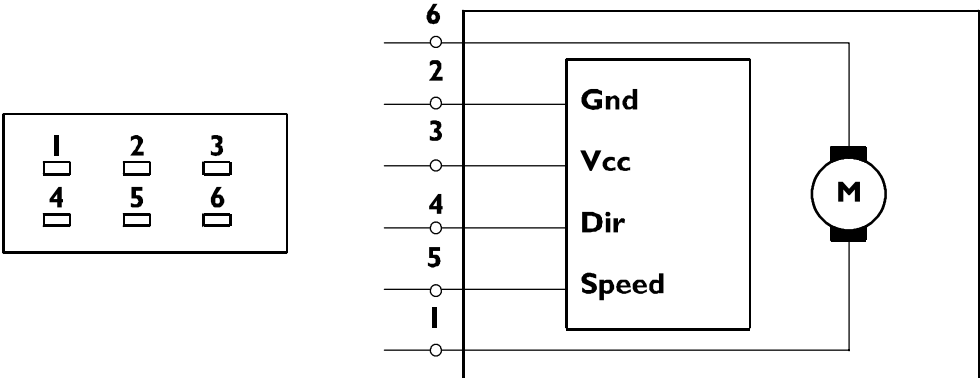
Figure 131/21



101872

1. Gear engaging electric motor - 2. Gear selection electric motor

Figure 131/22



101873

Wiring diagram

Pin	Function
1	Electric motor negative
2	Earth
3	Sensor voltage signal (5 V)
4	Sensor direction signal
5	Sensor speed signal
6	Electric motor positive

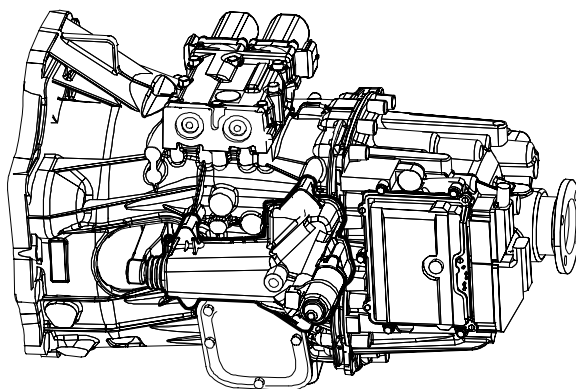
Characteristics of electric motor

Voltage	12 V
Output	95 W
Torque	0.72 Nm (at 125 °C)
n _{max}	5400 r.p.m.

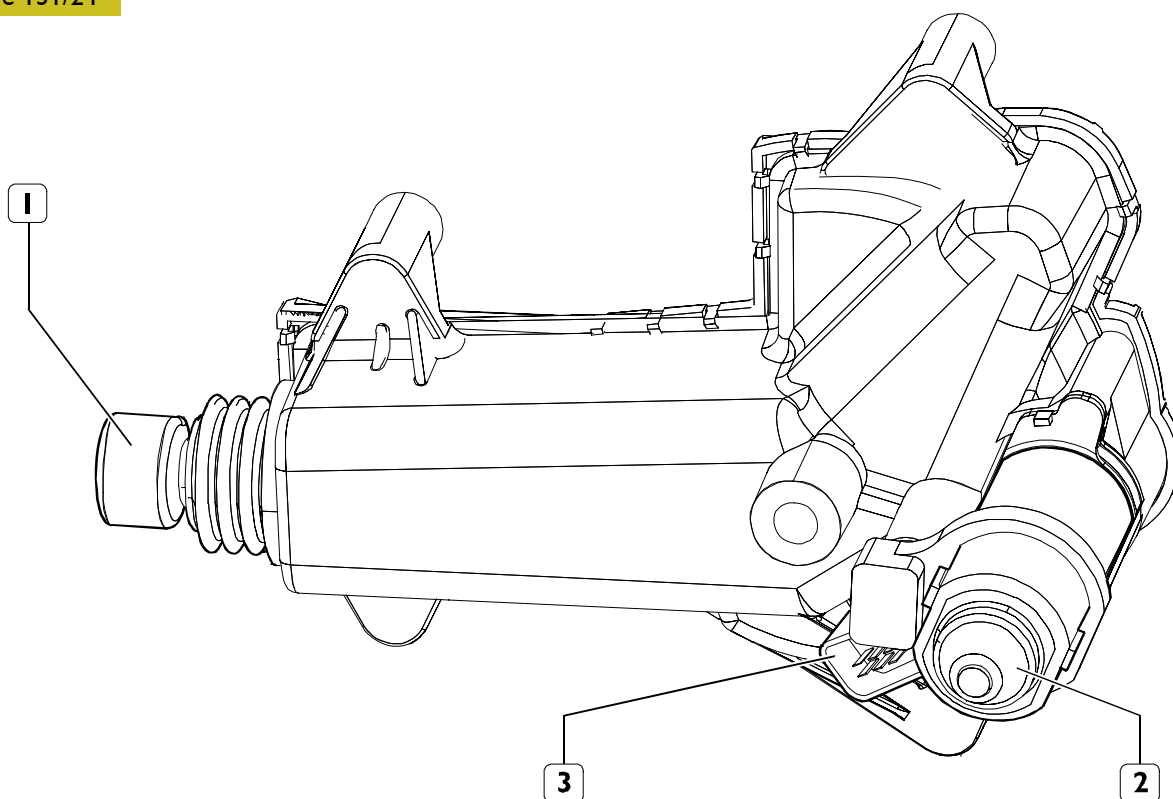
Clutch actuator

It is made up of the following:

- ☐ a cylinder acting on the clutch engaging/disengaging lever;
- ☐ a position sensor that detects the clutch lever stroke, by informing the electronic control unit of the actuating cylinder position and the clutch plate wear.

Figure 131/23

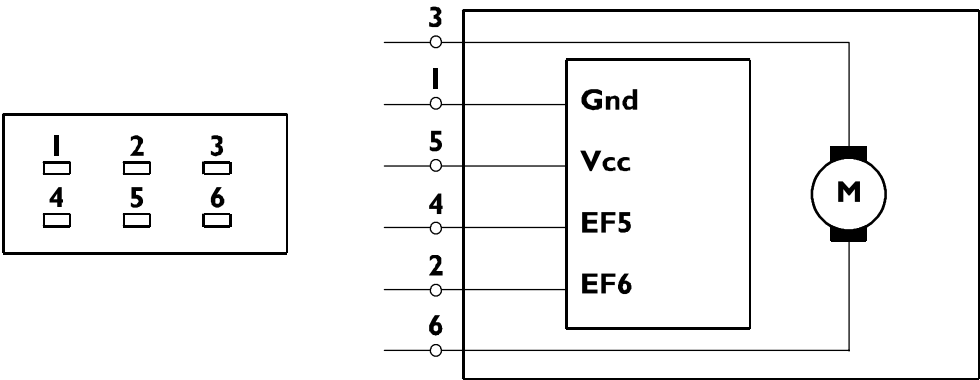
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Figure 131/24

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1. Actuating cylinder - 2. Electric motor - 3. Vehicle electric wiring junction block

Figure 131/25



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Wiring diagram

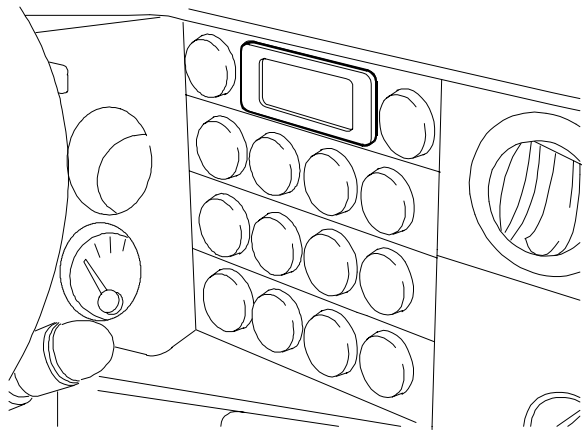
Pin	Function
1	Earth
2	Incremental sensor speed signal
3	Electric motor positive
4	Incremental sensor direction signal
5	Sensor voltage signal (5 V)
6	Electric motor negative

Characteristics of electric motor

Max. torque	0.65 Nm
No-load revs number	5800 r.p.m.
Output	80 W (0.3 Nm - 2,500 r.p.m.)

Display

Figure I31/26



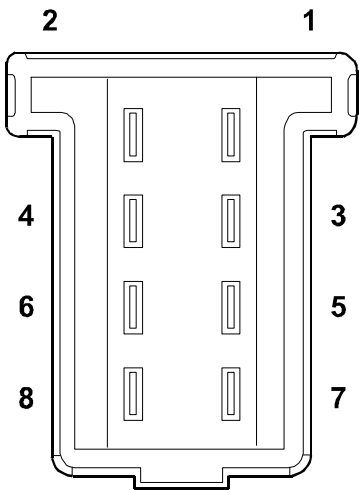
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The display, located on the board, makes it possible to show all the information required for correct use of the system, such as, for instance:

- ☐ mode (manual or automatic) and, in both cases, the gear engaged;
- ☐ the reverse gear/neutral position (R/N);
- ☐ faults/misuse with acoustic signal (e.g. clutch overheating);
- ☐ the fault code (troubleshooting) shown on page 47 can be displayed only by means of the diagnosis instrument;
- ☐ vehicle state condition: gear engaged.

Pin	Function
1	Backlighting ON/OFF control
2	
3	Analog backlighting control
4	-
5	CAN H
6	Earth
7	CAN L
8	+12 V

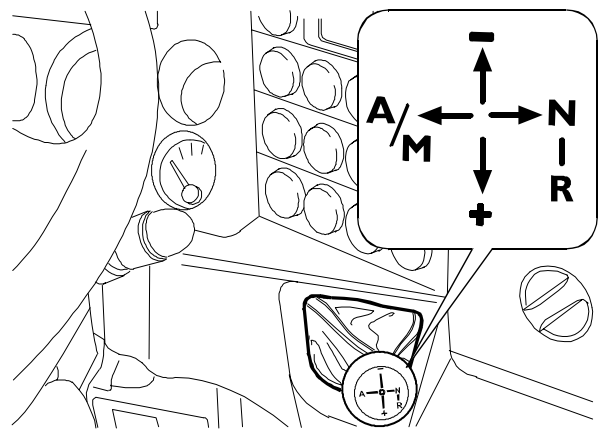
Figure I31/27



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Gear selector

Figure 131/28

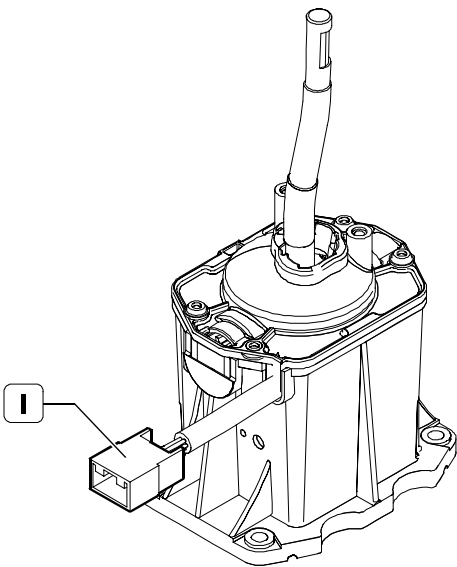


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The gear selector is an electronic component located in the cabin next to the driver. It is interfaced with the gearbox control unit, by means of the CAN communication line, in order to inform of the driver's intention of selecting and engaging the gears.

Pin	Function
1	KL 15
2	KL 30
3	CAN L
4	CAN H
5	-
6	-

Figure 131/29



90262

I. Gear selector connector

Accelerator pedal

To detect the engine idling position and allow the clutch to be engaged when the vehicle is about to start, the N.O. switch, incorporated into the position sensor, is used, with the pedal released.

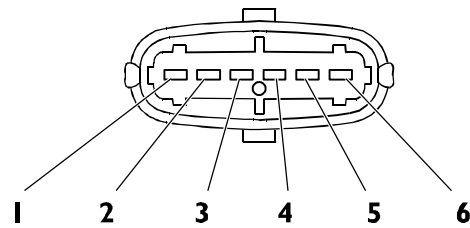
This signal reaches the EDC electronic control unit and is sent, through the CAN VDB (Vehicle Data Base) line, to the 6AS 300 VD gearbox control unit.

The "kick-down" function can be actuated during running, in the automatic mode.

When the pedal is pressed down almost fully (98%), e.g. when overtaking, the system will automatically shift down by one speed, thus making it possible to use the deflecting torque at its best.

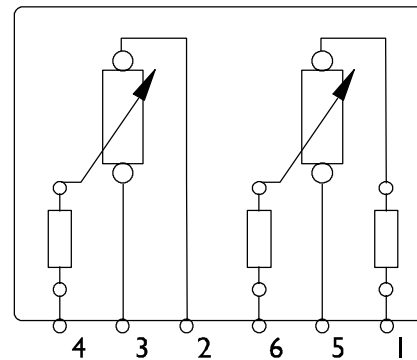
In practice, if the accelerator pedal is pressed fully down, the vehicle will ride at a running speed with a higher gear shift.

Figure I31/30



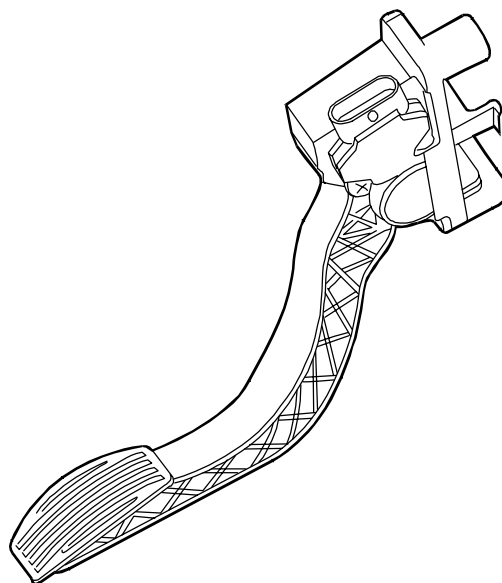
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Figure I31/31



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Figure I31/32



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PTO

One single button is used to actuate and deactivate the power take-off (PTO).

Operation under NON-STATIONARY conditions

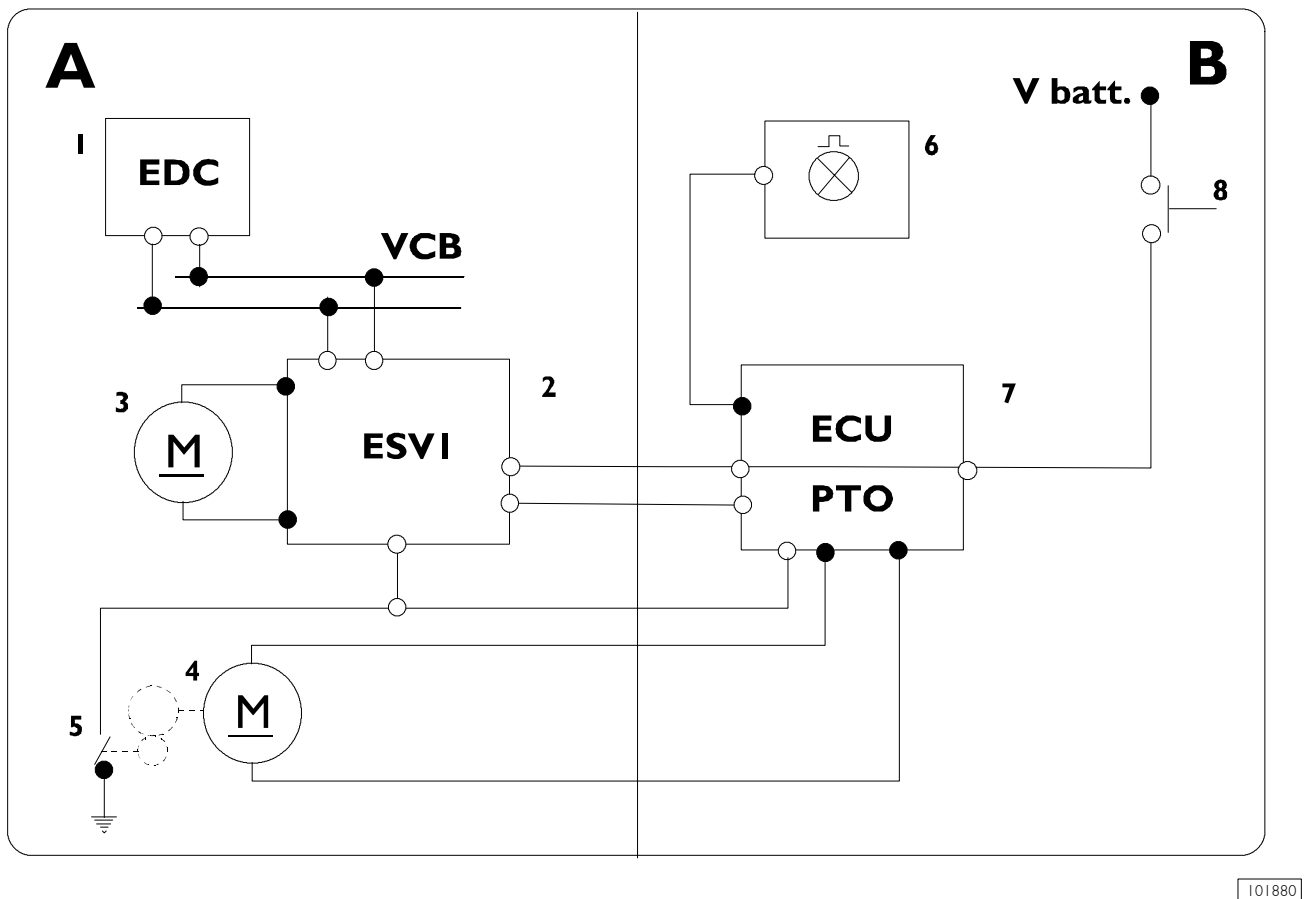
- ☐ The PTO can be engaged only when the gear shift is in neutral and the engine is ON;
- ☐ then, the first gear can be engaged, and the vehicle can be started;
- ☐ after setting off, the gear cannot be changed any more when the vehicle is running;
- ☐ the PTO can be disengaged both with the gears running and the neutral ON.

Operation under STATIONARY conditions

- ☐ The PTO can be engaged only when the gear shift is in neutral and the engine is ON.

PTO diagram

Figure 131/33



1. EDC control unit - 2. Gearbox control unit - 3. Clutch actuator electric motor - 4. PTO actuator motor - 5. PTO ON switch - 6. PTO ON warning light - 7. PTO control unit - 8. PTO actuation switch - A. Gearbox side - B. Vehicle side.

101880

PTO engaging procedure

To engage the power take-off, the following operations must be carried out:

- 1) press the PTO button;
- 2) wait at least 0.5 seconds;
- 3) release the PTO button.

NOTE

- ☐ The PTO will be actuated only if the PTO button is kept depressed for more than 0.5 seconds.
- ☐ The PTO button will be ignored until the next 5 seconds have elapsed.
- ☐ After such time, if the button is kept depressed for more than 30 seconds, a quick flash will be displayed, and PTO disengagement and lock will be commanded until the vehicle is switched on again.

PTO disengaging procedure

To disengage the power take-off, the following operations must be carried out:

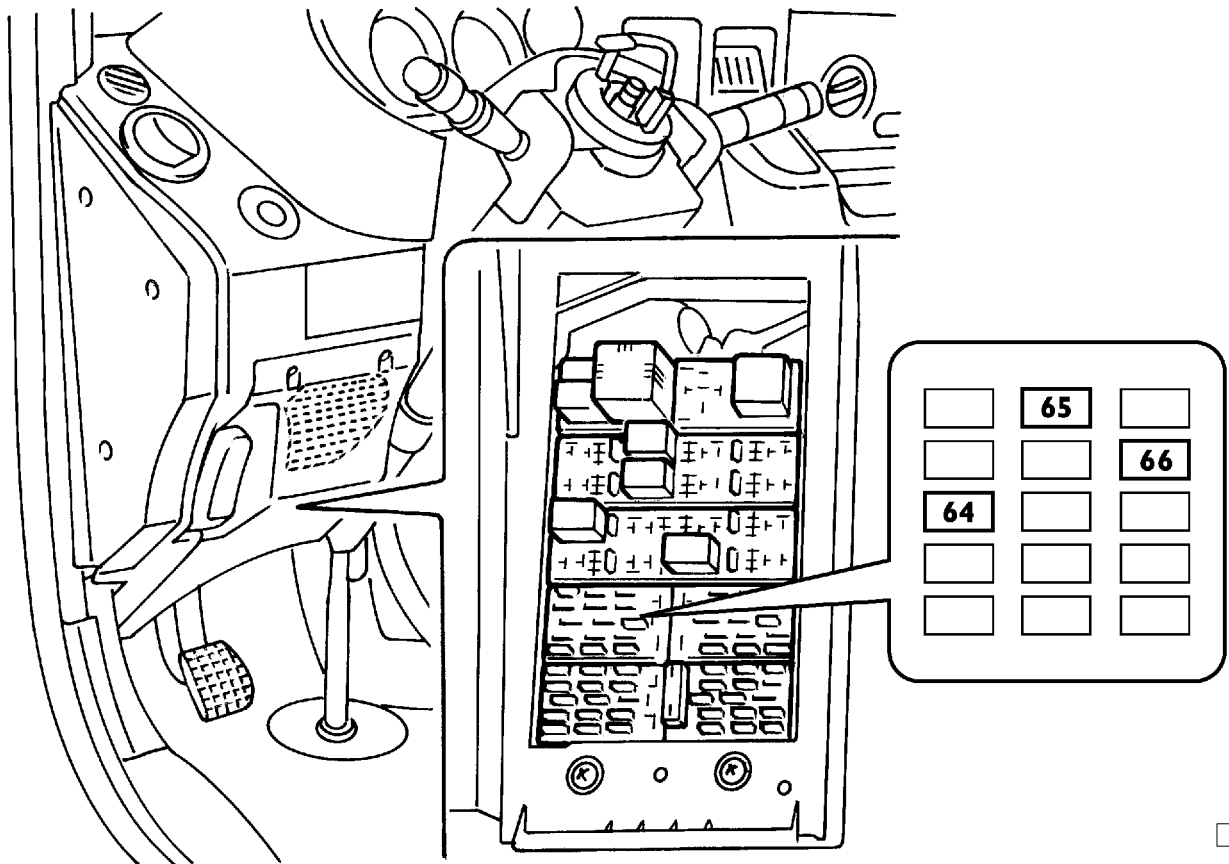
- 1) press the PTO button;
- 2) wait at least 0.5 seconds;
- 3) release the PTO button.

NOTE

- ☐ The operation will be carried out only if the PTO button is kept depressed for more than 0.5 seconds.
- ☐ The PTO button will be ignored until the next 5 seconds have elapsed. • In the event that the power take-off is not disengaged within 5 seconds, the operator may carry out the operation again, by pressing the button again.
- ☐ After such time, if the button is kept depressed for more than 30 seconds, a quick flash will be displayed, and PTO disengagement and lock will be commanded until the vehicle is switched on again.

Remote-control switch and fuse assembly

Figure 131/34



90264

The system's power supply is ensured by a set of fuses located in the cabin (see figure) in the marked positions.

64 - control unit power supply

65 - automatic transmission services

66 - start prevent

There are three more 25 A fuses (electric motor power supply) that are connected with control unit pins 4, 5, and 6. In the event that one or two of these fuses blow, the gearbox functions are not affected. If the third fuse is broken, the gearbox is not able to operate (i.e. both the clutch motion and the gear selection/engagement will be disabled).

Accelerator pedal sensor

The accelerator pedal position is provided to the gearbox control unit, through a suitable communication line, by the EDC electronic control unit which is informed about the position by the load transmitter fitted directly to the accelerator pedal.

Engine revs sensor

The sensor is mounted just next to the engine flywheel.

The engine running speed values are sent to the EDC electronic control unit which conveys them to the gearbox control unit by means of a special communication line.

Vehicle speed sensor

This sensor sends the pulses to the tachometer/electronic tachograph.

The signal is duplicated by the ABS/ESP sensors available on the wheels.

DIAGNOSIS INSTRUMENTS

DISPLAY

First-level diagnosis that provides coded displaying of the errors found in the control unit memory. The information is shown on the display only by means of diagnosis instruments.

MODUS

Computer-assisted diagnosis station used for diagnosis of braking systems, pneumatic suspensions, engines and electronically-controlled systems.

The station is equipped with auxiliary functions, such as electronic control unit programming, spare part catalogue referencing, time-charts, etc.

IT 2000

IT 2000 is a diagnosis instrument for all the Electronic Systems fitted to IVECO vehicles. It allows you to promptly operate on a vehicle by recognizing the latter by means of the chassis number.

It stores the results of the diagnostic work carried out.

It can also be used as a laptop PC and is set for remote diagnosis.

Using MODUS as a mother station allows you to update and configure IT 2000.

E.A.SY.

The E.A.SY. system allows you to easily diagnose and program the various electronic control units fitted to the vehicle. It is made up of an ECI module for communication with the electronic control units and a Panasonic PC.

The ECI module allows you, by taking advantage of the Panasonic PC, to easily carry out work on the road; in particular, diagnostic work can be assisted by a specialized remote centre, thanks to the wireless technology incorporated into the Panasonic PC (e.g. GPRS).

TROUBLESHOOTING

DTC DEC	DTC HEX	Description
10	A	ELECTRONIC CONTROL UNIT / TOO LOW TEMPERATURE
11	B	ELECTRONIC CONTROL UNIT / TOO HIGH TEMPERATURE
12	C	VEHICLE CAN LINE / FAULT
13	D	ZF CAN LINE / FAULT
30	1E	VEHICLE CAN LINE / DELAY IN RECEIVING THE EEC1 CAN MESSAGE
31	1F	VEHICLE CAN LINE / DELAY IN RECEIVING THE EEC2 CAN MESSAGE
32	20	VEHICLE CAN LINE / DELAY IN RECEIVING THE CCVS CAN MESSAGE
33	21	VEHICLE CAN LINE / DELAY IN RECEIVING THE EEC3 CAN MESSAGE
34	22	VEHICLE CAN LINE / DELAY IN RECEIVING THE EBC1 CAN MESSAGE
35	23	VEHICLE CAN LINE / DELAY IN RECEIVING THE ENGCONFMPM CAN MESSAGE
36	24	VEHICLE CAN LINE / DELAY IN RECEIVING THE HRW CAN MESSAGE
37	25	ZF CAN LINE / DELAY IN RECEIVING THE SL_2_TCU CAN MESSAGE
38	26	VEHICLE CAN LINE / DELAY IN RECEIVING THE VDC1 CAN MESSAGE
39	27	VEHICLE CAN LINE / DELAY IN RECEIVING THE VDC2 CAN MESSAGE
50	32	VEHICLE CAN LINE / ERROR ON ENGINE REVS SIGNAL - EEC1 CAN MESSAGE
52	34	VEHICLE CAN LINE / ERROR ON SIGNAL FOR REQUESTED ENGINE TORQUE - EEC1 CAN MESSAGE
54	36	VEHICLE CAN LINE / ERROR ON ACCELERATOR PEDAL POSITION SIGNAL - EEC2 CAN MESSAGE
56	38	VEHICLE CAN LINE / ERROR ON SIGNAL FROM BRAKE PEDAL - CCVS CAN MESSAGE
57	39	VEHICLE CAN LINE / ERROR ON ENGINE TORQUE AND BRAKING TORQUE CONFIGURATION SIGNAL - EEC1 CAN MESSAGE
58	3A	VEHICLE CAN LINE / ERROR ON ACTUAL ENGINE TORQUE SIGNAL - EEC1 CAN MESSAGE
60	3C	VEHICLE CAN LINE / ERROR ON ABSORBED ENGINE TORQUE SIGNAL - EEC2 CAN MESSAGE
62	3E	VEHICLE CAN LINE / ERROR ON REFERENCE ENGINE TORQUE SIGNAL - ENG_CONF_MPM CAN MESSAGE
66	42	VEHICLE CAN LINE / ERROR ON REAR LEFT WHEEL SPEED SIGNAL - HRW CAN MESSAGE
67	43	VEHICLE CAN LINE / ERROR ON REAR RIGHT WHEEL SPEED SIGNAL - HRW CAN MESSAGE
77	4D	VEHICLE CAN LINE / ERROR ON VEHICLE SPEED SIGNAL - CCVS CAN MESSAGE
100	64	SPEED ACTUATOR - SELECT / SHORT-CIRCUIT TO EARTH ON CONTROL MOTOR OUTPUT
101	65	SPEED ACTUATOR - SELECT / SHORT-CIRCUIT TO POSITIVE ON CONTROL MOTOR OUTPUT
102	66	SPEED ACTUATOR - SELECT / INTERRUPTION ON CONTROL MOTOR OUTPUT
103	67	SPEED ACTUATOR - SELECT / SENSOR SUPPLY VOLTAGE - TOO HIGH
104	68	SPEED ACTUATOR - SELECT / SENSOR SUPPLY VOLTAGE - TOO LOW
105	69	SPEED ACTUATOR - SELECT / SENSOR SHORT-CIRCUIT TO EARTH
106	6A	SPEED ACTUATOR - SELECT / SENSOR SHORT-CIRCUIT TO POSITIVE OR INTERRUPTION
107	6B	SPEED ACTUATOR - SELECT / NON-PLAUSIBLE MOTOR CONTROL SIGNAL
120	78	SPEED ACTUATOR - SELECT / TOO HIGH CURRENT
121	79	SPEED ACTUATOR - SELECT / TOO HIGH TEMPERATURE

DTC DEC	DTC HEX	Description
130	82	SPEED ACTUATOR - SELECT / MECHANIC FAULT
131	83	SPEED ACTUATOR - SELECT / MECHANIC FAULT
132	84	SPEED ACTUATOR - SELECT / POSITION CONTROL ERROR
180	B4	SPEED ACTUATOR - SELECT / REFERENCE EOL PARAMETER ERROR (LACKING MEASUREMENT VALUES)
181	B5	SPEED ACTUATOR - SELECT / REFERENCE EOL PARAMETER ERROR (OUT-OF-TOLERANCE MEASUREMENT VALUES)
200	C8	SPEED ACTUATOR - SHIFT / SHORT-CIRCUIT TO EARTH ON CONTROL MOTOR OUTPUT
201	C9	SPEED ACTUATOR - SHIFT / SHIFT / SHORT-CIRCUIT TO POSITIVE ON CONTROL MOTOR OUTPUT
202	CA	SPEED ACTUATOR - SHIFT / INTERRUPTION ON CONTROL MOTOR OUTPUT
205	CD	SPEED ACTUATOR - SHIFT / SENSOR SHORT-CIRCUIT TO EARTH
206	CE	SPEED ACTUATOR - SHIFT / SENSOR SHORT-CIRCUIT TO POSITIVE OR INTERRUPTION
207	CF	SPEED ACTUATOR - SHIFT / NON-PLAUSIBLE MOTOR CONTROL SIGNAL
220	DC	SPEED ACTUATOR - SHIFT / TOO HIGH CURRENT
221	DD	SPEED ACTUATOR - SHIFT / TOO HIGH TEMPERATURE
230	E6	SPEED ACTUATOR - SHIFT / MECHANIC FAULT
231	E7	SPEED ACTUATOR - SHIFT / MECHANIC FAULT
232	E8	SPEED ACTUATOR - SHIFT / POSITION CONTROL ERROR
252	FC	SPEED ACTUATOR - SHIFT / ENGAGING ERROR - 1 st SPEED
253	FD	SPEED ACTUATOR - SHIFT / ENGAGING ERROR - 2 nd SPEED
254	FE	SPEED ACTUATOR - SHIFT / ENGAGING ERROR - 3 rd SPEED
255	FF	SPEED ACTUATOR - SHIFT / ENGAGING ERROR - 4 th SPEED
256	100	SPEED ACTUATOR - SHIFT / ENGAGING ERROR - 5 th SPEED
257	101	SPEED ACTUATOR - SHIFT / ENGAGING ERROR - 6 th SPEED
258	102	SPEED ACTUATOR - SHIFT / ENGAGING ERROR - REVERSE GEAR
259	103	SPEED ACTUATOR - SHIFT / TEACHING REQUEST GEARBOX BOUNDARY OVERRIDE
260	104	SPEED ACTUATOR - SHIFT / TEACHING REQUEST GEARBOX NEUTRAL POSITION FAILURE
280	118	SPEED ACTUATOR - SHIFT / REFERENCE EOL PARAMETER ERROR (LACKING MEASUREMENT VALUES)
281	119	SPEED ACTUATOR - SHIFT / REFERENCE EOL PARAMETER ERROR (OUT-OF-TOLERANCE MEASUREMENT VALUES)
282	11A	SPEED ACTUATOR - SHIFT / SYSTEM GEOMETRY ERROR (ASYMMETRY)
283	11B	SPEED ACTUATOR - SHIFT / MECHANIC FAULT
300	12C	CLUTCH ACTUATOR / SHORT-CIRCUIT TO EARTH ON CONTROL MOTOR OUTPUT
301	12D	CLUTCH ACTUATOR / SHORT-CIRCUIT TO POSITIVE ON CONTROL MOTOR OUTPUT
302	12E	CLUTCH ACTUATOR / INTERRUPTION ON CONTROL MOTOR OUTPUT
303	12F	CLUTCH ACTUATOR / SENSOR SUPPLY VOLTAGE - TOO LOW
320	140	CLUTCH ACTUATOR / TOO HIGH CURRENT
321	141	CLUTCH ACTUATOR / TOO HIGH TEMPERATURE
330	14A	CLUTCH ACTUATOR / MECHANIC FAULT ON CONTROL MOTOR
331	14B	CLUTCH ACTUATOR / MECHANIC FAULT - LOCK AT END OF STROKE
350	15E	CLUTCH ACTUATOR / POSITION CONTROL ERROR
351	15F	CLUTCH ACTUATOR / NON-PLAUSIBLE POSITION SIGNAL AT THE END OF STROKE
352	160	CLUTCH ACTUATOR / NON-PLAUSIBLE POSITION SIGNAL
354	162	CLUTCH ACTUATOR / MULTIPLE INPUT SPEED PLAUSIBILITY ERROR
355	163	CLUTCH ACTUATOR / MULTIPLE CLUTCH TORQUE PLAUSIBILITY ERROR

DTC DEC	DTC HEX	Description
356	164	CLUTCH ACTUATOR / NO ENGINE REACTION UPON TORQUE REDUCTION REQUEST
380	17C	CLUTCH ACTUATOR / TOO SLOW OR JAMMED CLUTCH MOTION WHEN PERFORMING ACTIVE DIAGNOSIS
381	17D	CLUTCH ACTUATOR / OUT-OF-RANGE CLUTCH MOTION WHEN PERFORMING ACTIVE DIAGNOSIS
382	17E	CLUTCH ACTUATOR / CLUTCH MOTION BEYOND PERMITTED LIMITS
383	17F	ELECTRONIC CONTROL UNIT / DATA MEMORY (EEPROM) ACCESS ERROR WHEN PERFORMING ACTIVE DIAGNOSIS
384	180	CLUTCH ACTUATOR / ERROR DURING CLUTCH CLOSING
385	181	CLUTCH ACTUATOR / ERROR DURING CLUTCH OPENING
386	182	ELECTRONIC CONTROL UNIT / DATA MEMORY (EEPROM) ACCESS ERROR WHEN PERFORMING ACTIVE DIAGNOSIS (TOUCH-POINT)
400	190	DIGITAL OUTPUTS / SHORT-CIRCUIT TO POSITIVE ON ENGINE STARTING ENABLE OUTPUT
401	191	DIGITAL OUTPUTS / SHORT-CIRCUIT TO POSITIVE ON REVERSING LIGHT IGNITION OUTPUT
402	192	DIGITAL OUTPUTS / SHORT-CIRCUIT TO POSITIVE ON PTO ACTUATION ENABLE OUTPUT
403	193	SUPPLY VOLTAGE / SUPPLY VOLTAGE INSIDE THE ECU - TOO HIGH
404	194	SUPPLY VOLTAGE / SUPPLY VOLTAGE INSIDE THE ECU - TOO LOW
405	195	SUPPLY VOLTAGE / ECU SUPPLY VOLTAGE +30 HS - TOO HIGH
406	196	SUPPLY VOLTAGE / ECU SUPPLY VOLTAGE +30 HS - TOO LOW
407	197	SUPPLY VOLTAGE / KEY-CONTROLLED SUPPLY VOLTAGE +15 - TOO HIGH
408	198	SUPPLY VOLTAGE / KEY-CONTROLLED SUPPLY VOLTAGE +15 - TOO LOW
409	199	SUPPLY VOLTAGE / SUPPLY VOLTAGE INSIDE THE ECU - TOO HIGH
410	19A	SUPPLY VOLTAGE / SUPPLY VOLTAGE INSIDE THE ECU - TOO LOW
411	19B	SUPPLY VOLTAGE / ECU SUPPLY VOLTAGE +30 - TOO LOW
412	19C	SUPPLY VOLTAGE / ECU SUPPLY VOLTAGE +30 - TOO LOW
413	19D	GEARBOX OUTPUT REVS SENSOR / SHORT-CIRCUIT TO EARTH
414	19E	GEARBOX OUTPUT REVS SENSOR / NON-PLAUSIBLE SIGNAL
415	19F	GEARBOX OUTPUT REVS SENSOR / WRONG SIGNAL
434	1B2	SUPPLY VOLTAGE / ECU SUPPLY VOLTAGE (+30) DIFFERENT FROM KEY-CONTROLLED VOLTAGE (+15)
435	1B3	GEARBOX OUTPUT REVS SENSOR / NON-PLAUSIBLE SIGNAL
436	1B4	SPEED ACTUATOR - SELECT / NON-PLAUSIBLE SIGNAL FROM THE SENSOR
437	1B5	SPEED ACTUATOR - SHIFT / NON-PLAUSIBLE SIGNAL FROM THE SENSOR
438	1B6	DIGITAL OUTPUTS / SHORT-CIRCUIT TO EARTH ON ENGINE START-PREVENT SIGNAL
439	1B7	DIGITAL OUTPUTS / TOO LOW VOLTAGE ON START-PREVENT SIGNAL
440	1B8	GEAR SELECTOR LEVER / MINOR ANOMALY
441	1B9	GEAR SELECTOR LEVER / SERIOUS ANOMALY - LACKING SIGNAL
442	1BA	GEAR SELECTOR LEVER / GEARBOX LEVER POSITION NOT DEFINED
443	1BB	DIGITAL OUTPUTS / SHORT-CIRCUIT TO EARTH ON REVERSING LIGHT IGNITION SIGNAL

DTC DEC	DTC HEX	Description
444	IBC	DIGITAL OUTPUTS / SHORT-CIRCUIT TO EARTH ON PTO ACTUATION SIGNAL
450	IC2	ELECTRONIC CONTROL UNIT / GENERIC DATA MEMORY (EEPROM) ACCESS ERROR

ELECTRONIC INJECTION SYSTEM

Common rail

General

In order to reduce the emission of particulate, particularly high injection pressures are required.

The system on the new Daily makes it possible to inject the fuel at pressures reaching 1350 bar, while the pressure obtained with electronic system management optimises the work of the engine limiting emissions and consumption.

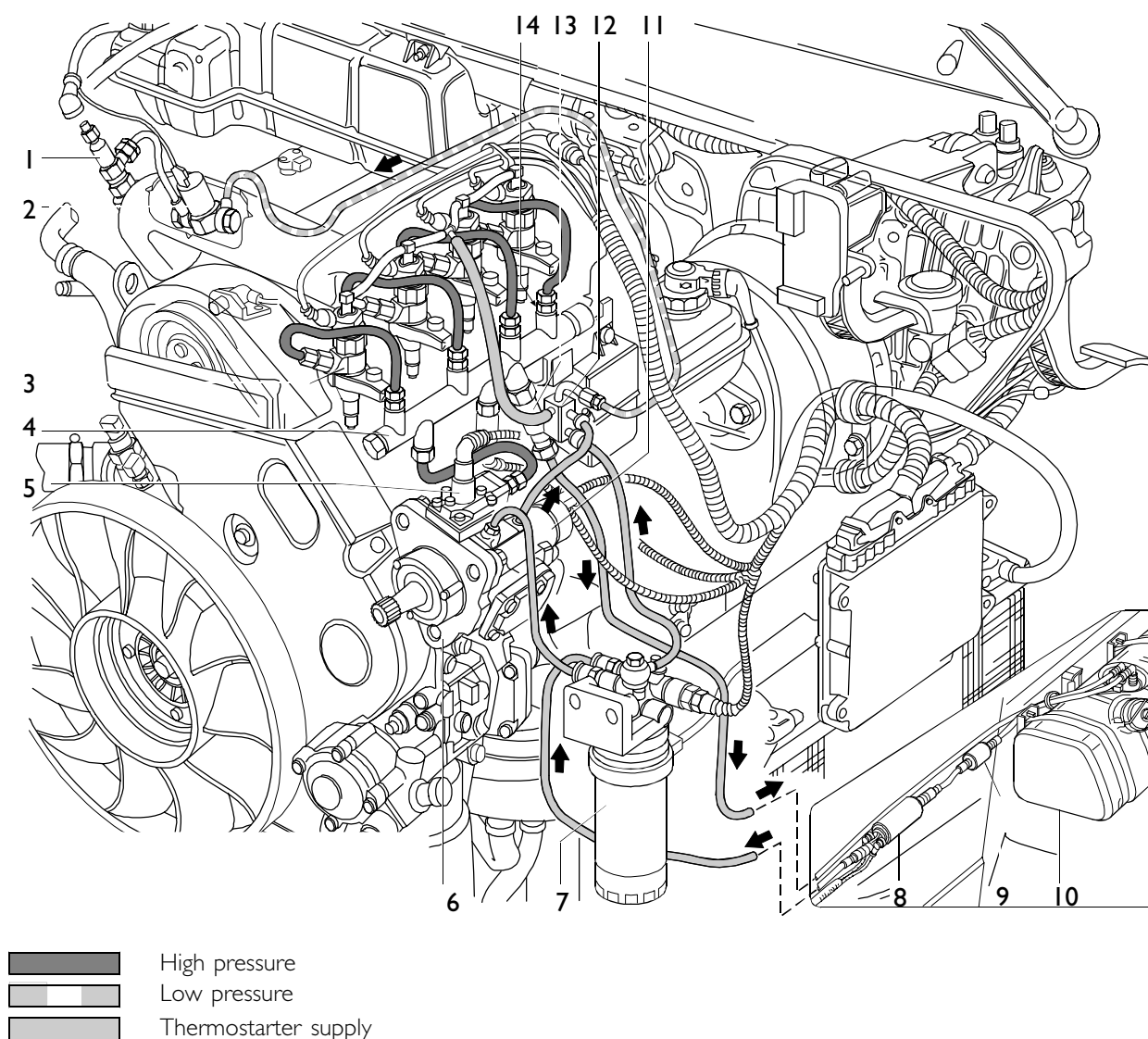
In other diesel systems the high pressure is generated only at the moment of injection for each single cylinder (injection pump or injector-pump).

In the common rail a special pump continuously keeps the fuel at high pressure, regardless of the stroke and of the cylinder that is to receive injection, and accumulates it in a common duct for all the injectors.

Therefore, fuel already at the injection pressure is always present at the inlet of the injectors.

When the solenoid valve of an injector is energised by the electronic control unit, injection takes place in the corresponding cylinder.

Figure 132



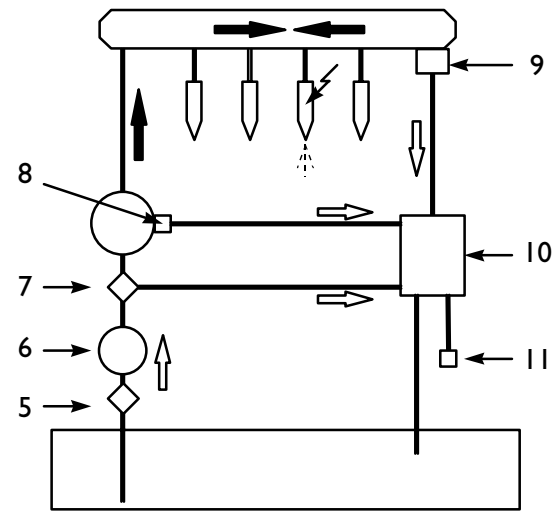
FUEL FEED AND RECIRCULATION DIAGRAM

1. Heat starter plug – 2. Heat starter electro valve – 3. Electrical injector – 4. Rail hydraulic accumulator – 5. 3rd pump cutout electro valve – 6. High-pressure pump – 7. Fuel filter – 8. Electrically driver pump – 9. Prefilter – 10. Reservoir – 11. Pressure regulator valve – 12. Connection – 13. Pressure limiter valve – 14. Flow limiter

In reality the system is much more complex, as the various control and adjustment components have to be integrated.

The MS6.3 electronic control centre monitors the entire system: it receives data from sensors and activates the actuators as required based on input signal processing.

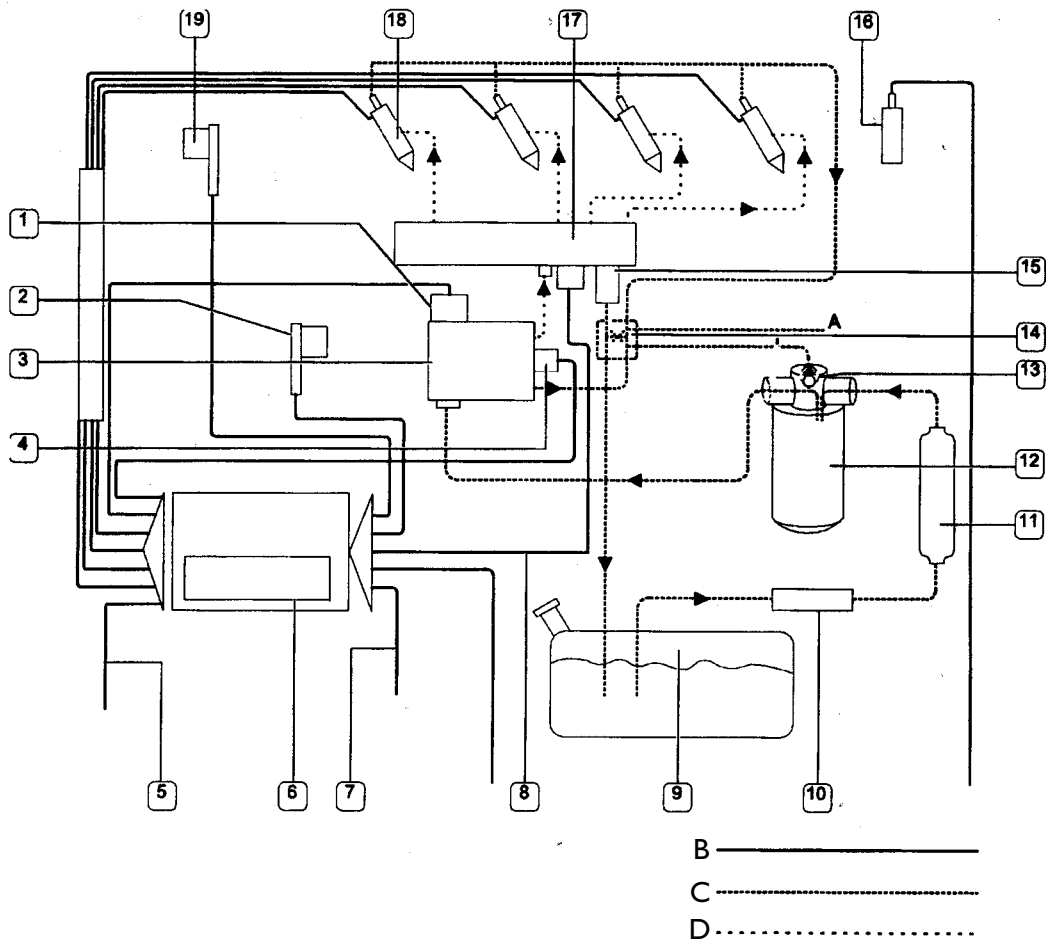
Figure I33



HYDRAULIC LAYOUT

5. Pre-filter – 6. Electric pump – 7. Filter with overpressure valve – 8. Pressure regulator – 9. Pressure limiter – 10. Fuel outlets unit – 11. Thermal starter

Figure I34



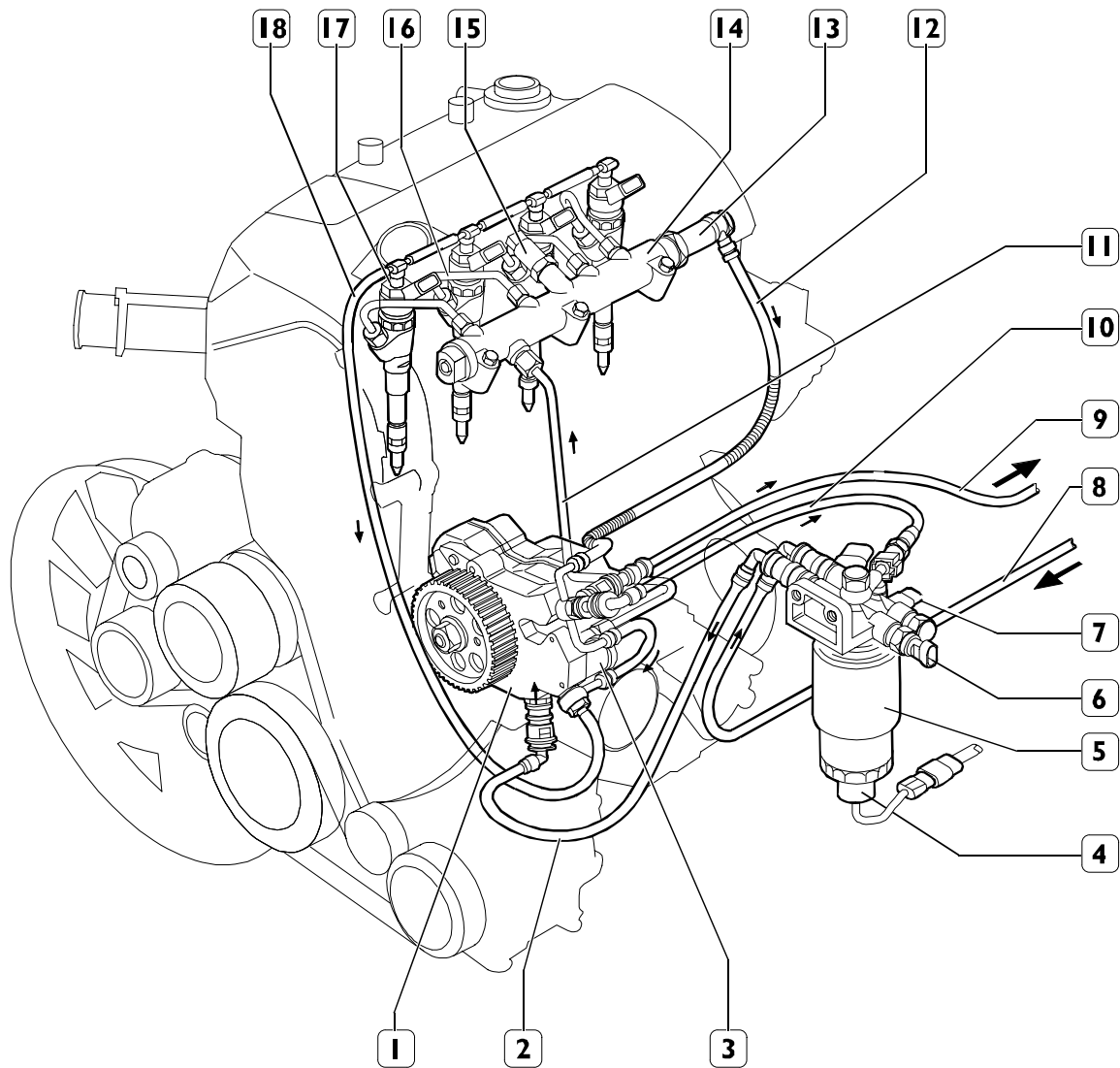
1. 3rd impeller cut-off device – 2. Flywheel rpm sensor – 3. High pressure pump – 4. Pressure regulator – 5. Other actuators (thermal starter, heated filter, fan control, ac control) – 6. Electronic control unit with built-in atmospheric pressure sensor – 7. Other sensors (accelerator, brake, clutch, vehicle speed, water temperature, air temperature) – 8. From rail pressure sensor – 9. Fuel tank – 10. Pre-filter – 11. Electric pump – 12. Fuel filter – 13. Filter overpressure valve – 14. Outlets unit with calibrated hole – 15. Rail pressure limiter – 16. Air flow meter – 17. Common rail – 18. Injectors – 19. Timing sensor on camshaft – A. To thermal starter – B. Electric circuit – C. Low pressure fuel – D. High pressure fuel

Common rail (FIA)

It differs from the Common rail system on previous engines in that injected fuel pressure is 1600 instead of 1350 bars.

Preheat plugs are provided for cold starts.

Figure 135



0003343t

1. High pressure pump – 2. Piping – 3. Pressure regulator – 4. Water in filter sensor – 5. Fuel filter with water separator – 6. Fuel filter clogging sensor – 7. Fuel temperature sensor – 8. Fuel pressure line to filter – 9. Fuel return line to tank – 10. Fuel return line to filter – 11. High pressure piping to common rail – 12. Low pressure return line from rail – 13. Over pressure valve – 14. Common rail – 15. Fuel pressure sensor – 16. High pressure line between rail and electro injectors – 17. Electro injectors – 18. Return line from electro injectors

Hydraulic system (Common Rail – FIA)

The hydraulic system consists of a low and high-pressure circuit, the former consisting of the following piping:

- ☐ connecting high-pressure output to the Rail;
- ☐ Common Rail;
- ☐ connecting the Rail and supplying electro injectors.

The low-pressure circuit consists of the following piping:

- ☐ fuel aspiration from tank to prefilter;
- ☐ supplying the pump and prefilter;
- ☐ supplying the high-pressure pump via the fuel filter.

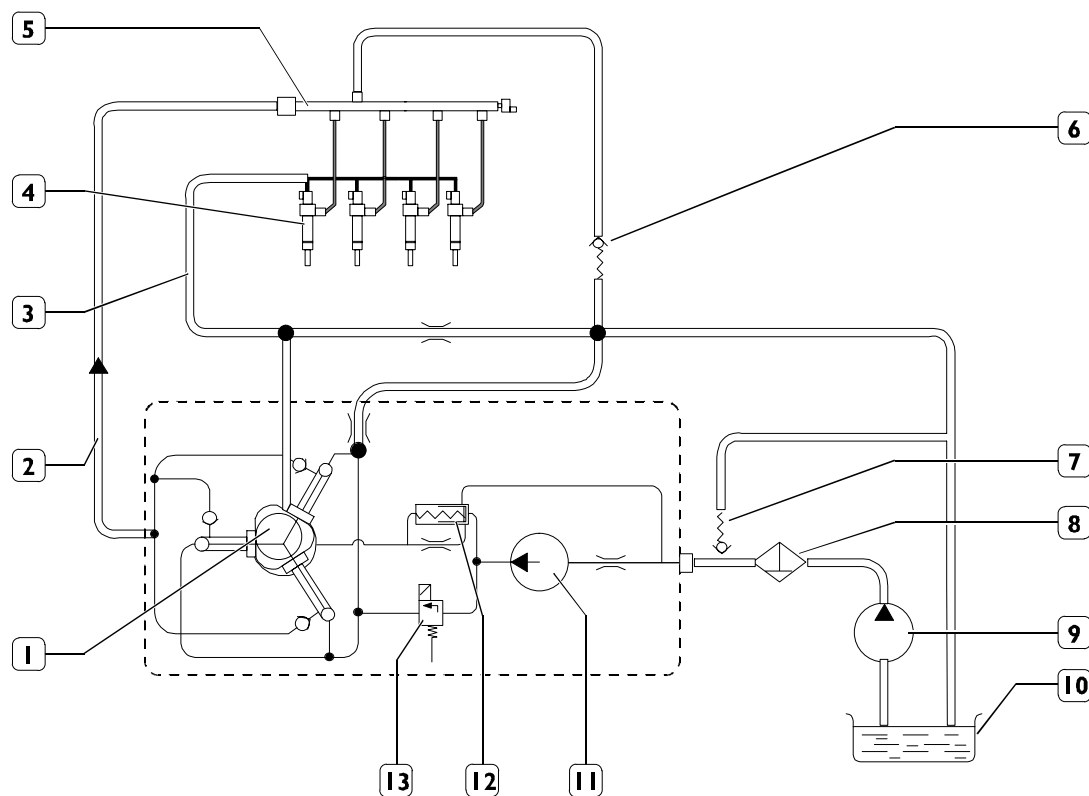
The Rail exhaust circuit and the electro injectors complete the supply circuit.

The following are mandatory for safety reasons because of high-pressure levels inside this hydraulic system:

- ☐ avoid improper connection of high pressure piping junctions;
- ☐ avoid connecting high pressure piping with the engine running (useless and hazardous attempts at purging must be avoided!)

Low-pressure circuit integrity is essential to proper system operation, so mishandling and modifications must be avoided, with immediate action required in case of leaks.

Figure 136

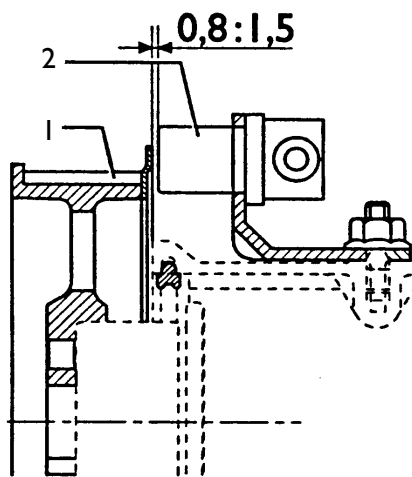
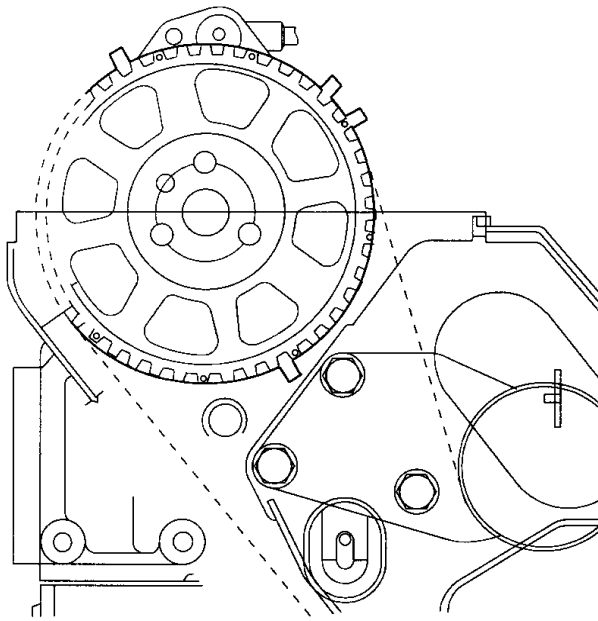


0003340t

1. High pressure pump – 2. High pressure line – 3. Electro injector recirculation line – 4. Electro injectors – 5. Common rail – 6. Over pressure valve – 7. Bypass valve – 8. Filter – 9. Priming pump – 10. Tank – 11. Mechanical supply pump – 12. High pressure pump limiter valve – 13. Pressure regulator

Figure I37

Motor 8040



TECHNICAL VIEWS OF PULLEY AND SENSOR
1. Camshaft pulley – 2. Sensor

SYSTEM COMPONENTS

Camshaft pulley and timing sensor

5 teeth (4 + 1 for timing detection) are machined on the pulley opposite an inductive sensor equal to the one on the flywheel.

5 holes (4 + 1 timing recognition hole) are provided on the FIA engine pulley.

The sensor is of the inductive type and is used by the electronic control unit to detect the engine stroke during synchronisation (when starting).

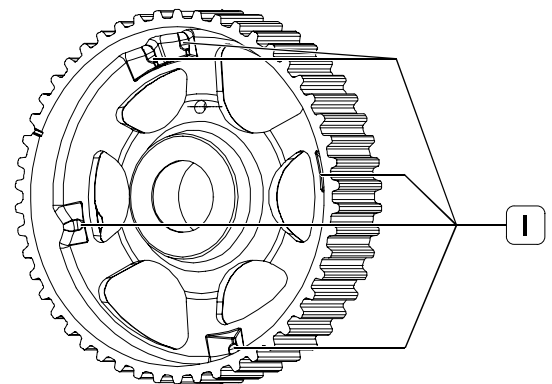
When assembling the sensor:

ensure the dimension illustrated working on the slots of the bracket and measuring with a thickness gauge

- ☐ make sure that the sensor is perpendicular in relation to the wheel tightening the nuts.

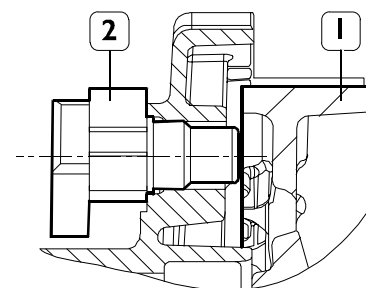
Figure I38

Motor FIA



0003320t

TIMING RECOGNITION HOLES



0003321t

1. Camshaft pulley – 2. Sensor

Flywheel and rpm sensor

The flywheel has a set of 58 holes (60 – 2), as illustrated, for generating the signal for the inductive sensor.

It is the same as the one on the camshaft and is positioned on the crankcase and detects the passage of the 58 holes on the flywheel.

The change in the signal caused by the lack of 2 holes (synchronism gap) which takes place at each rotation of the crankshaft is the reference signal that allows the control unit to recognise the advance in relation to the TDC of piston pair 1–4.

This signal is also used by the control unit to detect the engine rotation speed, the duration of injection and to control the rev counter.

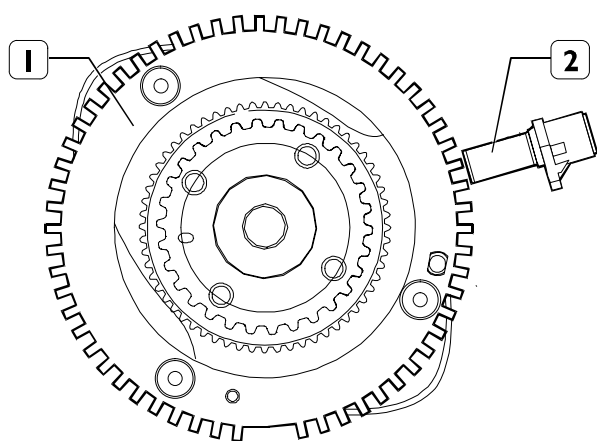
When assembling the sensor:

- ☐ ensure the dimension illustrated working on the slots of the bracket and measuring with a thickness gauge.
- ☐ make sure that the sensor is perpendicular in relation to the wheel tightening the nuts.

The FIA engine crankshaft sensor features a slatted sound wheel on the crankshaft front.

This features 58 (60–2) teeth and the sensor detects their passage.

Figure 139

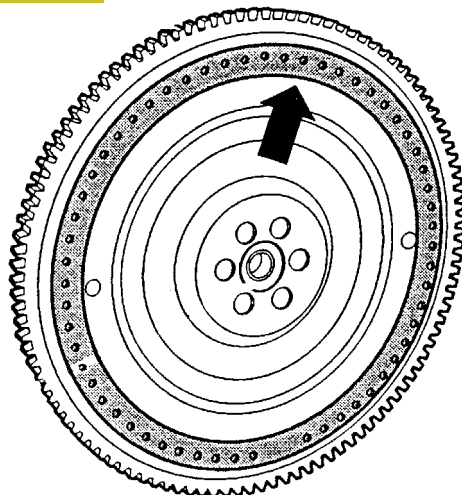


0003319t

TECHNICAL VIEW OF THE SOUND WHEEL AND SENSOR

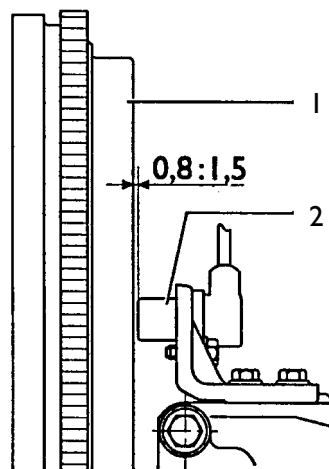
1. Sound wheel – 2. Crankshaft sensor

Figure 140



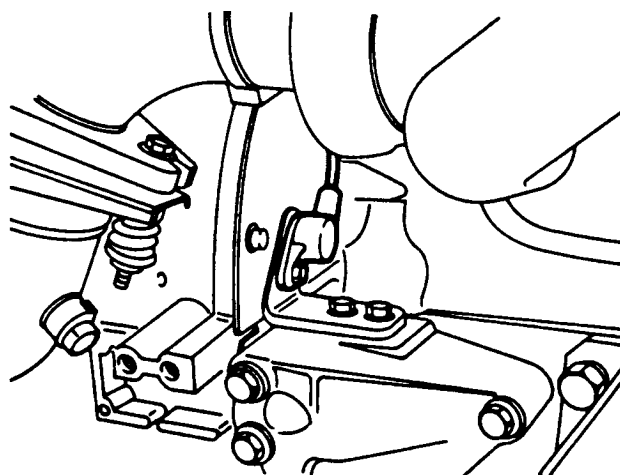
HOLES FOR GENERATING SIGNALS ON THE FLYWHEEL

Figure 141



TECHNICAL VIEW OF FLYWHEEL AND SENSOR
1. Flywheel – 2. Sensor

Figure 142



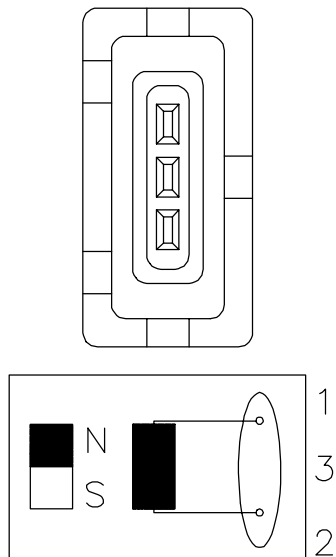
RPM SENSOR ON CRANKCASE

Figure 143



SENSOR AND CONNECTION CABLE

Figure 144



SENSOR CONNECTOR AND WIRING DIRAGRAM

Flywheel and camshaft sensor specifications

These are inductive sensors.

The flywheel sensor (48035) is connected at pins 29 and 37 of connector A of the control unit.

It is also used to control the electronic rev counter on the instrument cluster.

The timing sensor (48042) is connected at pins 4 and 31 of connector A of the control unit.

The resistance at 20 °C is approx. ~ 860 Ohm.

Pre-filter

This is a simple transparent filter like the one in petrol supply systems.

Replace it only when a sight check reveals traces of impurity.

Electric pump

Rotary volumetric pump with integrated by pass (Figure 145) located outside the fuel tank.

It is on the left-hand side of the frame in an accessible area.

There is no need to bleed the fuel supply system.

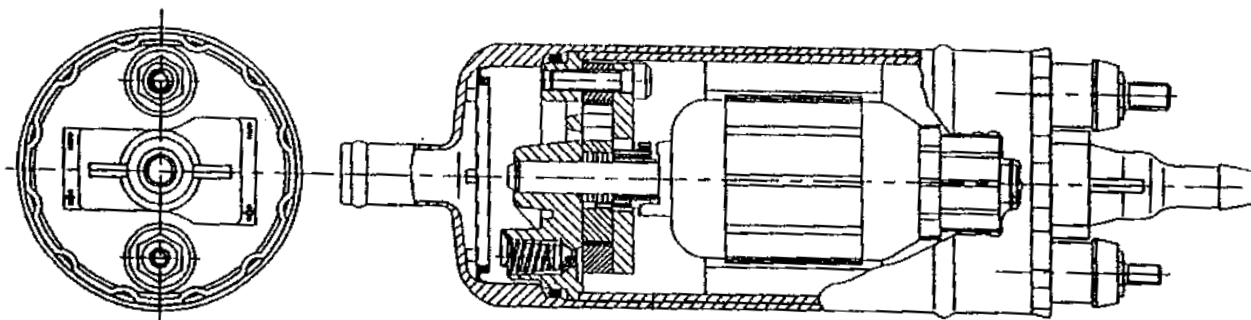
The pipes are fitted with quick couplings.

It is connected to pins 7 and 8 of connector A of the control unit.

Specifications

Delivery pressure:	2,5 bar
Flow rate:	> 155 litres/h
Supply:	13,5 V – < 5 A
Coil resistance at 20 °C:	28,5 Ohm

Figure 145



TECHNICAL VIEW AND CROSS SECTION OF ELECTRIC FUEL PUMP

Fuel filter

Cartridge degree of filtering: 5 micron

Differential operating pressure (obstruction indicator): 0.6 bar

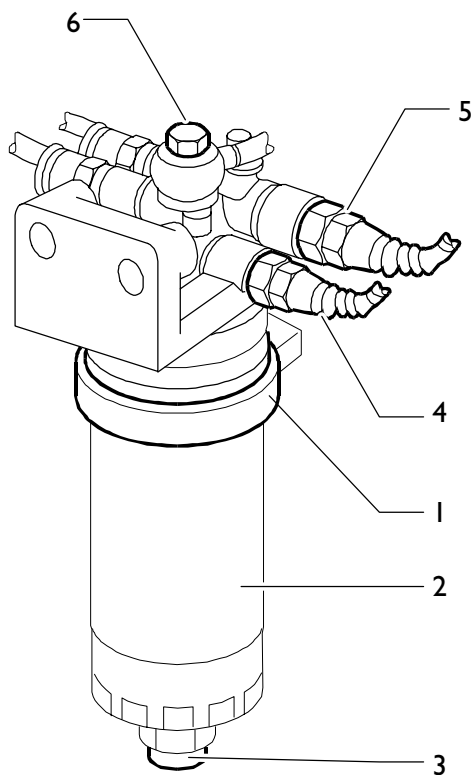
The fuel temperature sent by the sensor to the electronic injection control unit allows very accurate calculation of the flow rate of the fuel to be injected in the cylinders.

It is located in a fairly accessible position in the left front part of the engine compartment.

1. Actuated by ECU via relay with fuel temperature below 3°C.
2. Filtering element
3. Shows presence of water through a warning lamp on instrument panel.
4. Differential pressure sensor calibrated at 0,6 bar : any clogging is shown by warning lamp
5. It is a NTC sensor connected to EDC for fuel temperature reading enabling electronic control to calculate the amount of diesel oil to be injected into the cylinders.
6. Installed on filter support for excess fuel return to tank.

Figure 146

Engine 8040



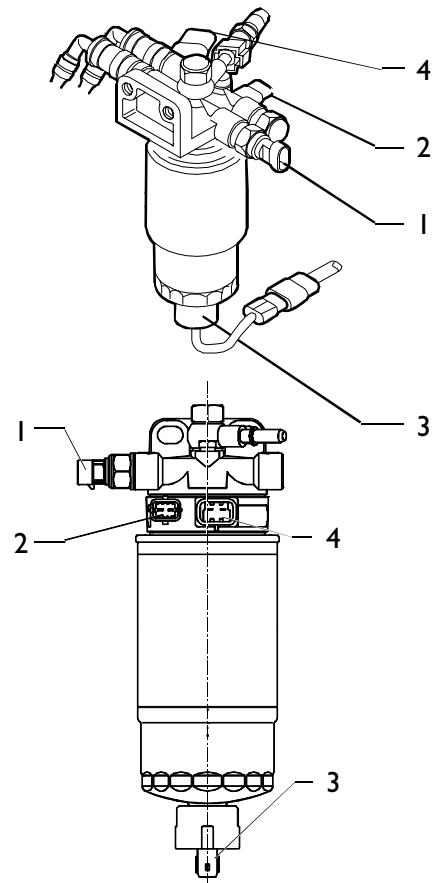
73731

FUEL FILTER

1. Heater – 2. Filtering element – 3. Condensation indicator – 4. Obstruction indicator – 5. Fuel temperature sensor – 6. Discharge

Figure 147

Engine F1A



003312t

1. Clogged filter sensor – 2. Fuel temperature sensor – 3. Water sensor – 4. Heater

High pressure pump

This pump has three radial pistons (total displacement 0,7 cc) controlled by the timing gear belt, with no need for calibration.

Each impeller unit comprises:

- ☐ a piston (5) operated by a cam (2) integral with the pump shaft (6)
- ☐ a plate inlet valve (3)
- ☐ a ball delivery valve (4)

The delivery pressure to the rail is modulated between 250 and 1350 bar by the electronic control unit, acting on the pressure regulator solenoid valve.

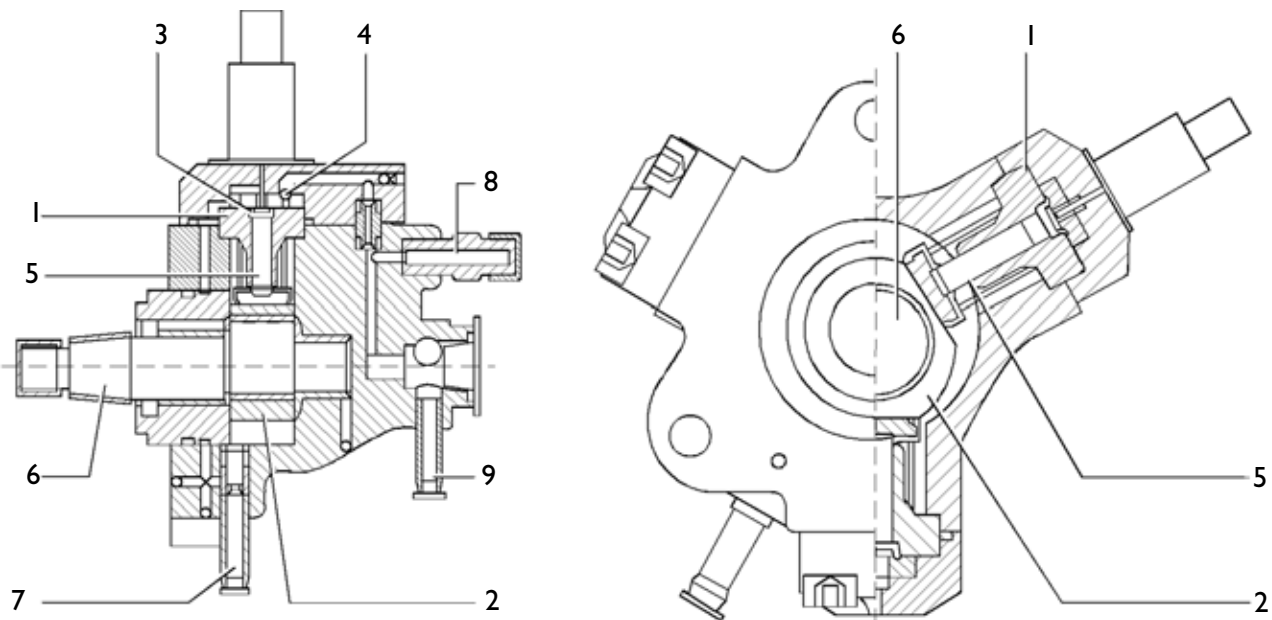
The pump is lubricated and cooled by the fuel itself.

The 3rd impeller cut-off solenoid valve (78015) serves to increase the part of fuel oil that lubricates the radialjet pump, it is used above 4200 rpm (when power is no longer needed) and keeps the intake valve open also when the cylinder is compressing. It sends one third of the capacity of the high pressure pump back to low pressure.

The on-off time is considerably lower than conventional injection pumps.

The control solenoid valve is connected to pins 20 and 21 of connector A of the control unit.

Figure 148



TECHNICAL VIEW AND CROSS SECTION OF HIGH PRESSURE PUMP

1. Cylinder – 2. Cam – 3. Plate inlet valve – 4. Ball delivery valve – 5. Piston – 6. Pump shaft – 7. Low pressure fuel inlet – 8. High pressure fuel delivery – 9. Fuel backflow

High-pressure pump (FIA engine)

This pump is similar to the 8140 engine one.

It differs in Rail line pressure, modulated at between 250 and 1600 bars by the electronic centre by acting on the pressure regulator electro valve and is not provided with a 3rd pump cutout electro valve (78015).

Pump (3) is oriented on the pump shaft cam.

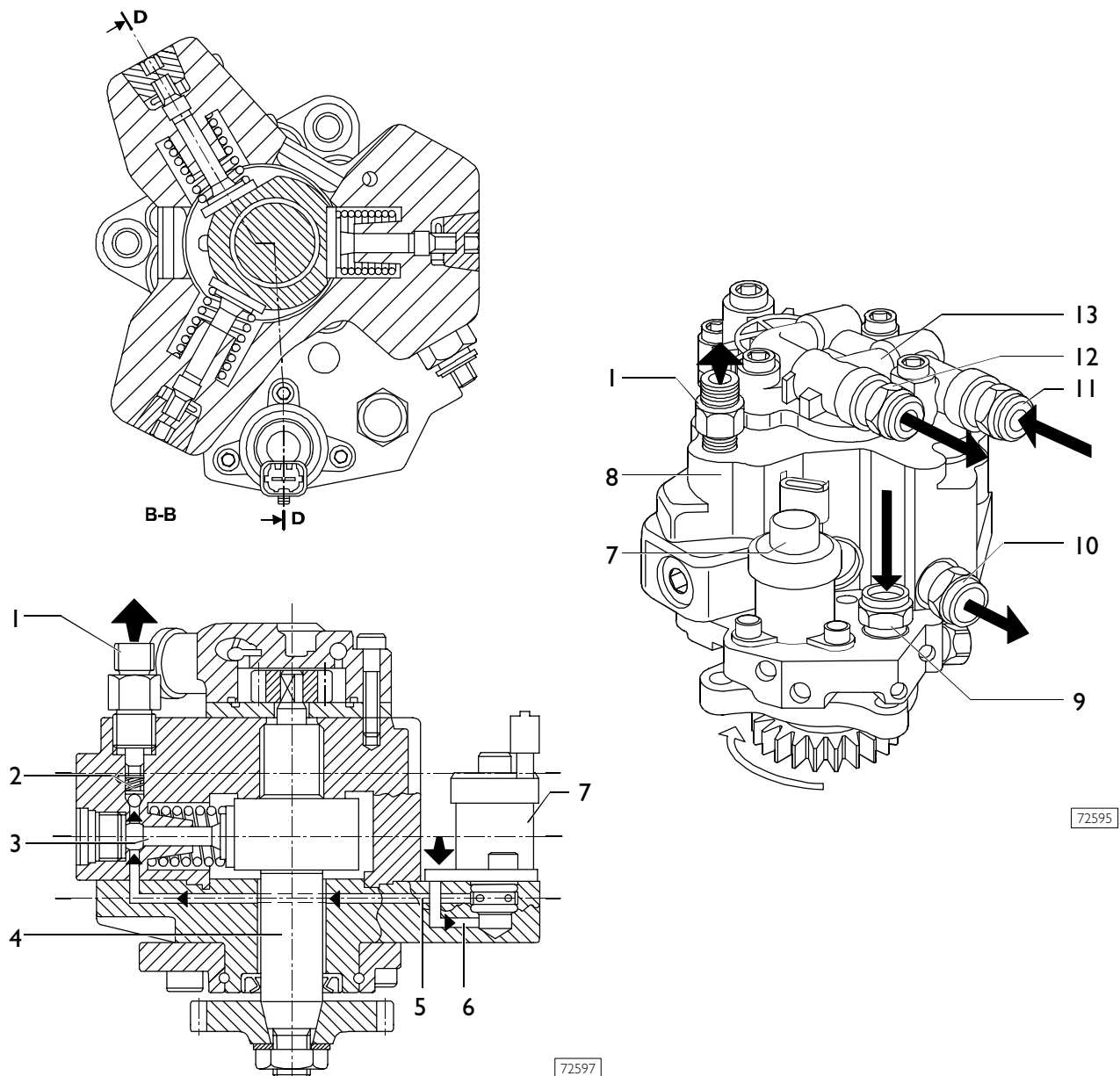
During aspiration the pump is powered via supply conduit (5).

The amount of fuel to be sent to the pump is set by pressure regulator (7).

The pressure regulator stops fuel inflow to the pump based on PWM input from the centre.

During pump compression, fuel reaches a pressure level sufficient to open the pressure valve to common rail (2), and supplies it via output (1).

Figure 149



1. Output for rail pressure - 2. Rail pressure valve - 3. Pump - 4. Pump shaft - 5. Pump supply conduit - 6. Pressure regulator supply conduit - 7. Pressure regulator - 8. High pressure pump - 9. Fuel input connection from filter - 10. Fuel output connection to filter support - 11. Fuel input connection from centre heat exchanger - 12. Fuel output connection from mechanical pump to filter - 13. Supply mechanical pump

Pressure regulator

This is located on the high pressure pump and modulates the fuel pressure to the rail (and thus to the injectors) on the basis of the commands received from the electronic control unit.

It mainly comprises:

- ☐ a ball shutter (1)
- ☐ a valve (3) control pin (2)
- ☐ a preloading spring (4)
- ☐ a coil (5)

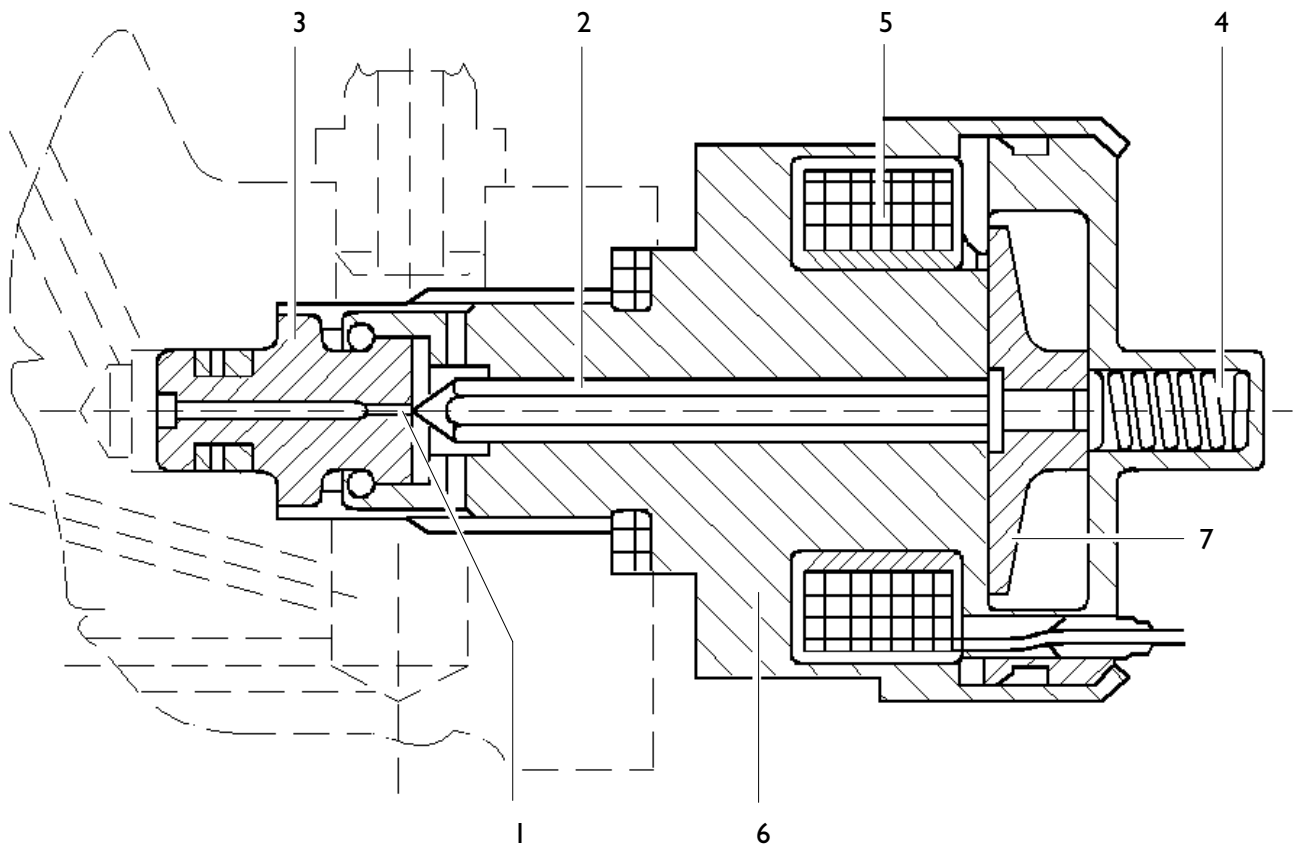
When the solenoid is not energised, the delivery pressure depends only on the spring preload.

When, after processing the various engine operating parameters, the electronic control unit determines the need for a different injection pressure, it modulates a command signal in PWM to the regulator solenoid valve which discharges the excess pressure in the pump backflow duct.

The modulated pressure is checked through the pressure sensor on the rail by the control unit, which, if necessary, suitably modifies the intensity of the command signal to reach the required result.

The control solenoid valve (78013) is connected to pins 9 and 20 of connector A of the control unit.

Figure 150



PRESSURE REGULATOR CROSS SECTION

1. Ball shutter – 2. Ppin – 3. Valve – 4. Preloading spring – 5. Coil – 6. Body – 7. Anchor

Pressure regulator (FIA)

Similar to the one installed on the 8140 engine and mounted on the CP3 pump low-pressure circuit.

When the engine control centre pilots the pressure regulator via the PWM signal, solenoid (1) is activated, which in its turn generates movement of magnetic core (2).

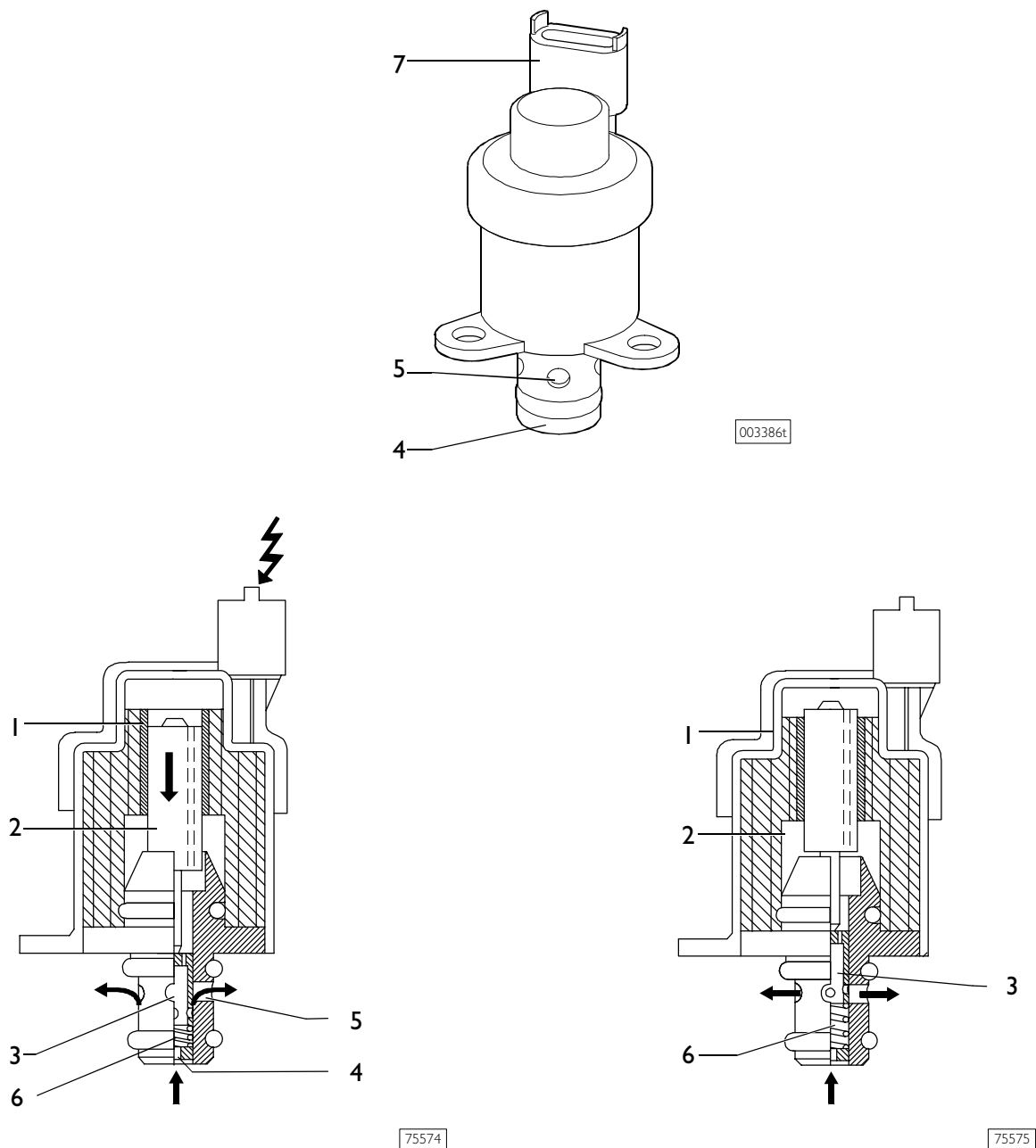
Core movement causes cylinder (3) axial displacement by fuel delivery partialization.

When solenoid (1) is not activated, the magnetic core is moved to its rest position by preload spring (6).

In these conditions, cylinder (3) is in a position to offer maximum fuel passage cross-section.

Control electro valve 78013 is connected to centre connector A pins 9 and 20.

Figure 151



1. Solenoid - 2. Magnetic core - 3. Cylinder - 4. Fuel input - 5. Fuel output - 6. Preload spring - 7. Connector

Rail (pressure accumulator)

The volume of the rail is small in size (29 cc) to allow quick pressurising when starting, at idle speed and in the event of high loads.

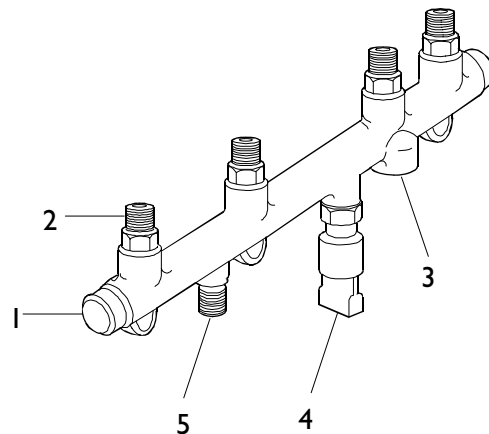
It has however sufficient volume to minimise the plenum effect caused by opening and closing of the injectors and operation of the high pressure pump.

Screwed onto the rail there is a fuel pressure sensor. The signal sent by this sensor to the electronic control unit is feedback information on the basis of which correct actuation of the rail pressure is checked.

Common Rail for engines with serial numbers over 3089322

Temporary solution with simulated flow limiters and over pressure valve connected to the fuel discharge unit.

Figure I52

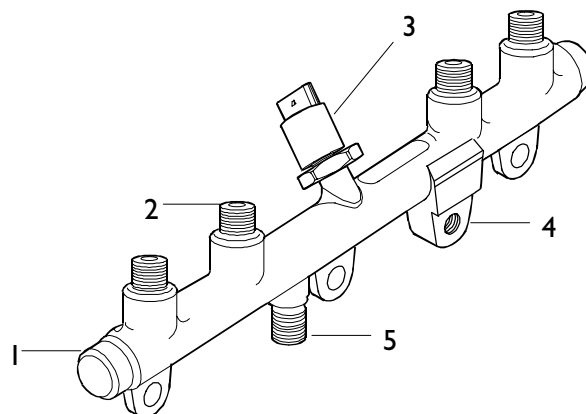


1. Common Rail – 2. Simulated flow limiter – 3. Simulated over pressure valve seat – 4. Pressure sensor – 5. Fuel input from high-pressure pump

Common Rail for engines produced since end-October 2000

Final Common Rail solution without flow limiters and over pressure valve. The new Common Rail features different pressure sensor layout and discharge unit connection predisposition.

Figure I53



1. Common Rail – 2. To electrical injectors – 3. Pressure sensor – 4. Fuel discharge unit attachment – 5. Fuel input from high-pressure pump

Rail (pressure accumulator – FIA)

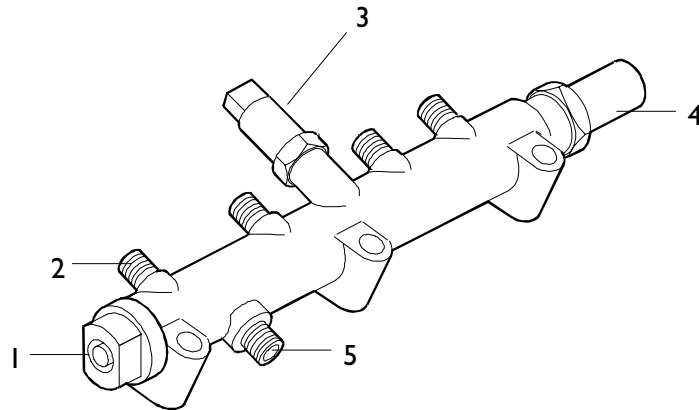
The hydraulic accumulator is mounted in the cylinder head on the side opposite aspiration.

Its some 22-cm³ volume dampens fuel pressure oscillations due to:

- ☐ high-pressure pump operation
- ☐ electro injector opening.

The fuel sensor and overpressure valve are positioned on the hydraulic accumulator.

Figure 154



75576

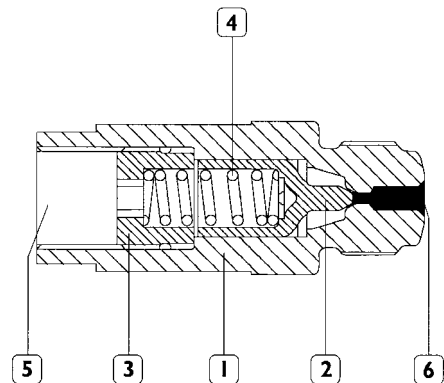
1. Common rail – 2. To electro injectors – 3. Fuel pressure sensor – 4. Overpressure valve – 5. Fuel input from high pressure pump

Over pressure valve

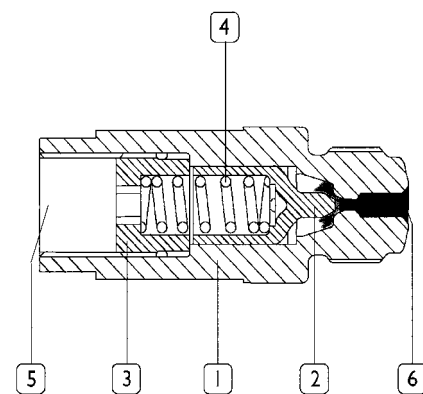
The over pressure valve protects system components in the event that fuel pressure exceeds rated 1750 bars.

- A The piston's conical end normally keeps discharge to tank open.
- B In the event of fuel pressure being over 1750 bars in the rail, the piston is relocated and excess pressure is discharged into the tank.

Figure 155



70500

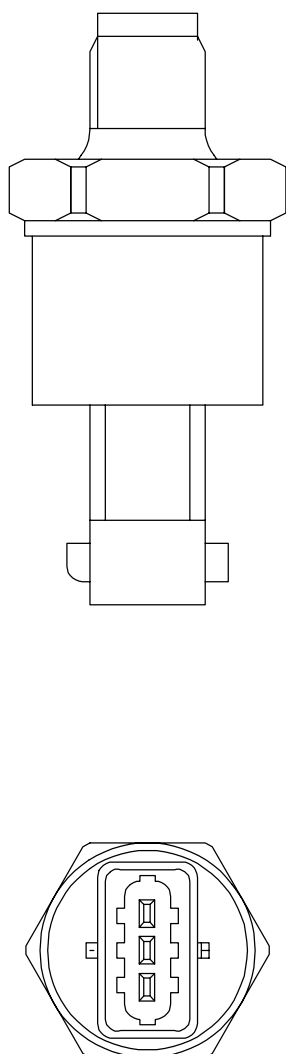


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OVER PRESSURE VALVE

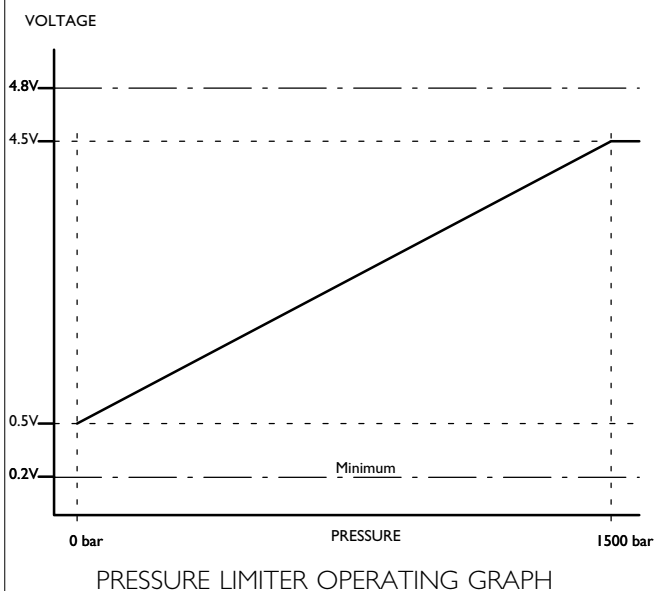
1. Body – 2. Piston – 3. Stop – 4. Spring – 5. Direct discharge to tank – 6. Seat on the rail

Figure 156



TECHNICAL VIEW OF COMPONENT AND LIMITER CONNECTOR

Figure 157



Flow limiters

These are located on the rail fuel outlet and their function is to protect the engine or vehicle in the event of internal leaks (for example, atomiser blocked open) and external leaks (example: high pressure pipe damaged).

Under these circumstances they also allow, within certain limits, the system to work through the components that are still intact.

The passage of fuel from the rail to the injectors takes place through the holes machined on the small diameter of the piston.

Under normal conditions, the fuel pressure is exerted on both sides of the piston, held in the open position by the spring.

In the event of a heavy loss of pressure downstream of the limiter, the inlet pressure becomes preponderant and moves the piston from the opposite side, obstructing the fuel outlet.

Pressure limiter

The pressure limiter (1500 bar) screwed in the rail, serves to protect the system components in the event of a failure to the pressure regulator on the pump.

If the pressure of the fuel in the rail exceeds 1500 bar the piston is raised and the excess pressure is eliminated.

Fuel pressure sensor

This is fitted at the centre of the rail and measures the existing fuel pressure in order to determine the injection pressure.

The injection pressure value is used as feedback for closed loop pressure control and to determine the duration of the electric command for injection.

It is connected to pins 6, 13 and 33 of connector A of the control unit.

It is supplied at 5 Volt.

Injectors

The solenoid valve controls the lift of the atomiser needle.

On the fuel inlet union a filter protects the injector for impurities. The injector is constructively the same as conventional ones, except that there is no needle return spring.

Access to the injectors is gained by releasing the side soundproof cover from the cylinder head. The fuel recovery pipe has a quick coupling.

The injector comprises two parts:

- ☐ actuator – atomiser composed of pressure rod (1), pin (2) and nozzle (3)
- ☐ control solenoid valve comprising a coil (4) and drive valve (5).

1st phase: rest position

The coil (4) is not activated and the shutter (6) is in the closed position.

The same fuel pressure acts in both the control area (7) and in the pressure chamber (8), but as the shutter (6) is closed, the needle (2) cannot be raised.

2nd phase: start of injection

The coil (4) is energised and causes the shutter (6) to move upwards.

The fuel of the control volume (9) flows towards the backflow duct (10) causing a drop in the pressure in the control area (7).

At the same time, the pressure of the fuel in the pressure chamber (8) causes the needle (2) to rise, resulting in fuel injection to the cylinder.

3rd phase: end of injection

The coil (4) is not activated and makes the shutter (6) return to the closed position, which re-creates a balance of forces that makes the needle (2) return to the closed position and consequently end injection.

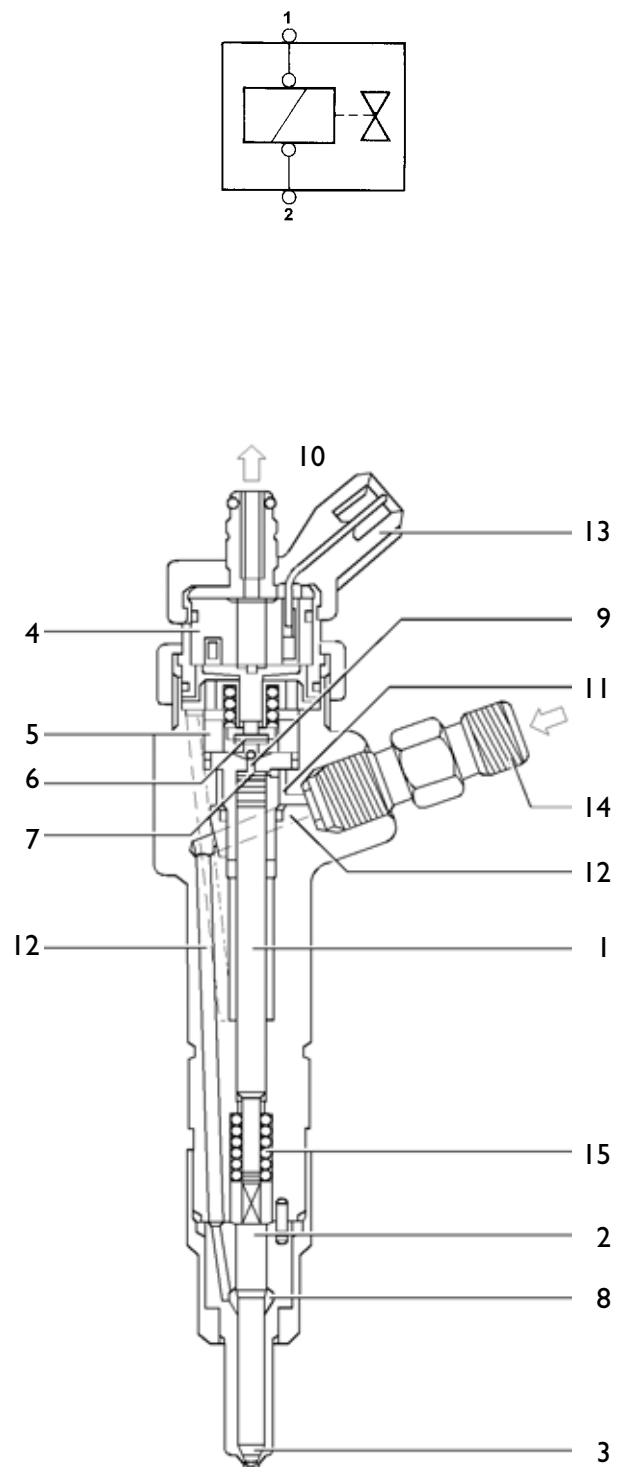
Injectors (78247)

The solenoid valve is of the N.C. type.

The injectors are connected individually to the control unit at the following pins:

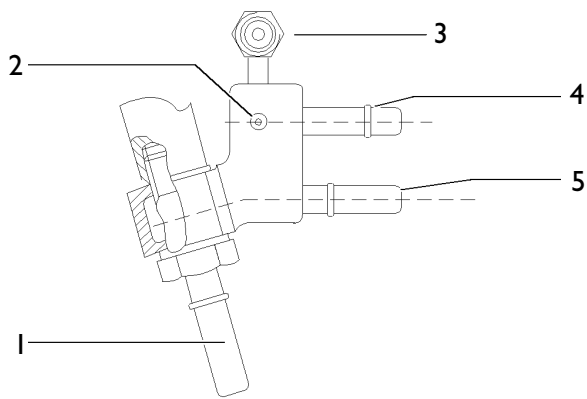
- ☐ A12 / A40 cylinder 1 injector
- ☐ A10 / A43 cylinder 2 injector
- ☐ A23 / A42 cylinder 3 injector
- ☐ A24 / A41 cylinder 4 injector

Figure 158



INJECTOR WIRING DIAGRAM AND CROSS SECTION
 1. Pressure rod – 2. Needle – 3. Nozzle – 4. Coil – 5. Pilot valve – 6. Ball shutter – 7. Control area – 8. Pressure chamber – 9. Control volume – 10. Backflow duct – 11. Control duct – 12. Supply duct – 13. Electrical connection – 14. High pressure fuel inlet – 15. Spring

Figure 159



FUEL OUTLET UNIT

1. Fuel outlet to tank – 2. Return from injectors –
3. Delivery to thermal starter – 4. Leading from high
pressure pump – 5. Arrival from filter

Fuel outlets unit

All the outlets, overpressure and backflows of the various hydraulic components converge in this unit.

Part of them is made available to the thermal starter, while the rest returns to the fuel tank.

In the unit there is a 2,3 mm diameter drain hole.

This hole regulates the whole outlet system to maintain a constant counter pressure of 0,5 bar, which is indispensable for correctly supplying the thermal starter and it ensures flow rates which prevent overheating of the fuel.



The calibrated hole is made in a union similar to the other unions. In the event of work on the system, take care not to fit another different union in its place as this would cause serious operating failures.

Air flow meter

This component incorporates a temperature sensor and a pressure sensor.

It is fitted on the engine intake manifold (Figure 162) and measures the maximum flow rate of the intake air which is used to accurately calculate the amount of fuel to be injected at each cycle.

It is connected to the control unit on pins A2 / A3 / A19 / A34.

Pin 1 sensor – Pin A19 ECU	–	earth –
Pin 2 sensor – Pin A2 ECU	–	temperature signal
Pin 3 sensor – Pin A3 ECU	–	5V – supply
Pin 4 sensor – Pin A34 ECU	–	0 ÷ 5V pressure signal –

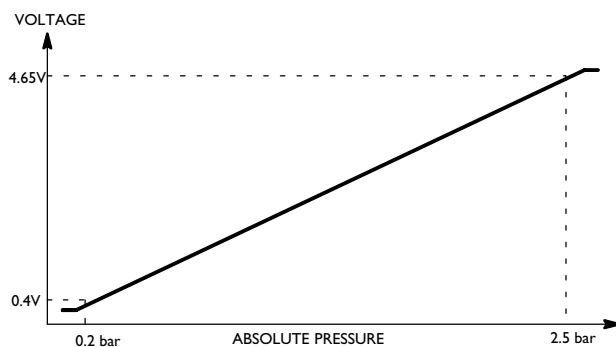
Course of sensor in relation to the temperature:

Temperature	Resistance
– 40 °C	48.50 kOhm
– 20 °C	15.67 kOhm
0 °C	5.86 kOhm
20 °C	2.50 kOhm
40 °C	1.17 kOhm
60 °C	0.59 kOhm
80 °C	0.32 kOhm
100 °C	0.18 kOhm
120 °C	0.11 kOhm

Course of sensor in relation to the pressure:

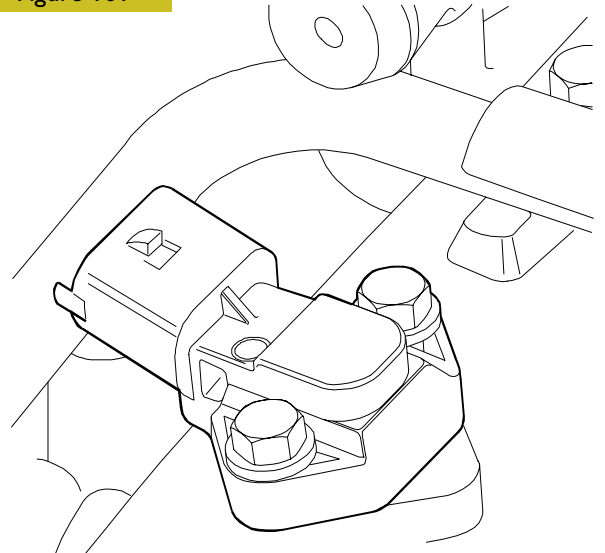
See graph opposite.

Figure 160



AIR FLOW METER OPERATING GRAPH

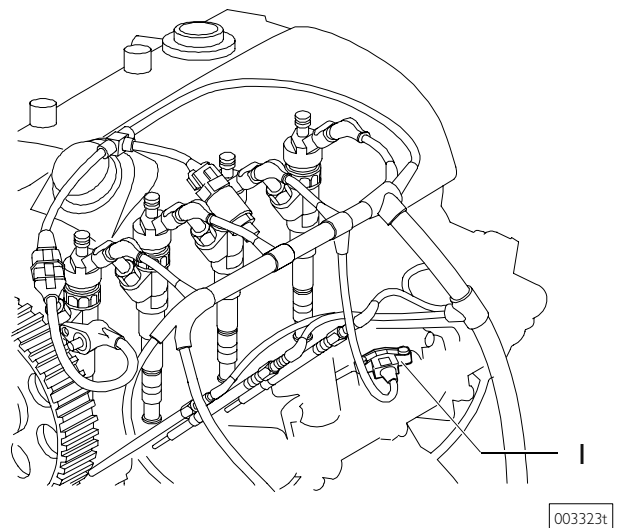
Figure 161



AIR FLOW METER

8660

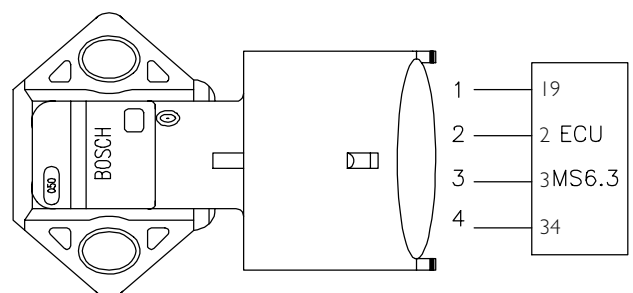
Figure 162



I. Air flow meter location

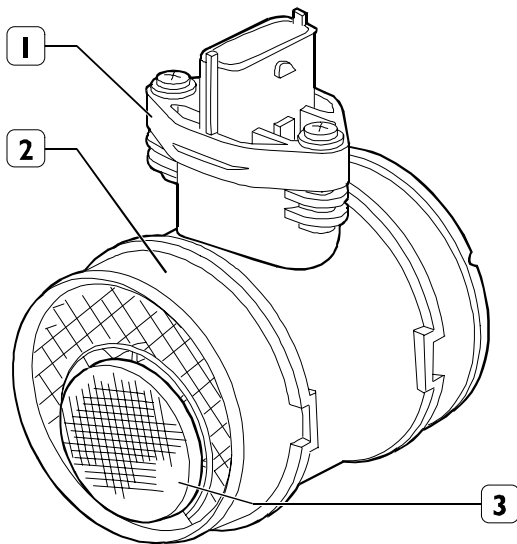
003323t

Figure 163



AIR FLOW METER CONNECTION

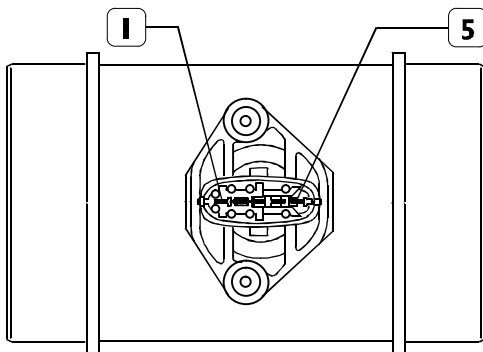
Figure 164



003333t

1. Connector – 2. Gauge body – 3. Air input grid

Figure 165



003334t

TECHNICAL VIEW OF GAUGE CONNECTOR

Air delivery gauge

Used in the EGR version to replace the one mounted on the engine aspiration manifold.

The gauge is of the heated film type and is located on their aspiration conduit between the turbine and the air filter.

The gauge contains the aspired air temperature sensor.

It is connected to the centre on pins **A5 / A17 / A18 / A26 / A28**.

- Pin 1 sensor – Pin A5 ECU – temperature signal
- Pin 2 sensor – Pin A17 ECU – 5V power supply
- Pin 3 sensor – Pin A18 ECU – mass
- Pin 4 sensor – Pin A26 ECU – reference voltage
- Pin 5 sensor – Pin A28 ECU – pressure signal

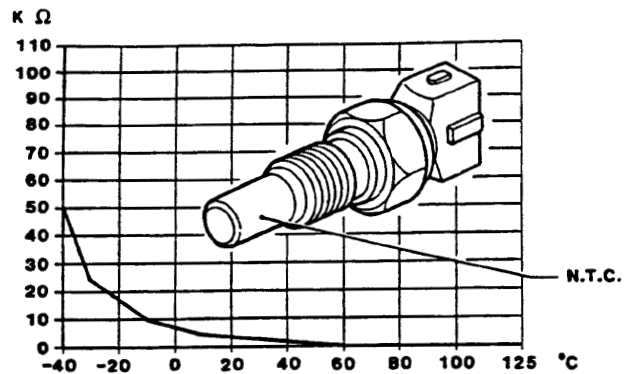
The operating principle is based on a heated membrane inserted in a measurement canal through which air to the engine flows.

The hot film membrane is kept at a constant temperature some 120 °C above incoming air level by the heating resistor.

The air mass traversing the measurement canal tends to subtract heat from the membrane so current must cross the resistor to maintain constant film temperature.

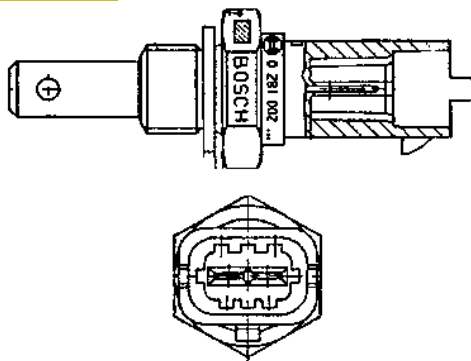
Current absorbed is proportional to the air mass flowing to the engine and is measured with a Wheatstone bridge and the signal is forwarded to the electronic centre.

Figure 166



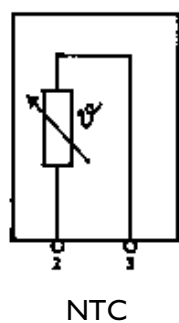
COURSE OF SENSOR RESISTANCE IN RELATION TO TEMPERATURE

Figure 167



TECHNICAL VIEW OF ENGINE COOLANT TEMPERATURE SENSOR

Figure 168



WIRING DIAGRAM

Atmospheric pressure sensor

This is integrated inside the control unit.

It measures the atmospheric pressure to correct the flow rate in relation to the altitude.

Engine coolant temperature sensor

This is an NTC sensor located on the thermostat box.

It detects the temperature of the coolant fluid to give the control unit information about the engine temperature conditions.

It is connected to pins 1 and 30 of connector A of the control unit.

Course of the sensor in relation to the temperature:

Temperature	Resistance
- 40°C	48.30 kOhm
- 20°C	15.46 kOhm
0°C	5.89 kOhm
20°C	2.50 kOhm
40°C	1.17 kOhm
60°C	0.59 kOhm
80°C	0.32 kOhm
100°C	0.19 kOhm
120°C	0.11 kOhm

Fuel temperature sensor

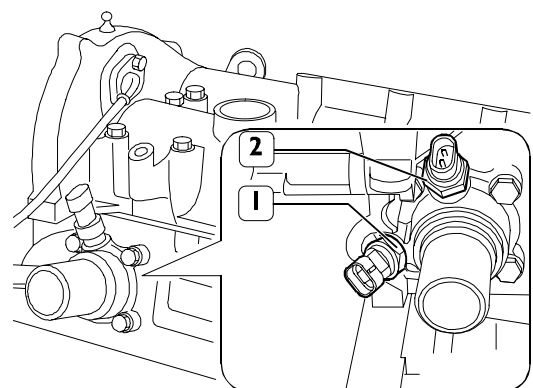
This is an NTC sensor located on the fuel filter.

It detects the temperature of the fuel to give the control unit information about the fuel oil temperature conditions.

It is connected to pins 15 and 30 of connector A of the control unit.

It is exactly the same as the engine coolant temperature sensor.

Figure 169



003324t

LOCATION OF FIA ENGINE COOLANT TEMPERATURE SENSOR
1. EDC – 2. Signal instrument panel signal

The accelerator pedal position sensor is of the potentiometric type with incorporated N.O. minimum switch.

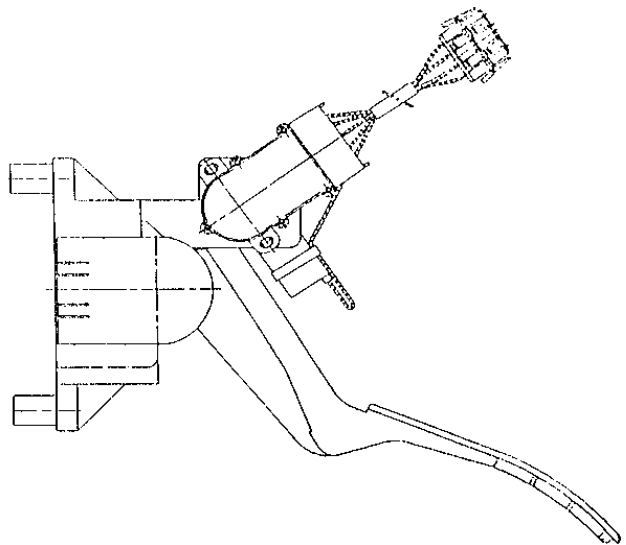
It gives the electronic control unit a value proportionate with the operating angle of the pedal itself determining fuel delivery.

It is supplied at 5 Volt.

The potentiometer resistance is approx. $\sim 1 \text{ k}\Omega$.

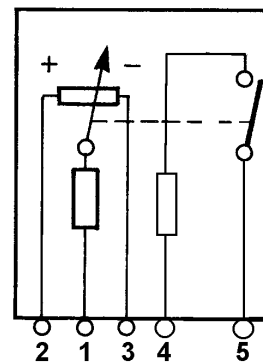
It is connected at pins 2, 13, 27, 29 and 35 of connector B of the control unit.

Figure 170



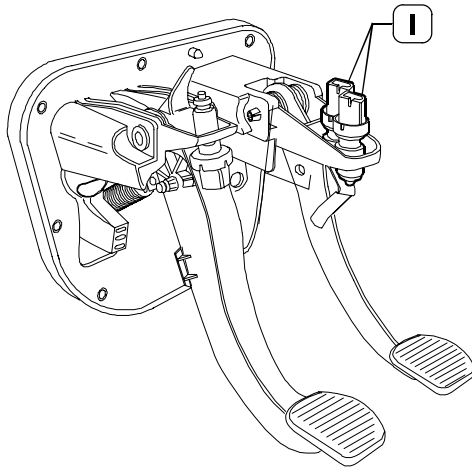
LOCATION OF SENSOR ON ACCELERATOR PEDAL

Figure 171



SENSOR WIRING DIAGRAM

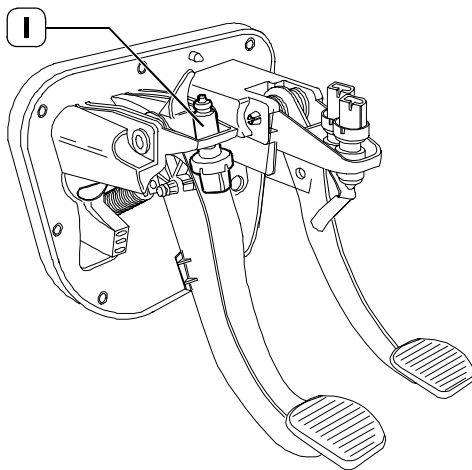
Figure 172



003326t

I. Brake pedal switches

Figure 173



003327t

I. Clutch pedal switch

Brake pedal switches

Two switches are located on the brake pedal; N.C. 53565 is connected directly to electronic centre pin B31 and 53501 is connected to the centre via remote switch 25006 for stop light actuation and to generate a redundant signal on the centre increasing system reliability degree.

Clutch pedal switch

An N.C. switch connected to electronic centre pin B38 is mounted on the clutch pedal.

The "clutch pedal actuated signal" is used by the centre to identify gear condition selected and gear shifts.

In absence of the pedal pressed switch signal, the centre disengages the Cruise Control function.

Electromagnetic junction fan

The fan is provided with an electromagnetic junction monitored by the electronic centre pin A39 that activated the junction remote control switch, to optimise water cooling.

The electrical fan remote control switch is activated or deactivated by the centre according to the temperature of:

- ☐ the coolant
- ☐ over supply air
- ☐ the fuel

Engine coolant temperature

Activated at over 96 °C and deactivated at under 84 °C

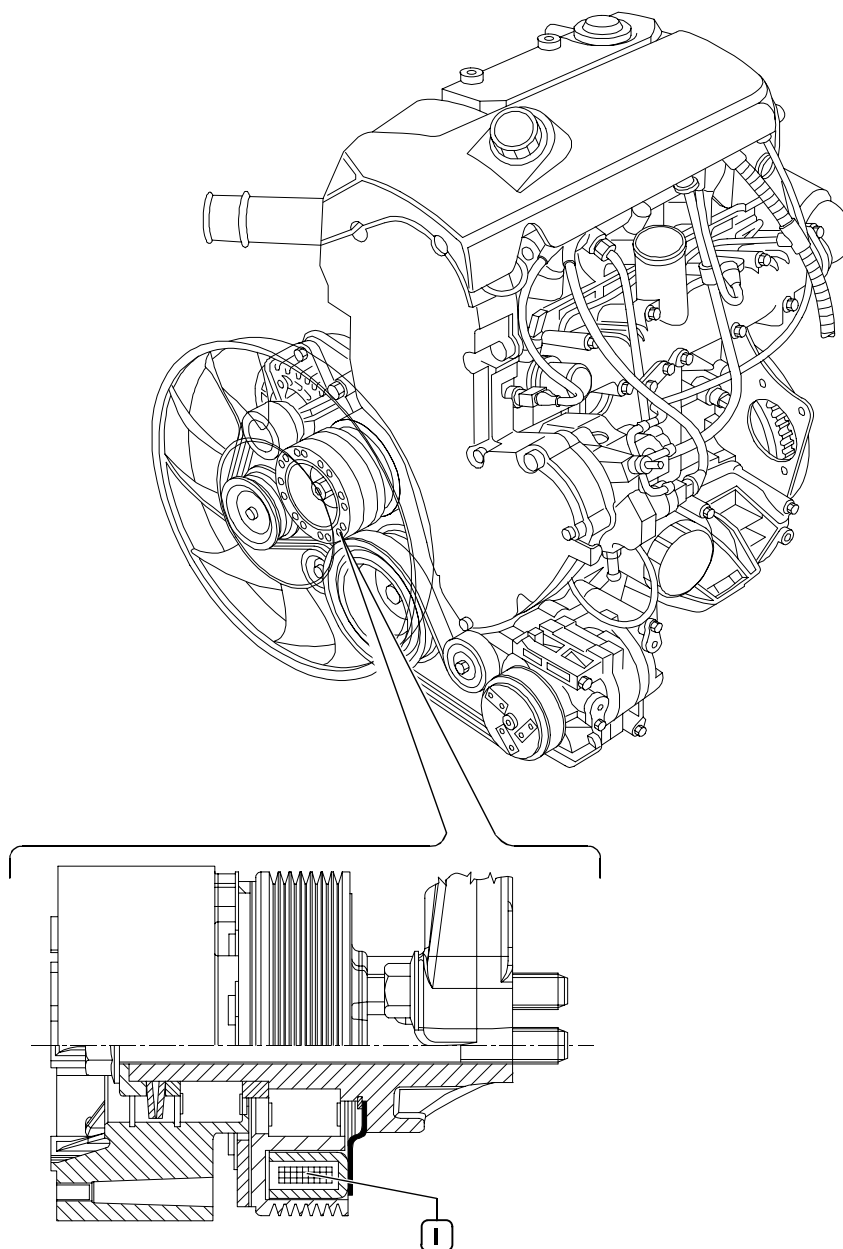
Over supply air temperature

Activated at over 75 °C and deactivated at under 65 °C

Fuel temperature

Activated at over 20 °C and deactivated at under 10 °C

Figure 174



ELECTROMAGNETIC JUNCTION TECHNICAL VIEW0 (F1A)

I. Coil

003328t

Preheat plug electronic centre (FIA/FIC engine)

EDC central unit effects the timing of the functioning of glow plugs pre-heating central unit depending on engine temperature, which, in turn, activates the glow plugs.

The preheat centre contains an "intelligent" remote control switch that sends a feed-back to the control centre for information on any preheat centre defect or plug earth short circuit.

Preheat centre pin-out

- 31 – Mass
- 86 – Start switch (+15)
- ST – EDC electronic centre (pin B42)
- DI – EDC electronic centre (pin B37)
- 30 – Battery positive (+30)
- G1 – Preheat plugs
- G2 – Preheat plugs
- G3 – Preheat plugs
- G4 – Preheat plugs

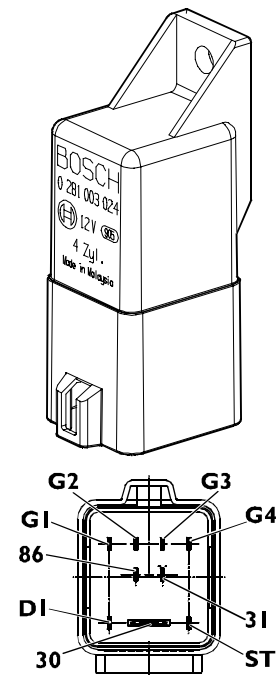
Preheat plugs

CONTROL VALUES

With constant di 11 V power supply:

- | | |
|---|----------------|
| ☐ maximum current absorbed | 18 A |
| ☐ in 5" | $11 \pm 1,5$ A |
| ☐ in 30" | $6 \pm 0,9$ A |
| ☐ temperature after 7" | 850°C |
| ☐ torque | 8–10 Nm |

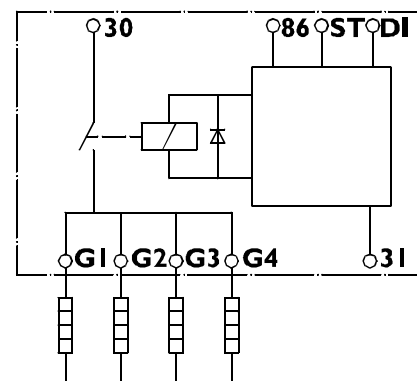
Figure 175



003332t

PREHEAT CENTRE

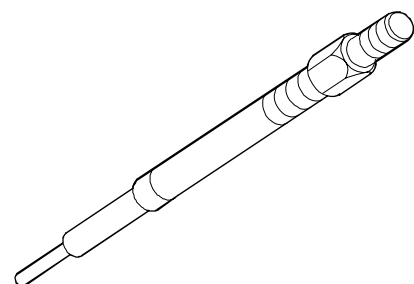
Figure 176



003331t

ELECTRICAL DIAGRAM

Figure 177



75579

PREHEAT PLUS

EDC MS6.3 / EDC I6

Electronic injection control

The system calculates injection on the basis of the processing of the following parameters:

- ☐ Engine rpm
- ☐ Engine coolant fluid temperature
- ☐ Intake air capacity
- ☐ Battery voltage
- ☐ Fuel pressure
- ☐ Accelerator pedal position

Fuel pressure ranges from 400 to 1350 bars (1600 for FI engines), according to engine rpm and load operating conditions.

The lower pressure is compensated by longer injection times and vice versa, always taking account of the loads required.

Up to 2800 rpm pre-injection is also carried out in order to reduce the typical noise of direct injection.

Pre-injection advance angles, the distance between pre-injection and main injection and advance angles of main injection vary according to the instantaneous engine operating conditions.

System diagnosis is performed by means of diagnostic instruments (no Blink Code is used).

Immobilizer recognition

When the control unit receives the signal of the key on "MAR" it communicates with the immobilizer control unit to enable starting.

Checking fuel temperature

With the fuel temperature greater than 75°C, detected by the sensor on the fuel filter, the control unit operates the pressure regulator to decrease the line pressure (injection times are not changed). If the temperature exceeds 90°C, the power is reduced to 60%.

Checking engine coolant temperature

The control unit, depending on the temperature:

- ☐ of the engine coolant, turbocharging air and fuel, operates the electromagnetic fan (Baruffaldi) and switches on the coolant temperature warning light.

Checking quantity of fuel injected

According to the signals from the sensors and the mapped values, the control unit:

- ☐ operates the pressure regulator;
- ☐ varies the "pilot" injection time to 2200 rpm;
- ☐ varies the "main" injection time.
- ☐ Checking idling adjustment

The control unit processes the signals from the various sensors and regulates the amount of fuel injected:

- ☐ it operates the pressure regulator;
- ☐ it varies the injection times of the electro-injectors.

Within certain thresholds the speed takes account of the battery voltage.

Fuel cut-off in release phase

In the phase of releasing the throttle pedal the control unit actuates the following logic elements:

- ☐ it cuts off supply to the electro-injectors;
- ☐ it partially reactivates supply to the electro-injectors before reaching idling speed;
- ☐ it operates the fuel pressure regulator.

Checking cylinder balancing on idling

According to the signals received from the sensors, the control unit controls the regularity of the torque at idling speed:

- ☐ it varies the amount of fuel injected into the single electro-injectors (injection time).

Checking regular engine rotation (anti-sawing)

It ensures regular engine rotation at a constant rate while increasing revs.

The control unit processes the signals received from the sensors and determines the amount of fuel to be injected via:

- ☐ the pressure regulator;
- ☐ the electro-injector opening time.

Checking smokiness at exhaust on acceleration

With heavy acceleration, on the basis of the signals received from the air introduction meter and engine speed sensor, the control unit determines the optimum amount of fuel to inject:

- ☐ it operates the pressure regulator;
- ☐ it varies the electro-injector injection time.

Checking exhaust gas recirculation (E.G.R. if present)

Depending on the engine load and the signal from the accelerator pedal sensor, the control unit limits the amount of air taken in, actuating partial suction of the exhaust gases.

Checking top speed limit

Depending on the number of revs, the control unit actuates two action strategies:

- ☐ at 4250 rpm it cuts off the fuel, decreasing the electro-injector opening time. Over 5000 rpm it deactivates the electro-injectors.

Checking regular rotation on acceleration

Regular progression is assured in all conditions by the control of the pressure regulator and the electro-injector opening time.

Preheat plug centre control (FI Engine)

During:

- ☐ the start step
- ☐ the after start step

the injection centre times the heater starter (or preheat plugs for the FI Engine) according to engine temperature.

Checking activation of air-conditioning system

The control unit operates the air-conditioning compressor:

- ☐ switching it on/off when the relative switch is pressed;
- ☐ momentarily turning it off (approximately 6 sec.) if the engine coolant reaches the set temperature.

Checking fuel pump

Irrespective of the speed, the control unit:

- ☐ supplies the auxiliary fuel pump with the key on MAR;
- ☐ cuts off auxiliary pump supply if the engine is not started up within a few seconds.

Checking diesel warming

It times operation of diesel warming in relation to ambient temperature.

Checking cylinder position

During each turn of the engine, the control unit recognizes which cylinder is in the power stroke and operates the injection sequence for the appropriate cylinder.

Checking pilot and main injection timing

According to the signals from the various sensors, including the absolute pressure sensor built into the control unit, the control unit determines the optimum point of injection according to internal mapping.

Checking injection pressure closed cycle

Depending on the engine load, determined by processing the signals from the various sensors, the control unit operates the regulator to obtain optimum line pressure.

Fuel supply

The fuel supply is calculated in relation to:

- ☐ accelerator pedal position
- ☐ engine speed
- ☐ quantity of air introduced.

The outcome may be corrected in relation to:

- ☐ the water temperature.

Or to avoid:

- ☐ noise
- ☐ smoke
- ☐ overloading
- ☐ overheating
- ☐ turbine over-revving.

The delivery can be modified in the case of:

- ☐ action of external devices (ABS), ABD, EDB
- ☐ serious trouble decreasing the load or stopping the engine.

After determining the mass of air introduced by measuring its volume and temperature, the control unit calculates the corresponding mass of fuel to inject into the relevant cylinder (mg per delivery) also taking into account the temperature of the diesel.

The mass of fuel calculated in this way is first converted into volume (mm³ per delivery) and then into degrees of throw, or duration of injection.

Correcting flow rate according to water temperature

A cold engine meets with greater resistance during operation: friction is high, the oil is still very viscous, and the various clearances are not yet optimized.

In addition, the injected fuel tends to condense on the metal surfaces that are still cold.

The fuel supply for a cold engine is therefore greater than for a warm one.

Correcting flow rate to avoid noise, smoke or overloading

The behaviour that could lead to this kind of trouble is well known.

The designer has therefore included special instructions in the control unit to avoid it.

De-rating

In the event of the engine overheating, injection is modified, decreasing the delivery to a varying degree, in proportion to the temperature reached by the coolant.

Turbine rpm setting (.15 – .17) (for the variable geometry turbo-compressor)

By changing its geometry, adjusted turbine speed is monitored by the electronic centre via an electrical signal feeding the compressed air actuator electro valve. Based on signals from the accelerator pedal position and suction manifold air temperature/pressure engine rpm sensors, the electronic centre processes the field-back signal to properly modulate turbine actuator pilot electro valve opening.

Injection timing electronic test

The advance (start of delivery, expressed in degrees) may be different from one injection to the next, also differentiated from one cylinder to another. It is calculated, similarly to the delivery, in relation to the engine load (accelerator position, engine speed and air introduced).

The advance is appropriately corrected:

- ☐ in phases of acceleration;
- ☐ according to the water temperature.

And also to obtain:

- ☐ lower emissions, noise and overloading;
- ☐ better vehicle acceleration.

An extremely high advance is set on starting, depending on the water temperature.

Feedback from the start of delivery is supplied by the change in impedance of the injector solenoid valve.

Speed governor

The electronic speed governor has both features of governors:

- ☐ idling and top speed
- ☐ all speeds

It is stable in ranges where conventional, mechanical governors are imprecise.

Engine starting

During the first few turns of the engine, the timing and cylinder no. 1 recognition signals (flywheel sensor and camshaft sensor) are synchronized.

The accelerator pedal signal is ignored on starting. Starting delivery is set only according to water temperature, by a special map.

When the control unit detects such speed and acceleration of the flywheel as to be able to consider the engine started up and no longer driven by the starter motor, it re-enables the accelerator pedal.

Cold starting

If even just one of the three temperature sensors (water, air or diesel) records a temperature lower than 10°C, pre-post heating is activated.

On inserting the key contact, the pre-heating indicator goes on and remains on for a period varying depending on temperature (air is heated by the pre-heating glow plugs that are located on cylinder head for FI engines), then it blinks. Thereafter, the engine can be started up.

When the motor is running this indicator light goes out, while the glow plugs continue to be powered for a certain length of time (variable) for post-heating.

If, with the indicator light flashing, the engine is not started up within 20–25 seconds (inattention time), the operation is cancelled so as not to run down the batteries pointlessly. The pre-heating curve is also variable in relation to the battery voltage.

Warm starting

If the reference temperatures all exceed 10°C, when the key makes contact the indicator light comes on for approximately 2 sec., for a short test, and then goes out. It is now possible to start up the engine.

Run up

When the key makes contact, the control unit transfers the information stored in memory when the engine was last stopped into the main memory (see After Run) and makes a diagnosis of the system.

After run

Whenever the engine is switched off with the key, the control unit stays powered for a few seconds by the main relay.

This makes it possible for the microprocessor to transfer some data from the main memory (volatile) to a non-volatile memory, which can be erased and written over (EEPROM), so as to make it available at the next start up (see Run Up).

These data basically consist of:

- ☐ various settings (engine idling adjustment, etc.);
- ☐ settings of some components;
- ☐ fault memory.

The process lasts a few seconds, typically from 2 to 7 (depending on the amount of data to save), after which the ECU sends a command to the main relay and makes it disconnect from the battery.



It is extremely important for this procedure not to be broken off, for example by switching off the engine with the battery cut-out, or by disconnecting the battery cut-out before 10 seconds have passed since switching off the engine.

If this happens, the functioning of the system is ensured, but repeated interruptions may damage the control unit.

Cut – off

This function cuts off fuel delivery when the vehicle is decelerating (accelerator pedal released).

Cylinder balancing

Individual cylinder balancing contributes to increasing comfort and handling.

This function permits individual, customized control over the delivery of fuel and the start of delivery for each cylinder, even differently from one cylinder to another, to compensate for the hydraulic tolerances of the injector.

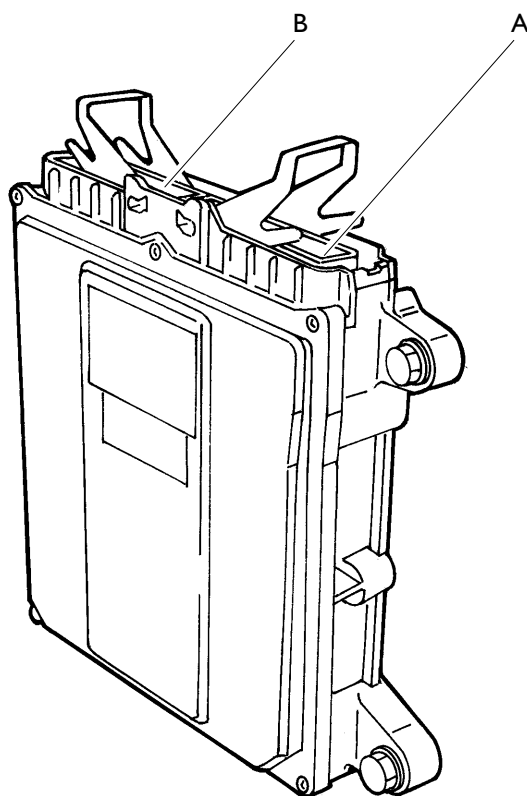
The differences in flow (delivery specifications) between the various injectors cannot be evaluated directly by the control unit. This information is supplied by Modus reading the bar code of each injector at the time of assembly.

Synchronisation search

If there is no signal from the camshaft sensor, the control unit is anyhow able to recognize the cylinders into which the fuel is to be injected.

If this occurs when the engine is already running, the combustion sequence has already been acquired, so the control unit continues with the sequence on which it has already been synchronized.

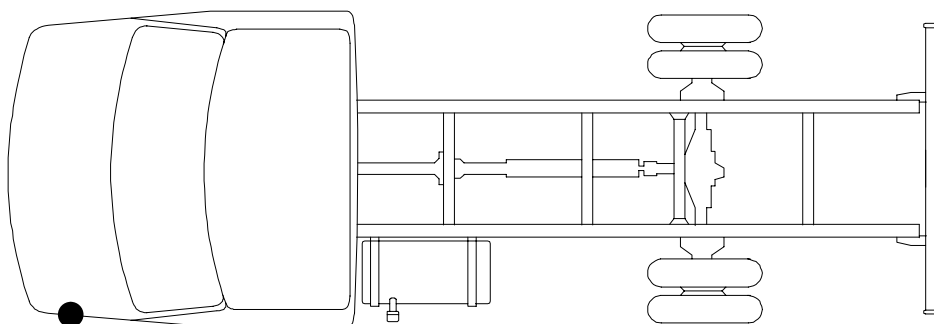
If this occurs when the machine is at a standstill, the control unit energizes a single solenoid valve. Within at most 2 turns of the crankshaft, injection will take place in that cylinder, so the control unit just needs to get synchronized on the firing sequence and to start up the engine.

Bosch MS6.3 control unit**Figure 178****85150**

7420

PERSPECTIVE VIEW

A. Housing for injection cable connector – B. Seat for cabin-bonnet cable (UNIJET motor drive)

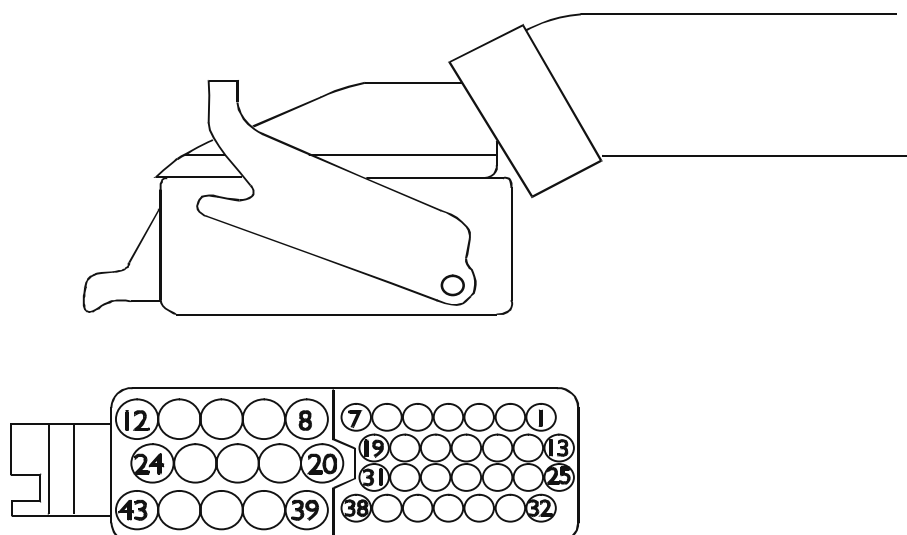
Figure 179

7406

LOCATION ON THE VEHICLE

Control unit connection to the injection cable on engine side (housing A)

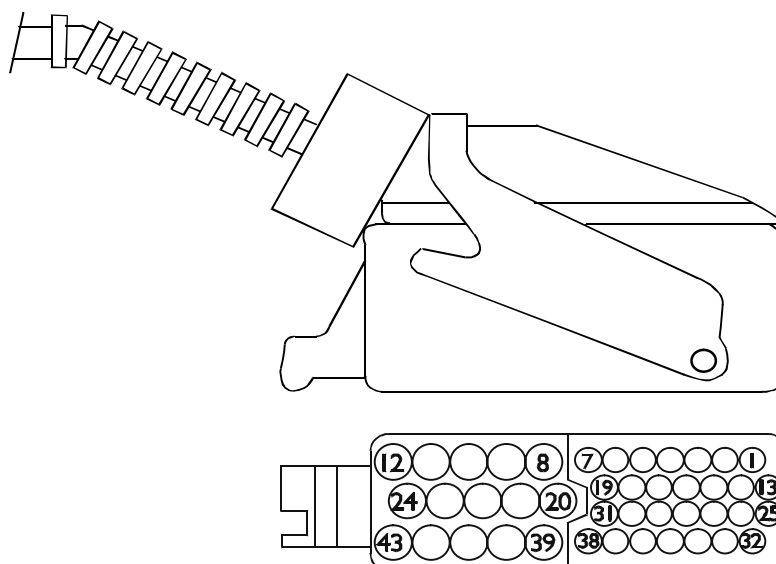
Figure 180



VIEW OF BOSCH 43-WAY CONNECTOR FROM CABLE INPUT SIDE

7421

Pin	Function	Cable colour code
1	To engine coolant temperature sensor	5154
2	To turbo-blower air pressure and temperature sensor for EDC (without EGR)	5151
3	To turbo-blower air pressure and temperature sensor for EDC (without EGR)	5153
4	To engine rpm sensor on camshaft (cams)	White
5	To temperature sensor and ambient air pressure for EDC (with EGR)	5151
6	Fuel pressure adjustment sensor earth	0000
7	Control to relay for switching on fuel pump	8150
8	Common EDC centre mass – Centre monitored remote control switches – EGR electro valve	0000
9	To solenoid valve for pressure adjustment	9925
10	To solenoid valve for electronic injection (injector 2 – cylinder 3)	—
11	Spare	—
12	To solenoid valve for electronic injection (injector 1 – cylinder 1)	—
13	To solenoid valve for pressure adjustment	5590
14	Spare	—
15	Sensor (fuel temperature) for switching on fuel warming	5592
16	Spare	—
17	To ambient air temperature and pressure sensor for EDC (with EGR)	8150
18	To ambient air temperature and pressure sensor for EDC (with EGR)	8151
19	Air temperature and pressure sensor earth (without EGR)	0165
20	Earth shared by control unit and radialjet and pressure solenoid valves (no radialjet for FIA)	0000
21	To solenoid valve for radialjet pump control (not for FIA)	9917
22	Spare	—
23	To solenoid valve for electronic injection (injector 3 – cylinder 4)	—
24	To solenoid valve for electronic injection (injector 4 – cylinder 2)	—
25	To solenoid valve controlling anti-pollution system (EGR) (.13 – FIA)	5577
26	To ambient air temperature and pressure sensor for EDC (with EGR)	8152
27	To speed limiter adjustment sensor	8847
28	To ambient air temperature and pressure sensor for EDC (with EGR)	8153
29	To sensor for engine rpm	White
30	Earth shared by control unit and temperature sensors	0150
31	To engine rpm sensor on camshaft (cams)	Black
32	Control to relay for heated fuel oil filter	8159
33	To sensor for fuel pressure adjustment	5591
34	To turbo-blower air pressure and temperature sensor for EDC (without EGR)	5152
35	Control to relay for engaging conditioner compressor	9990
36	To solenoid valve actuator for VGT (.15) or to WASTE GATE solenoid valve (.13)	5000
37	To engine rpm sensor (.13 – .15 – FIA)	Black
38	Spare	—
39	Control to relay for engine cooling joint	7740
40	To solenoid valve for electronic injection (injector 1 – cylinder 1)	—
41	To solenoid valve for electronic injection (injector 4 – cylinder 2)	—
42	To solenoid valve for electronic injection (injector 3 – cylinder 4)	—
43	To solenoid valve for electronic injection (injector 2 – cylinder 3)	—

Control unit connection to cab-bonnet cable (housing B)**Figure 181**

VIEW OF BOSCH 43-WAY CONNECTOR FROM CABLE INPUT SIDE

7422

Pin	Function	Cable colour code
1	To cruise Control	8156
2	To load sensor on accelerator for EDC	5157
3	Spare	—
4	To instrument panel module A1 rpm indicator repeater	5155
5	Spare	—
6	Compressor engaged signal to EDC	8162
7	To diagnostic socket	2299
8	To alarm control unit	Green
9	EDC control unit supply	8150
10	EDC control unit supply	8150
11	Control to relay for connection with fuel tank for thermal starter (not for FIA)	0000
12	To earth signal (battery negative)	0000
13	To load sensor on accelerator for EDC	5156
14	Instrument panel module A20 rpm indicator repeater mass	0000
15	Spare	—
16	To instrument panel module A17 rpm signal	5614
17	Spare	—
18	Spare	—
19	To diagnostic socket	1199
20	To key-operated fuse 2	8051
21	To instrument panel module A1 A30 engine preheat warning leds	0000
22	EDC control unit supply	8150
23	To instrument panel module A1 EDC A29 defect warning leds	5156
24	To earth signal (battery negative)	0000
25	To Cruise Control	8155
26	Supply with stop lights on	8153
27	To load sensor on accelerator for EDC	0150
28	To diagnostic socket	9932
29	To load sensor on accelerator for EDC	0159
30	Spare	—
31	Supply when brake pedal is pressed	8158
32	To cruise Control	8154
33	To cruise Control	8157
34	Spare	—
35	To load sensor on accelerator for EDC	5158
36	Spare	—
37	FIA preheat centre control	1310
38	To clutch pressed signal relay for EDC	0160
39	To alarm control unit	White
40	Control to relay for EDC engagement	8150
41	EDC control unit supply	8150
42	Heater starter insertion remote control switch control/FIA preheat centre control	0000 / 1311
43	To earth signal (battery negative)	0000

EDC system components

Ref.	Component code	Description
1	58918	Instrument panel tachometer
2	58918	Instrument panel rev counter
3	42374	Clutch pedal switch
4	53565	Brake pedal switch
5	85152	Idling switch and accelerator pedal position sensor
6	47106	Fuel temperature sensor (FIA engine)
7	47106	Fuel temperature sensor Fuel filter clogging sensor (8140 engine)
8	85157	Fuel pressure sensor (FIA engine)
9	85157	Fuel pressure sensor (8140 engine)
10	47035	Coolant temperature sensor
11	85156	Air delivery sensor
12	48042	Distribution sensor
13	48035	Crankshaft sensor (FIA engine)
14	48035	Flywheel sensor (8140 engine)
15	78247	Electro injectors
16	19005	Hot starter
	78000	Hot starter electro valve (8140 engine)
17	25231	Plug preheat centre (FIA engine)
18	19010	Preheat plug (FIA engine)
19	85151	Fuel electro pump
20	78015	3rd pump cutout electro valve (8140 engine)
21	78013	Pressure regulator (8140 engine)
22	78013	Pressure regulator (8140 engine)
23	12012	AC compressor
24	78209	EGR modulating electro valve
25	85159	Air delivery sensor
26	58701	EDC warning light
27	58702	Preheat warning light
28	54032	Cruise Control/PTO controls
29	85130	Start key with Immobilizer
30	72027	Diagnosis connection
31	78248	VGT control electro valve (for 8140.43N engine)
32	85022	Fan electromagnetic connection
(*)	On version without EGR	
(**)	On version with EGR	
■	Only on 8140 engines	
●	Only on FIA engines	

Print 603.93.281

Blink Code (up to chassis no. 5383302/ DI87233)

With the ignition switch off, press the diagnostic button.

Two series of blinks of the EDC warning light, at intervals with a brief pause, will indicate the code of the first error memorised.

Press the button again to move to the next error.

Each time the last error is reached, the first one will be repeated.

The list of the errors contains all the errors memorised and not only the active ones.

The order in which the errors are shown is the one in which they occurred.

To remove the error list from the memory, proceed as follows:

- ☐ With the ignition key off, press the diagnostic button
- ☐ keeping the button pressed, switch the ignition on
- ☐ keep the button pressed for 5 seconds
- ☐ release the button
- ☐ Switch the ignition off.

The error codes are given in the following table.

Common Rail 8140.43B – 8140.43S – 8140.43N – EDC MS6.3

(Software version 5.5)

Blink-Code	Warning light EDC	Warning light	Reduction level (*)
VEHICLE			
1.1	on	Vehicle speed	
1.3	off	Cruise Control	
1.4	flashing	Accelerator pedal	X
1.5	off	Clutch switch	
1.6	on	Brake switch	
1.7	off	Brake/accelerator plausibility	Idling setting
1.8	off	Diagnostic warning light / Main EDC warning	
1.9	off	Conditioner control relay	
ENGINE 1			
2.1	flashing	H ₂ O temperature sensor	X
2.2	off	Air temperature sensor	
2.3	on	Fuel oil temperature sensor	
2.4	flashing	Boosting pressure sensor	X
2.5	off	Atmospheric pressure sensor	
2.7	flashing	Fuel pump	
2.8	off	Fuel filter heater control relay	
2.9	on	Fan control relay	
ENGINE 2			
3.1	flashing	Cylinder 1 balancing	
3.2	flashing	Cylinder 2 balancing	
3.3	flashing	Cylinder 3 balancing	
3.4	flashing	Cylinder 4 balancing	
3.5	off	Battery voltage	
3.6	off	Thermal starter warning light	
3.7	off	Thermal starter	
3.8	off	Thermal starter solenoid valve	
3.9	off	Preheat monitoring	
TURBINE			
4.4	flashing	VGT monitoring	
4.5	on	VGT electro valve	

Blink-Code	Warning light EDC	Warning light	Reduction level (*)
INJECTORS			
5.1	flashing	Cylinder injector 1	
5.2	flashing	Cylinder injector 2	
5.3	flashing	Cylinder injector 3	
5.4	flashing	Cylinder injector 4	
5.7	on	Bearing 1 (cylinders 1–4)	
5.8	on	Bearing 2 (cylinders 2–3)	
ENGINE RPM			
6.1	flashing	Engine rpm sensor	X
6.2	flashing	Camshaft revolution sensor	X
6.4	flashing	Over-revving	
INTERFACE			
7.1	off	PWM signal	
7.2	off	CAN bus inactive	
7.3	off	CAN bus control	
7.4	off	CAN message control	
FUEL PRESSURE/EGR			
8.1	flashing	Fuel pressure control	X or Engine cutout
8.2	flashing	Fuel pressure sensor	X
8.3	flashing	Pressure regulator	
8.4	off	3 rd impeller cut-off solenoid valve	
8.5	on	EGR control	
8.6	on	EGR valve	
8.7	on	Debit meter	X
8.8	off	EGR atmospheric pressure sensor	
CENTER			
9.1	flashing	Control unit error	X or Engine cutout
9.2	on	Control unit EEPROM fault	
9.3	flashing	Immobiliser – EDC communication	
9.4	on	Main relay	
9.5	off	After run Test	
9.6	flashing	Engine stop test (ECU) Engine stop test (ECU)	
9.7	flashing	Sensor supply	X or Engine cutout
9.8	flashing	Control unit error	Start impossible
9.9	flashing	ECU operating system	Engine cutout

FIA Common Rail

Differs for the following codes:

3.6	off	Preheat warning light	
3.7	off	Preheat plug centre	
3.8	off	Preheat plug centre	
3.9	off	Preheat monitoring	
8.4	off	(Not used)	

EDC16

The EDC.16 control unit offers a few innovations compared to the control unit used with system EDC MS6.3.

- ☐ latest-generation hardware and software;
- ☐ possibility of processing a greater number of signals (both input and output);
- ☐ preset for the control of new additional functions and devices, which will be adopted later for technological improvement of the product range or compliance with forthcoming anti-pollution rules.

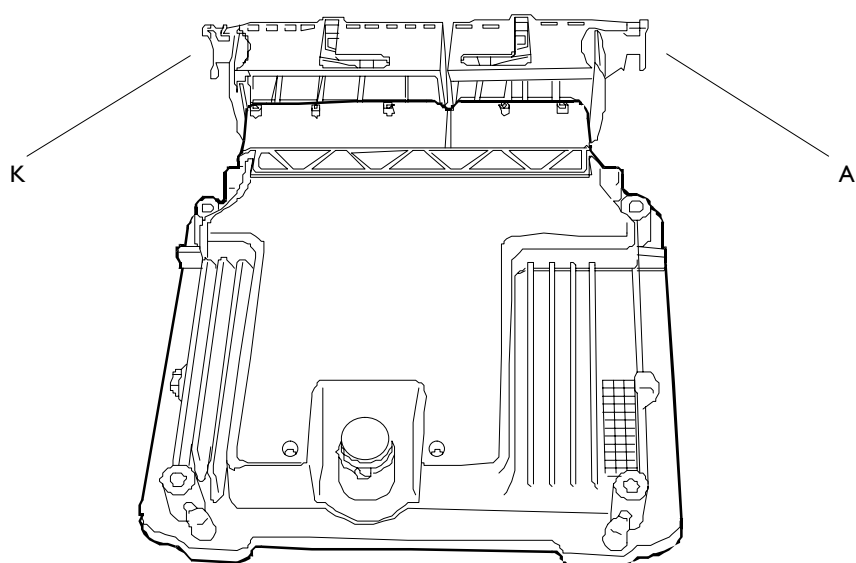
The variants compared with the MS6.3 system are as follows:

- ☐ new sensor on the accelerator pedal;
- ☐ new sensor on the distributing shaft (phase);
- ☐ engine cable;
- ☐ bonnet/cab cable;
- ☐ new connectors for control unit connection.

The EDC.16 system replaces the MS6.3 system on engines F1A (.10 – .12). It is present on F1C engines.

Diagnosis is only performed by using the diagnostic instruments available. No Blink Code is provided.

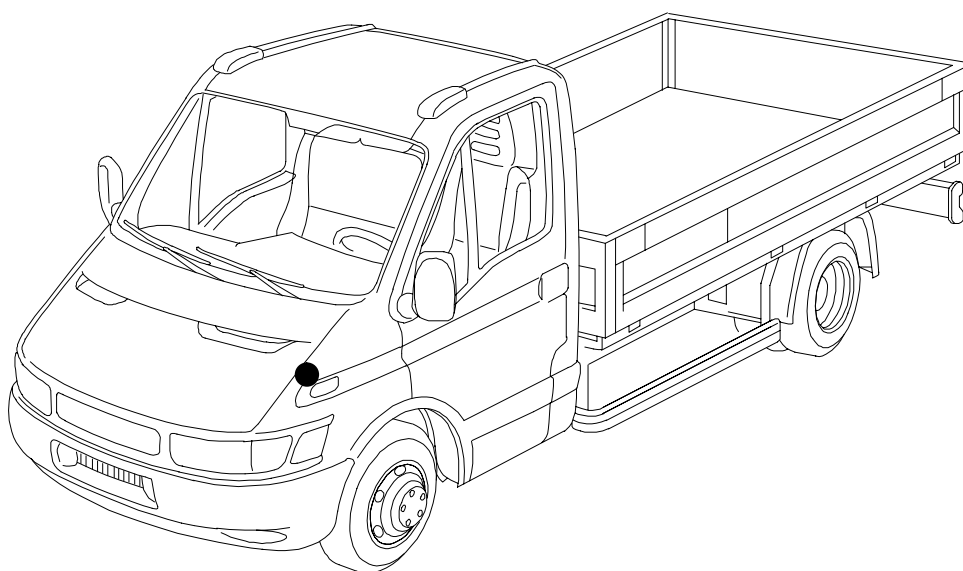
The other features are the same as the 6.3 system.

Bosch EDC16 control unit**Figure 183**

85711

PERSPECTIVE VIEW

A. Engine side injection cable connector – K. Bonnet/cab cable connector

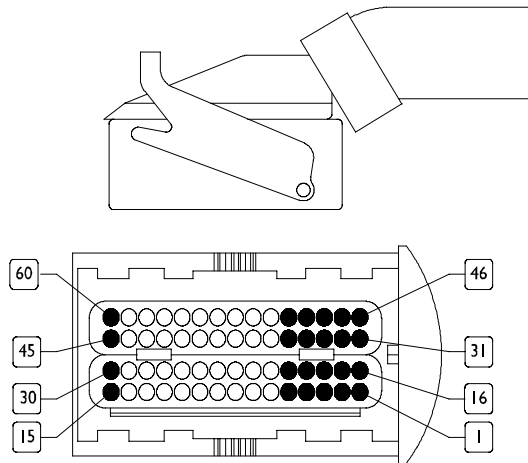
Figure 184

7373

LOCATION ON THE VEHICLE

Control unit connection to the injection cable on engine side (housing A)

Figure 185



85708

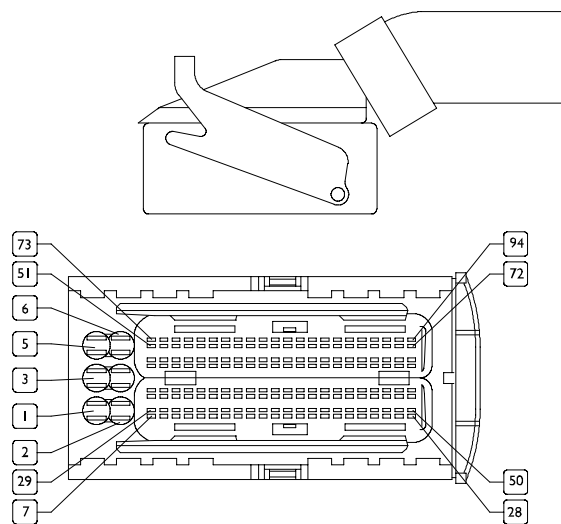
85710

Pin	Cable colour code	Function
1	0000	Cylinder injector 3
2	0000	Cylinder injector 2
8	0000	Rail pressure sensor negative
11	0174	Distributing shaft sensor negative (phase)
12	red	Drive shaft sensor
13	5153*	Boosting air pressure and temperature sensor power supply
16	9924	Cylinder injector 1
17	9924	Cylinder injector 4
19	0000	Pressure regulator negative
20	7158	Distributing shaft sensor positive
21	—	Drive shaft sensor braided wire
23	0165*	Boosting air pressure and temperature sensor negative
27	white	Drive shaft sensor
28	5591	Rail sensor power supply
29	8152	Air flow meter power supply (available with EGR)
31	9924	Cylinder injector 2
33	0000	Cylinder injector 4
37	5151	Air flow meter air temperature signal (available with EGR)
40	5152*	Boosting air pressure sensor signal
41	0150	Water temperature sensor negative
42	8153	Air flow meter signal
43	5591	Rail pressure signal
44	8151	Air flow meter negative (available with EGR)
46	9924	Cylinder injector 3
47	0000	Cylinder injector 1
49	9925	Pressure regulator
50	9160	Distributing shaft sensor signal (phase)
51	0150	Fuel temperature sensor negative

Pin	Cable colour code	Function
52	5592	Fuel temperature sensor signal
53	5151*	Boosting air temperature sensor signal
58	5154	Water temperature sensor signal
60	8150	EGR solenoid valve
●	Power seats	
○	Signal seats	
(*)	Available when the EGR is not provided	
–	Pins not highlighted are not used	

Control unit connection to cab–bonnet cable (housing K)

Figure 186



85708

85709

Pin	Cable colour code	Function
1	–	+30 (main relay)
2	0000	Earth
4	0000	Earth
5	8150	+30 (main relay)
6	0000	Earth
8	0150	Accelerator pedal sensor negative (pin 5)
9	5157	Accelerator pedal sensor signal (pin 4)
13	–	Signal from power takeoff (if any) state selector
16	–	Negative from power takeoff (if any) state selector
17	–	Signal from brake pedal pressed for stop light ignition
25	2299	K line
28	8051	+15
30	0159	Accelerator pedal sensor negative (pin 3)
31	5157	Accelerator pedal sensor signal (pin 6)
38	8155	Cruise Control (resume)
42	–	Speed limiter button
45	5158	Accelerator pedal sensor power supply (pin 2)
46	5158	Accelerator pedal sensor power supply (pin 1)
48	5614	Engine speed sensor (revs counter)
52	1310	To preheating spark plug actuation remote–control switch pin D1
54	8162	Signal from air–conditioning ON compressor remote–control switch
56	8157	Cruise Control (set +)
57	–	Auxiliary speed limiter (where available)
58	–	Signal from clutch switch
61	–	CAN L line
62	–	CAN H line
68	8150	Fuel filter heating remote–control switch positive

Pin	Cable colour code	Function
70	9990	Positive to the remote-control switch for engine water recirculation shut-off solenoid valve control with auxiliary heater ON
71	5156	EDC warning light negative
72	8150	Main relay (negative)
75	5155	Vehicle speed signal (tachometer)
77	8154	Cruise Control (off)
78	8156	Cruise Control (set -)
80	8158	Brake pedal signal
90	7740	Positive for engine cooling electromagnetic joint control
91	-	Fuel electric pump remote-control switch negative
92	0000	Pre-heating warning light negative
93	1311	To pre-heating spark plug actuation remote-control switch pin ST
-	Pins not highlighted are not used	

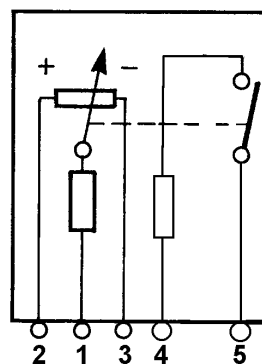
Accelerator pedal sensor

A new sensor which incorporates two potentiometers (no idling switch is provided) is available on the accelerator pedal. The ratio between the signals from the two potentiometers is 2:1 (one potentiometer exhibits a twofold resistance value compared with the other). Both of these signals (V) are detected by the control unit that processes them according to stored threshold values and manages the

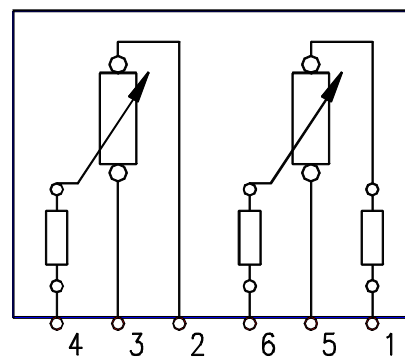
injection system as an accelerator pedal position set by the driver. (At the output of these potentiometers, a variable voltage is available which corresponds to the potentiometer resistance value.)

It is connected to the EDC control unit connector K pins 9–30–45–31–8–46. The potentiometers are powered with 5 V voltage supplied by the control unit itself.

Figure 187

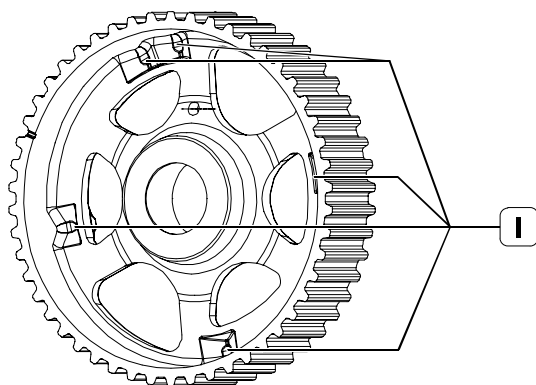


EDC MS6.3



EDC 16

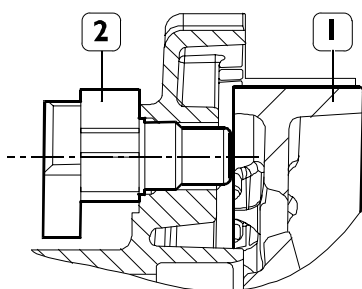
Figure 188



0003320t

I. Phase identification holes

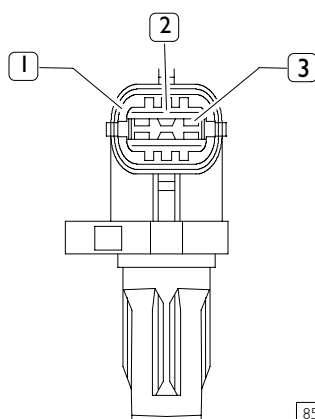
Figure 189



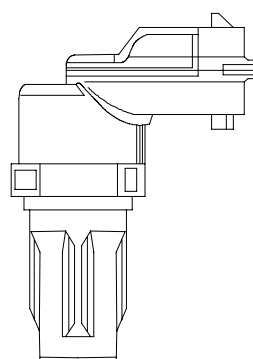
0003321t

I. Distributing shaft pulley – 2. Sensor

Figure 190



85712



85713

PERSPECTIVE VIEW

I. Power supply positive – 2. Signal output – 3. Earth

Camshaft sensor (FIA)

A semiconductor layer, immersed in a magnetic field and through which current flows, generates a potential difference (called Hall voltage) at its ends.

If current intensity remains constant, the generated voltage depends only on the magnetic field strength: periodical variation of field strength is enough to obtain a modulated electric signal.

The smooth portion of the phonic wheel (distributing shaft pulley) covers, while moving, the sensor, thus blocking the magnetic field with resulting low output signal.

On the contrary, the sensor generates a high signal next to the openings and when a magnetic field is available.

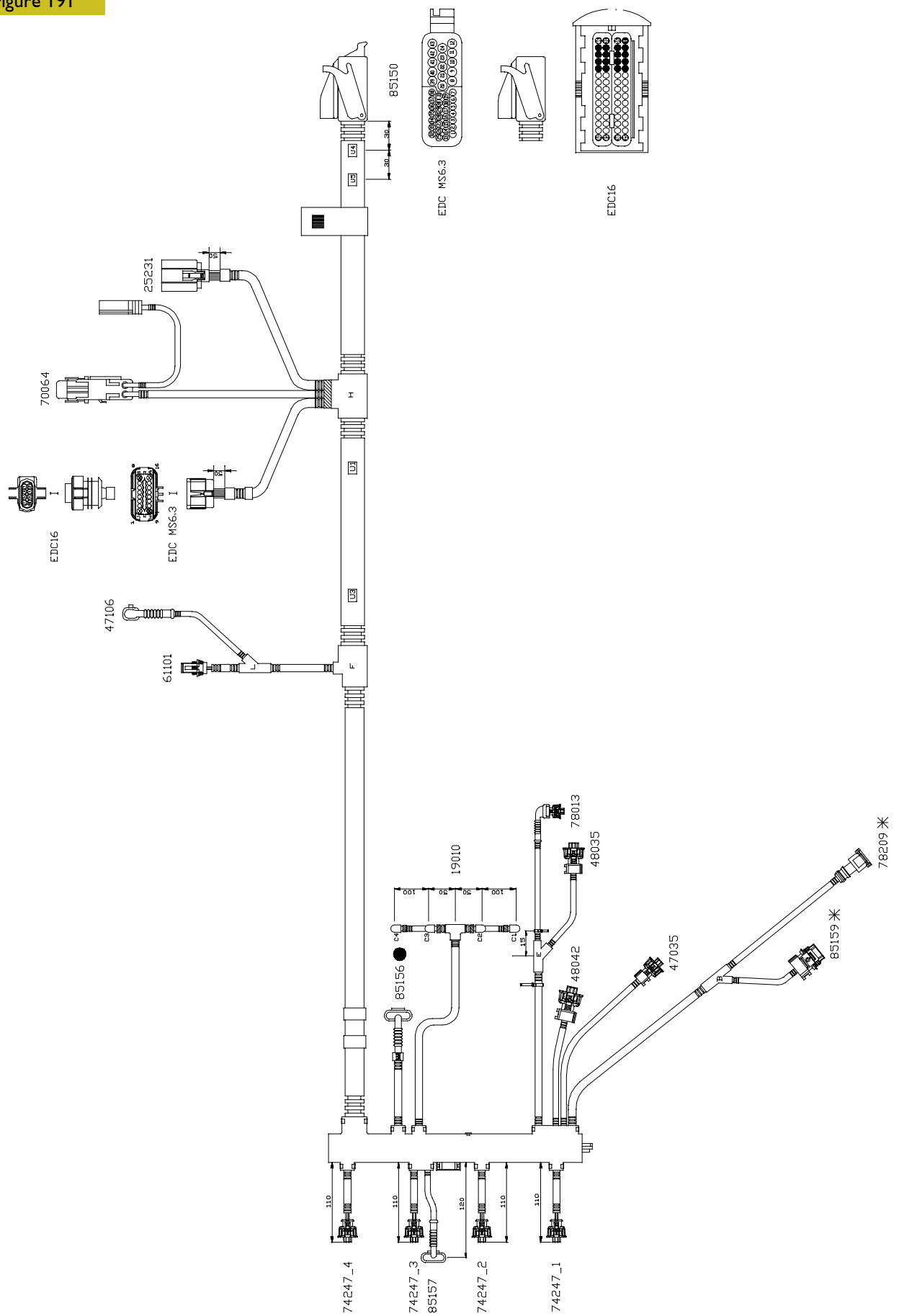
Phase sensor signals are acquired, and the engine position is recognized according to the sequence of the phonic wheel notches.

The mounting function makes it possible to identify signal errors and interferences (if any).

The resulting signal is supplied to the processor that controls the injection system.

Injection cable FIA (.10 - .12)

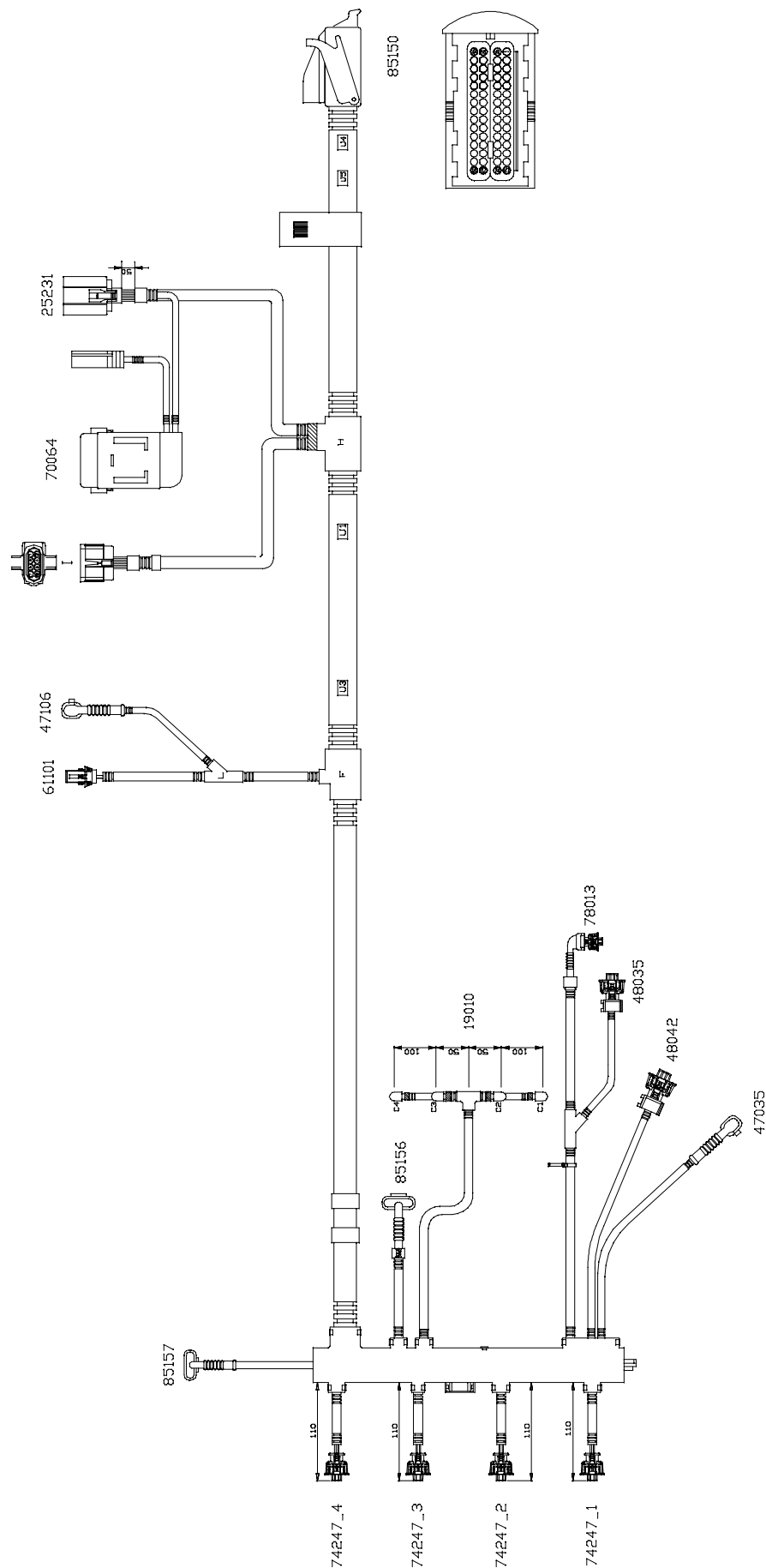
Figure 191



85723

Injection cable FIA (.14)

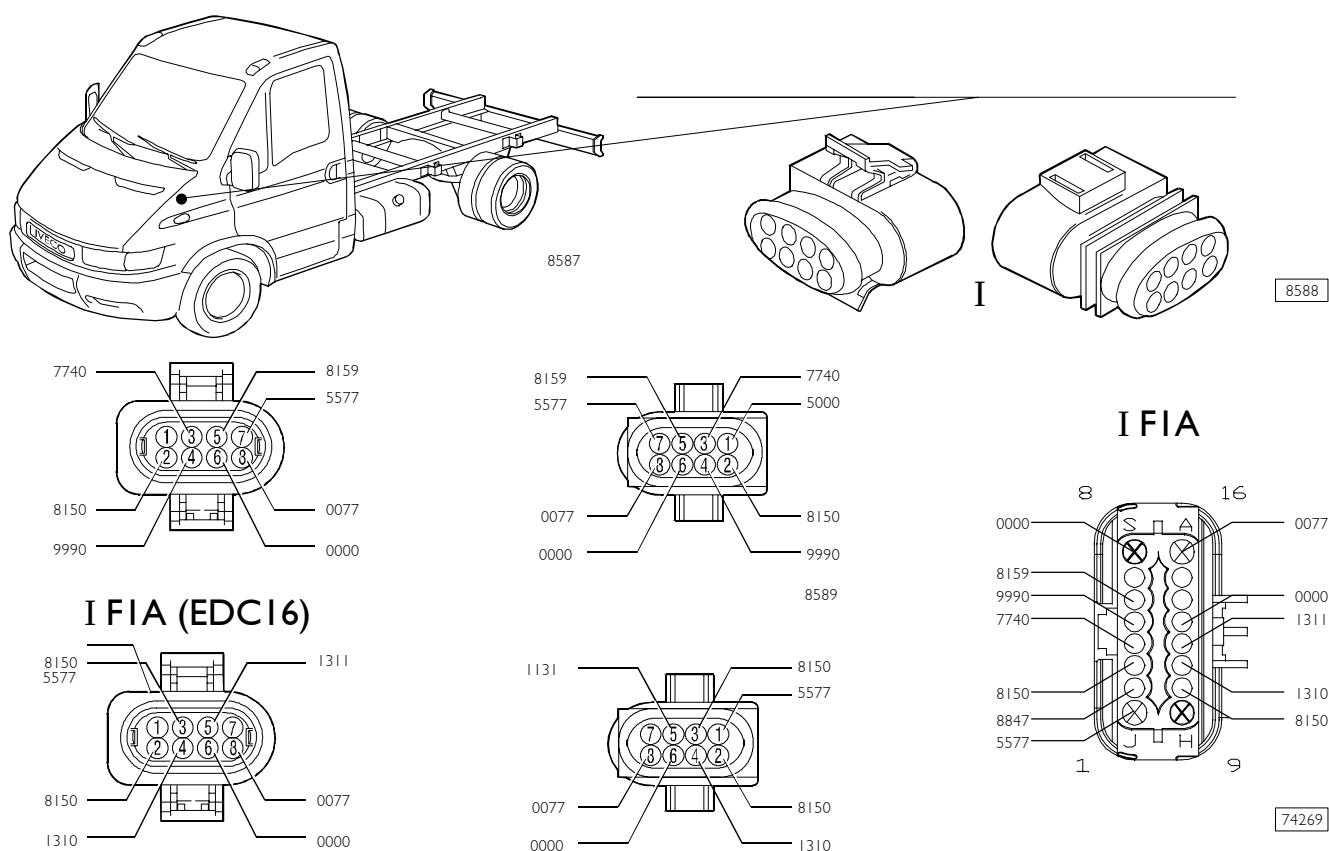
Figure 191/I



Component code	Description
I	Connection to hood cab cable
19010	Preheat plug
25231	Plug insert centre
47035	Coolant temperature sensor
47106	Fuel heat on switch
48035	Engine rpm sensor
48042	rpm sensor on distributor
61101	Fuel heat resistor
70064	I-way fuse holder
78013	Pressure adjustment electro valve
78247	Electrical injection electro valve
85150	EDC center
85157	Fuel pressure sensor
85156	EDC blower air pressure sensor

Joint connector

Connection between cab/bonnet cable and injection cable (Unijet)

Figure 192

Ref.	Ref. (FIA)	Ref. (FIA EDCI6)	Cable colour code	Function
I 1			5000	To pin 36 of EDC electronic control unit
2	3		8150	Fuel pump engagement relay
3	4		7740	Relay engagement for engine cooling electromagnetic joint control
4	5		9990	Air conditioning compressor switching off
5	6		8159	Fuel oil heating circuit switching on
6	8		0000	Earth
7	1	1	5577	Fuel oil heating resistance supply
8	16	8	0077	Fuel oil heating resistance earth
	10	2	8150	Preheat plug centre positive (pin-86)
	11	4	1310	Preheat plug centre control (pin-D1), (EDC pin-B37)
	12	5	1311	Preheat plug centre control (pin-ST), (EDC pin-B42)
	13	6	0000	Preheat plug centre mass (pin-31)
	2		8847	Speed adjustment sensor positive
	7		-	Free
	9		-	Free
	14		-	Free
	15		-	Free
		3	8150	Positive after key for air flow meter

Diagnostic connector

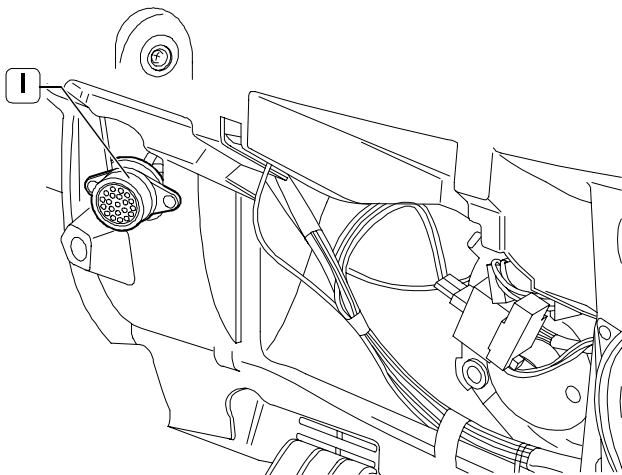
Near the positive terminal of the battery there is a diagnostic connector. It is fastened by 2 screws to a special bracket fitted on the body and access to it is gained by unscrewing the protective cap; suitably connected to the diagnostic system (IWT, MODUS), it allows quick identification of the cause of faults in the various electronic devices.

The table on the following page shows the correspondence between the various electronic systems and the connector pins referring to them.

NOTE The connector shown is seen from the pin side.

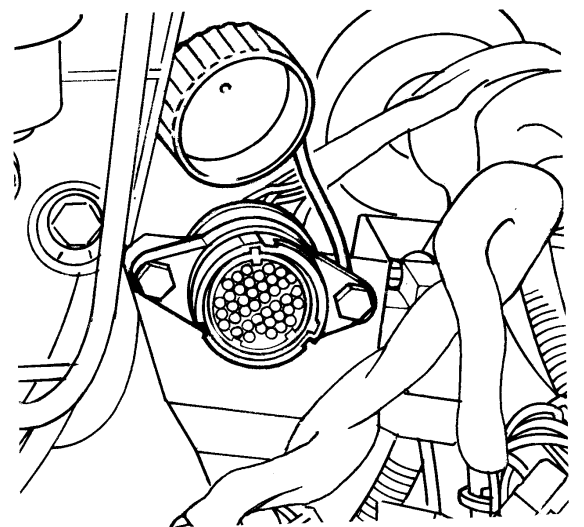
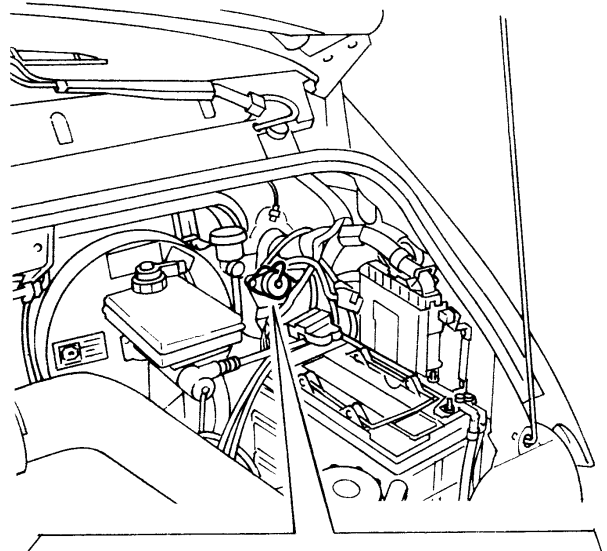
The diagnosis connector on FIA engine vehicles is located on the conveyor unit lower party on the passenger side.

Figure 193



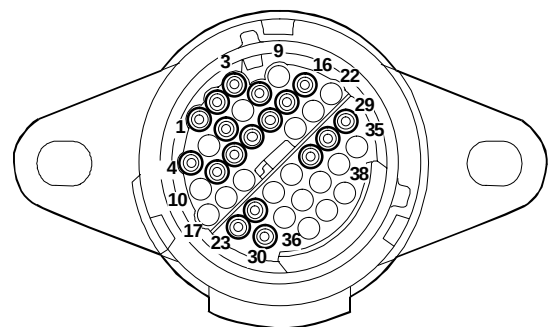
38-POLE DIAGNOSIS TAKEOFF

Figure 194



38-PIN DIAGNOSTIC CONNECTOR

Figure 195



IDENTIFICATION OF THE PINS OF THE 38-PIN DIAGNOSTIC CONNECTOR

Diagnostic connector pin description table

System	Pin	Function
EDC/EGR/Methane	1* 2	L K
ABS/EBD/ABD	3 4	L K
Air Bag/Retarder	6	K
Tachometer (connector A cell 2 instrument cluster)	8	K
Supply +15/A services	11	Input
Immobilizer/Central door locking/Alarm	12	K
Climate control system	13 14	L K
Self-levelling suspension	15 16	L K
Engine timing signal (connector B pin 28 EDC control unit) Engine timing earth (battery -)	23* 24	— Screening
Supply +30	27	Battery +V
Engine rpm	28	RPM
Vehicle speed	29	—
Earth	30	—

(*) In vehicles equipped with FI engine and EDC 16 electronic control unit, diagnosis connector pins 1 and 23 are free.
Pin 21/22 CAN line

High-pressure pump (FIA engine)

This pump is similar to the 8140 engine one.

It differs in Rail line pressure, modulated at between 250 and 1600 bars by the electronic centre by acting on the pressure regulator electro valve and is not provided with a 3rd pump cutout electro valve (78015).

Pump (3) is oriented on the pump shaft cam.

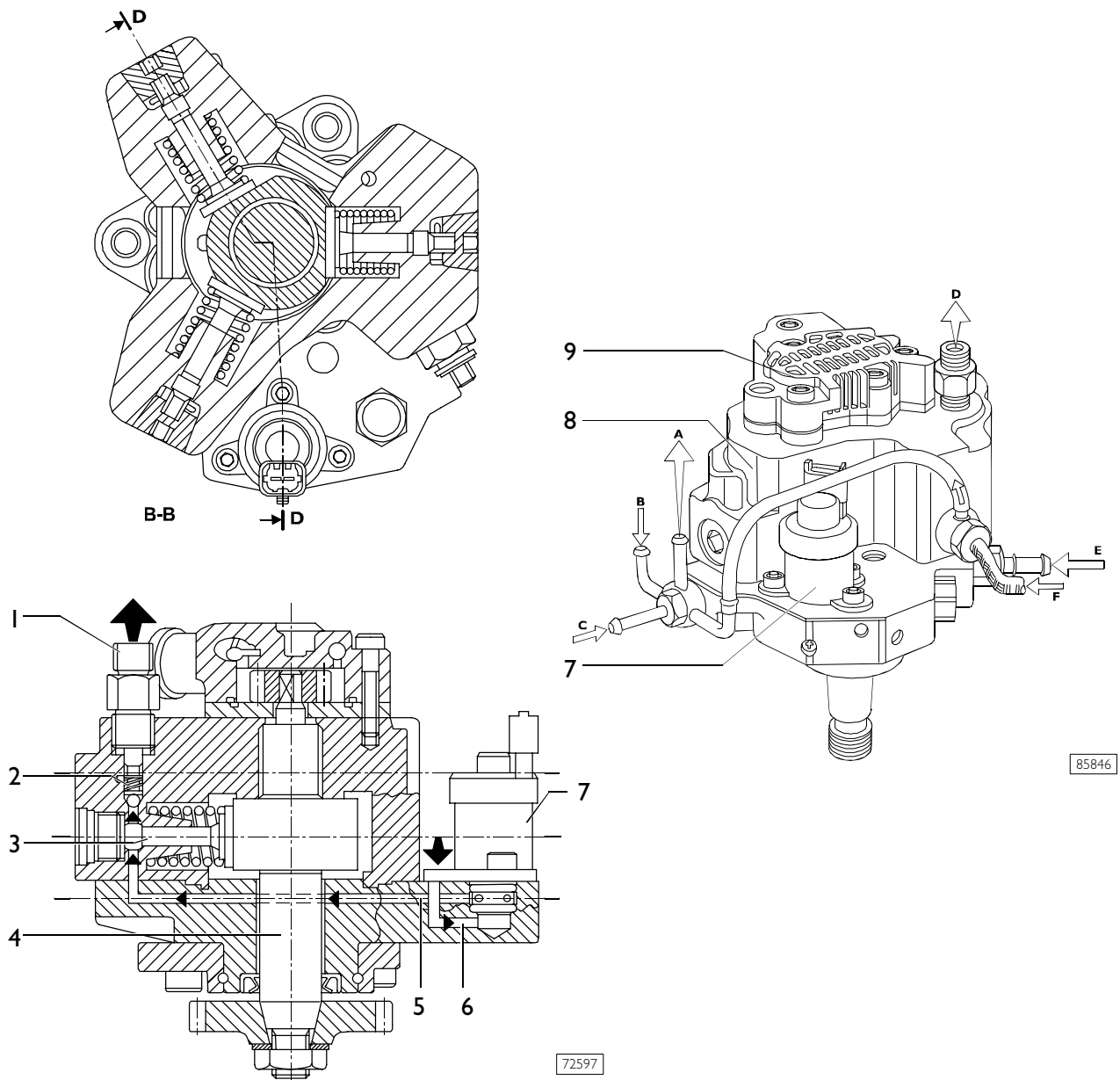
During aspiration the pump is powered via supply conduit (5).

The amount of fuel to be sent to the pump is set by pressure regulator (7).

The pressure regulator stops fuel inflow to the pump based on PWM input from the centre.

During pump compression, fuel reaches a pressure level sufficient to open the pressure valve to common rail (2), and supplies it via output (1).

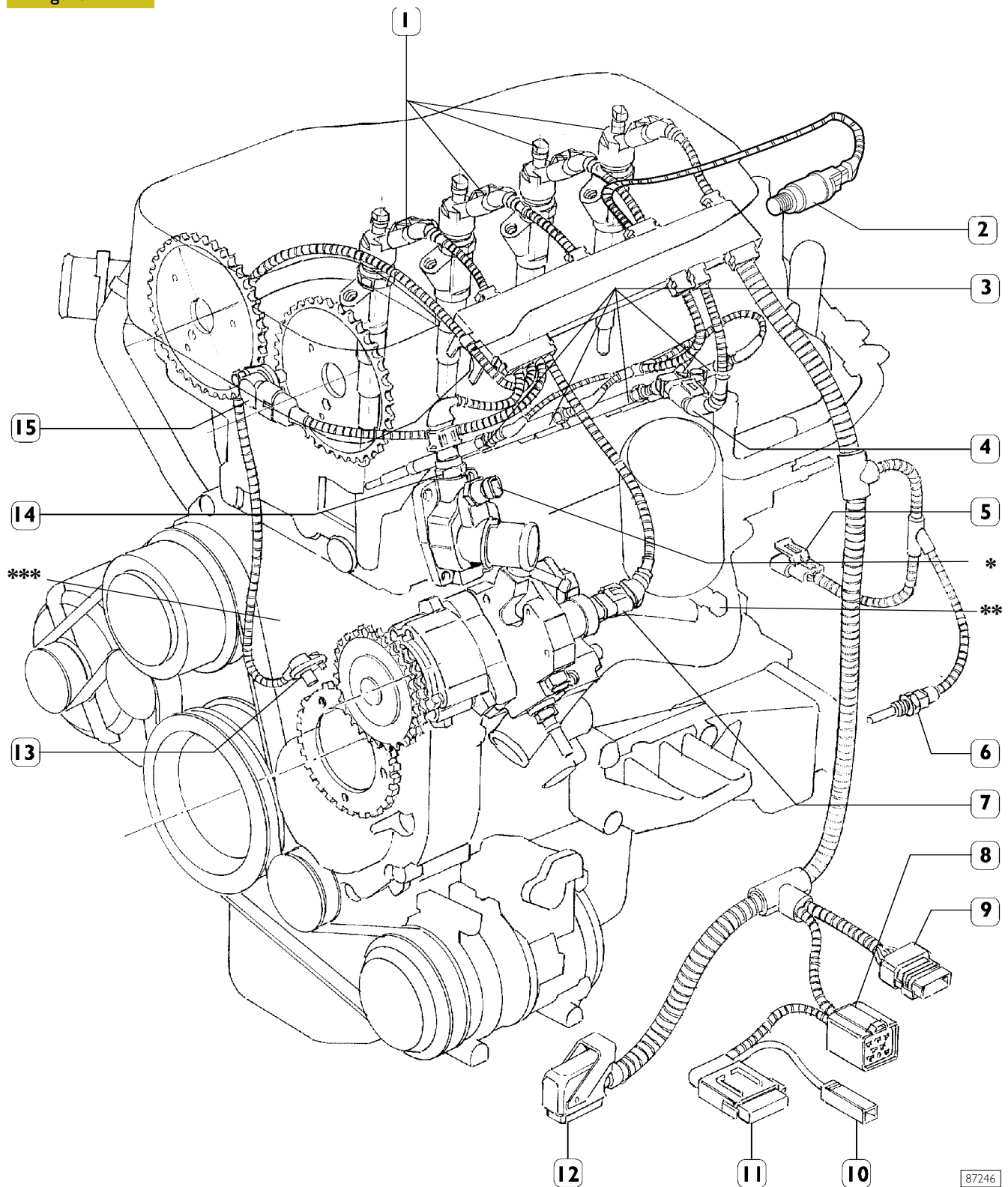
Figure 196



1. Rail delivery output – 2. Rail delivery valve – 3. Plunger – 4. Pump shaft – 5. Plunger supply duct – 6. Pressure regulator supply duct – 7. Pressure regulator – 8. High-pressure pump – 9. Mechanic feed pump – A. To the tank – B. Return from hydraulic accumulator (rail) – C. Fuel filter return – D. Delivery to hydraulic accumulator (rail) – E. From the tank – F. Return from injectors.

FIC ENGINE

Figure 197

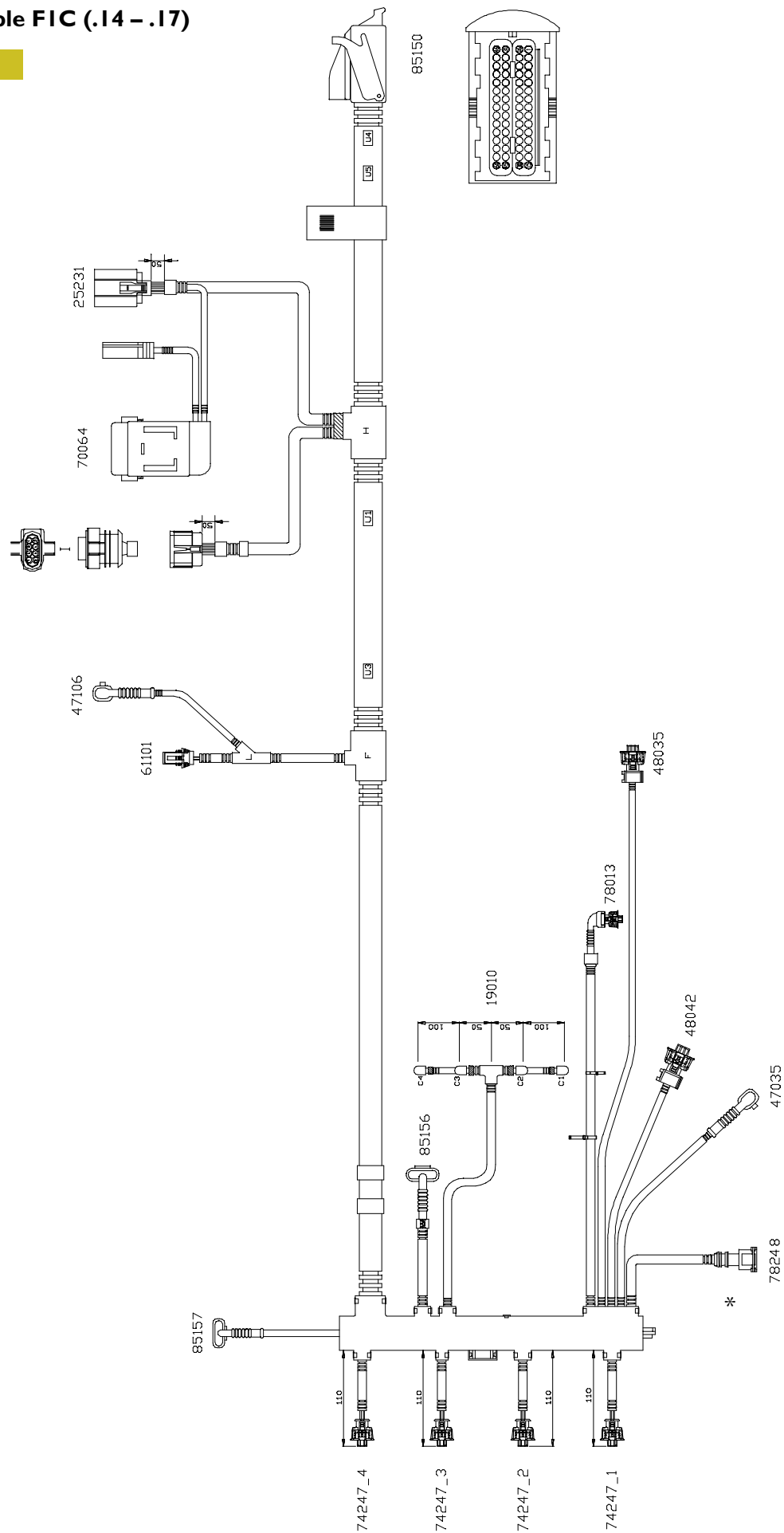


1. Injectors – 2. Common-Rail pressure sensor – 3. Pre-heating glow plugs – 4. Air temperature sensor – 5. Pre-heated diesel oil filter – 6. Diesel oil temperature sensor – 7. High pressure sensor – 8. Pre-heating glow plugs central unit – 9. Sectioning by cab cable – 10. Pre-heating central unit positive – 11. Fuse – 12. Central unit – 13. Engine r.p.m. sensor – 14. Water temperature sensor for EDC – 15. Camshaft r.p.m. sensor – *. Water temperature transmitter for thermometer – **. Oil filter clogged sensor – ***. Engine oil level sensor

87246

Injection cable FIC (.14 – .17)

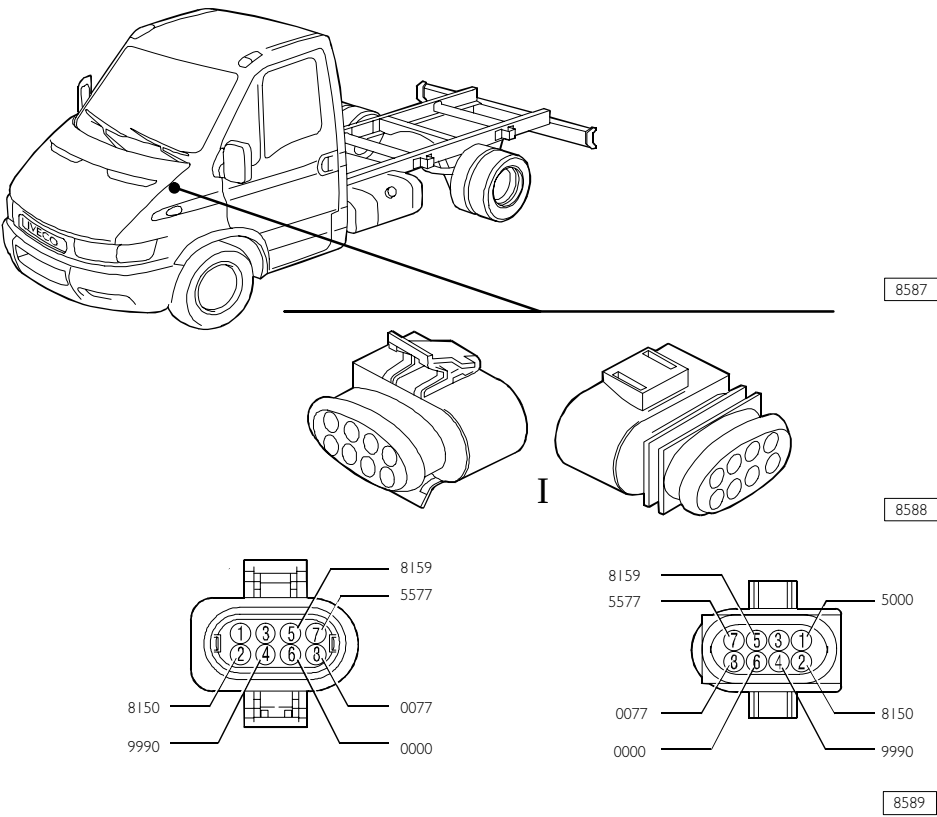
Figure 198



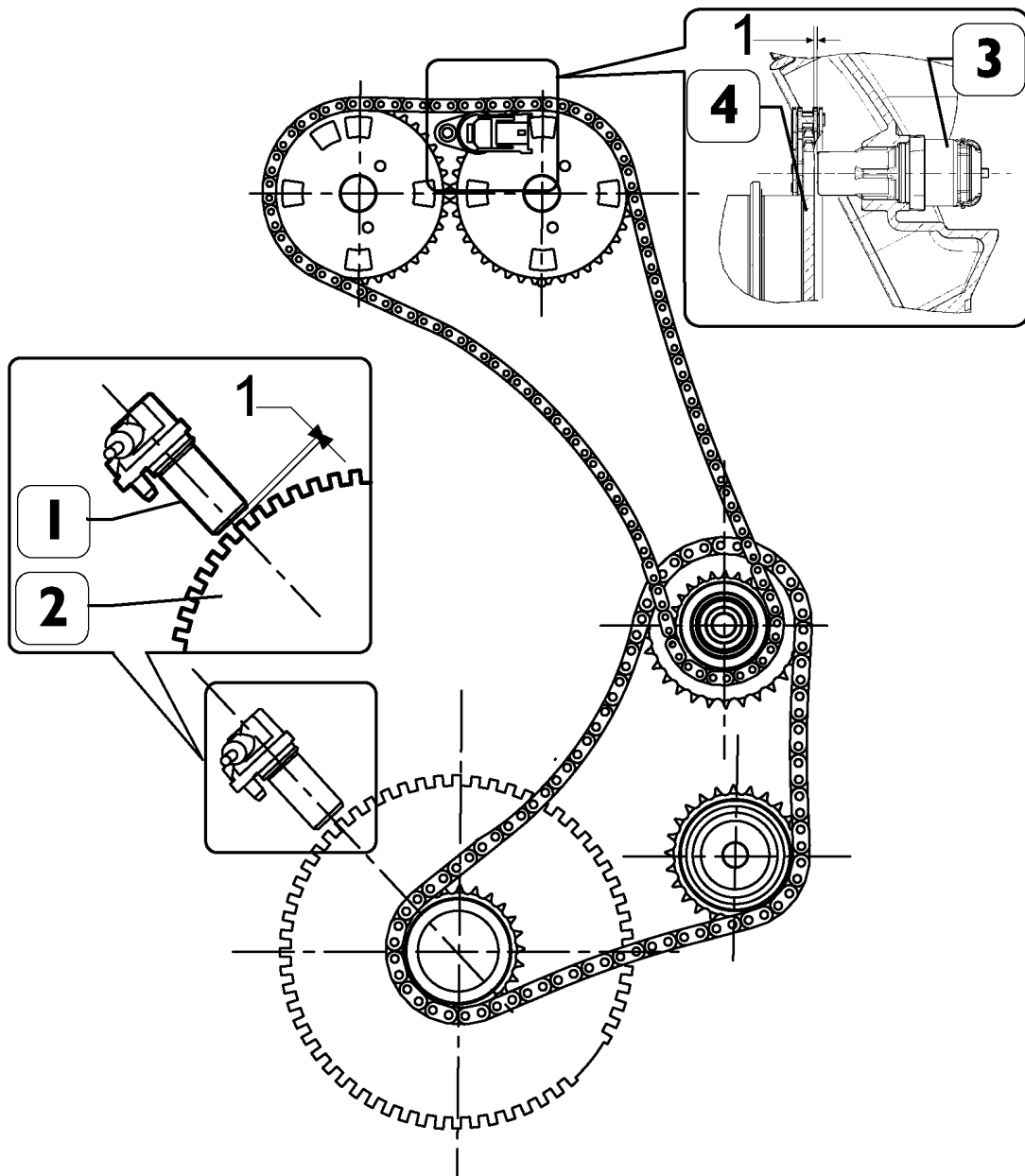
*with .17 (VGT)

Connection between cab/bonnet cable and injection cable

Figure 199

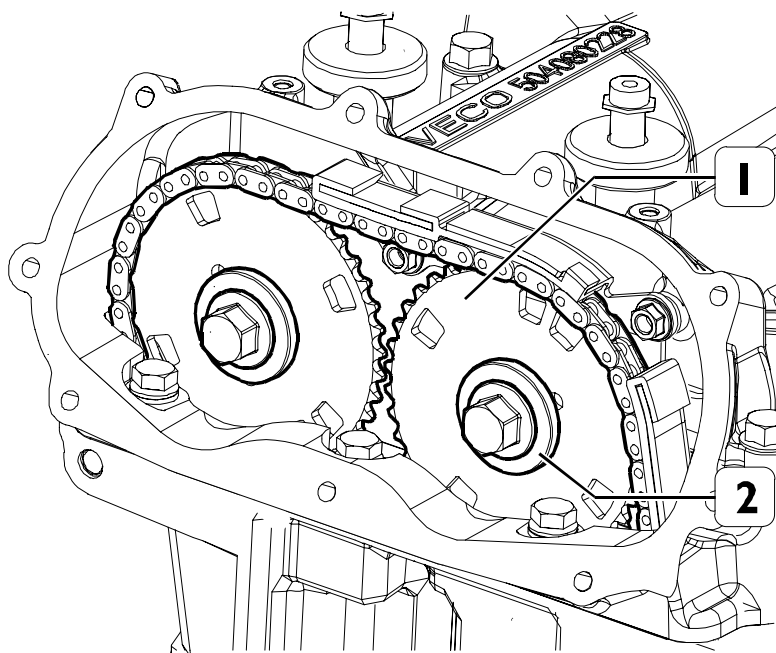


Ref.	Cable colour code	Function
I	1 5577	Heated diesel oil filter positive
	2 8150	Pre-heating central unit (pin 86)
	3 -	-
	4 1310	Pre-heating central unit (pin-DI)
	5 1311	Pre-heating central unit (pin-ST)
	6 0000	Pre-heating central unit earth (pin-31)
	7 -	-
	8 0077	Heated diesel oil filter earth

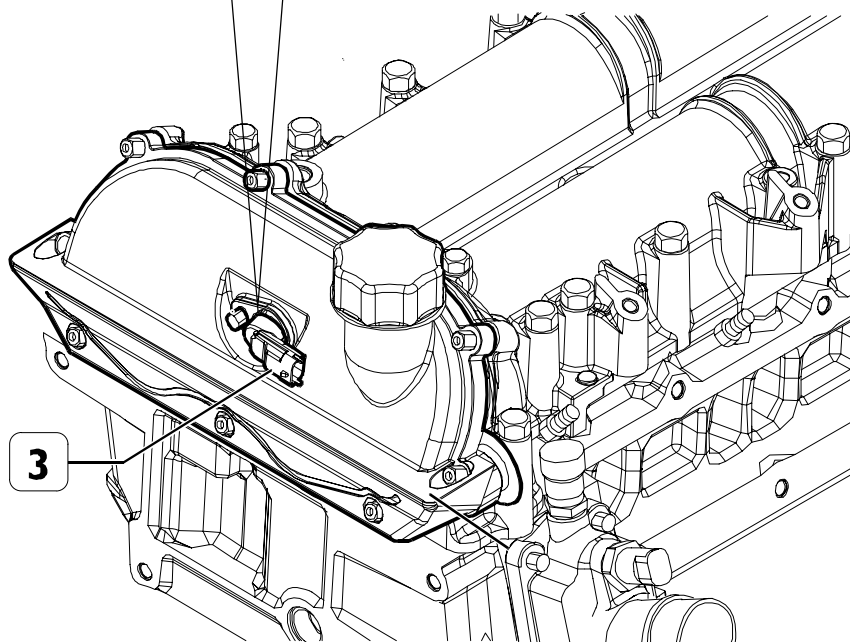
R.p.m. / timing sensors**Figure 200**

88056

1. R.p.m. sensor – 2. Phonic wheel on drive shaft – 3. Timing sensor – 4. Phonic wheel on camshaft

Timing sensor (stroke)**Figure 201**

88264



88267

1. Phonic wheel on camshaft – 2. Identification slots – 3. Sensor

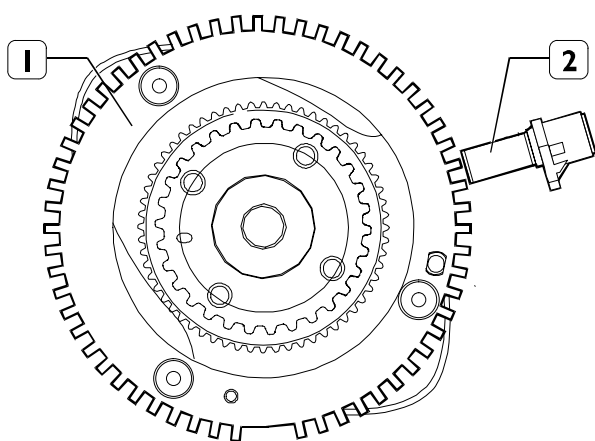
RPM sensor

A phonic wheel is fitted on the drive shaft. As the sensor detects existing teeth passing, it provides the central unit with the signal that is necessary to determine engine r.p.m.'s.

The variation of the signal generated by the lack of some teeth (synchronisation gap) occurring at each drive shaft turn is the reference signal which enables the central unit to detect the lead of the pair of pistons 1–4 with respect to PMS.

This signal is also used by the control unit to detect the engine rotation speed, the duration of injection and to control the rev counter.

Figure 202

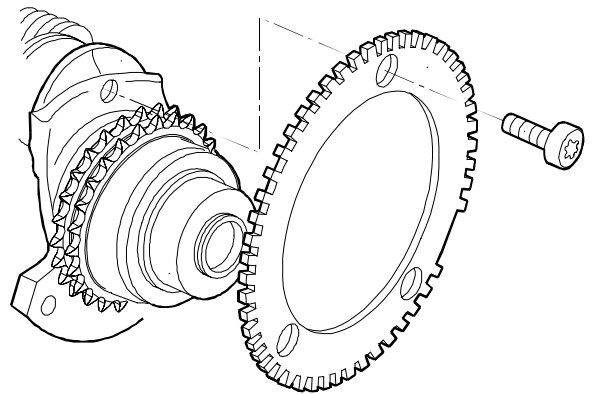


0003319t

TECHNICAL VIEW OF THE SOUND WHEEL AND
SENSOR

1. Sound wheel – 2. Sensor

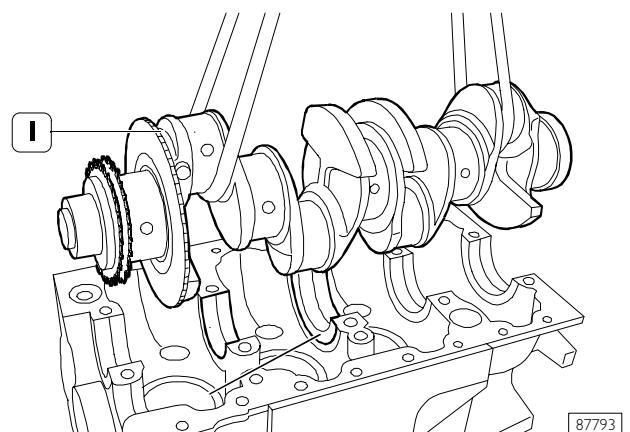
Figure 203



87792

Phonic wheel mounting

Figure 204



87793

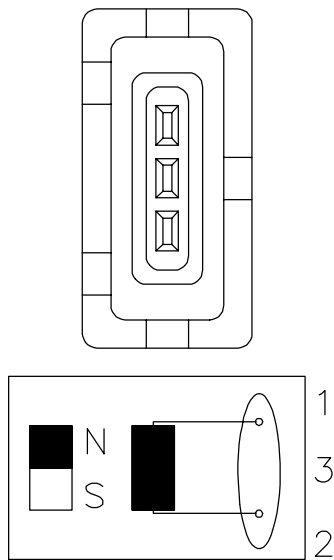
1. Phonic wheel

Figure 205



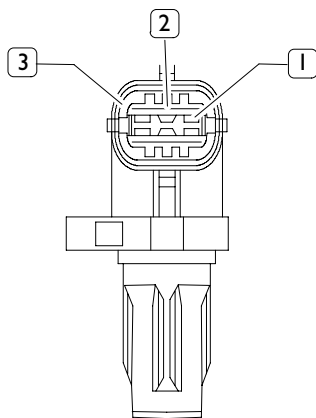
RPM SENSOR AND CONNECTION CABLE

Figure 206



SENSOR CONNECTOR AND WIRING DIAGRAM

Figure 207



85712

TIMING SENSOR

1. Earth – 2. Signal output – 3. Power supply positive

RPM sensor

These are inductive sensors.

The flywheel sensor (48035) is connected at pins 27 and 12 of connector A of the control unit.

Timing sensor

A semiconductor layer, immersed in a magnetic field and through which current flows, generates a potential difference (called Hall voltage) at its ends.

If current intensity remains constant, the generated voltage depends only on the magnetic field strength: periodical variation of field strength is enough to obtain a modulated electric signal.

The smooth portion of the phonic wheel (distributing shaft pulley) covers, while moving, the sensor, thus blocking the magnetic field with resulting low output signal.

On the contrary, the sensor generates a high signal next to the openings and when a magnetic field is available.

Phase sensor signals are acquired, and the engine position is recognized according to the sequence of the phonic wheel notches.

The mounting function makes it possible to identify signal errors and interferences (if any).

The resulting signal is supplied to the processor that controls the injection system.

The sensor (48042) is connected to the central unit at pins A20/50/11.

High-pressure pump

Pressure delivered to the rail is modulated between 250 and 1600 bars by the electronic central unit by operating on the solenoid valve of pressure regulator.

Pump (3) is oriented on the pump shaft cam.

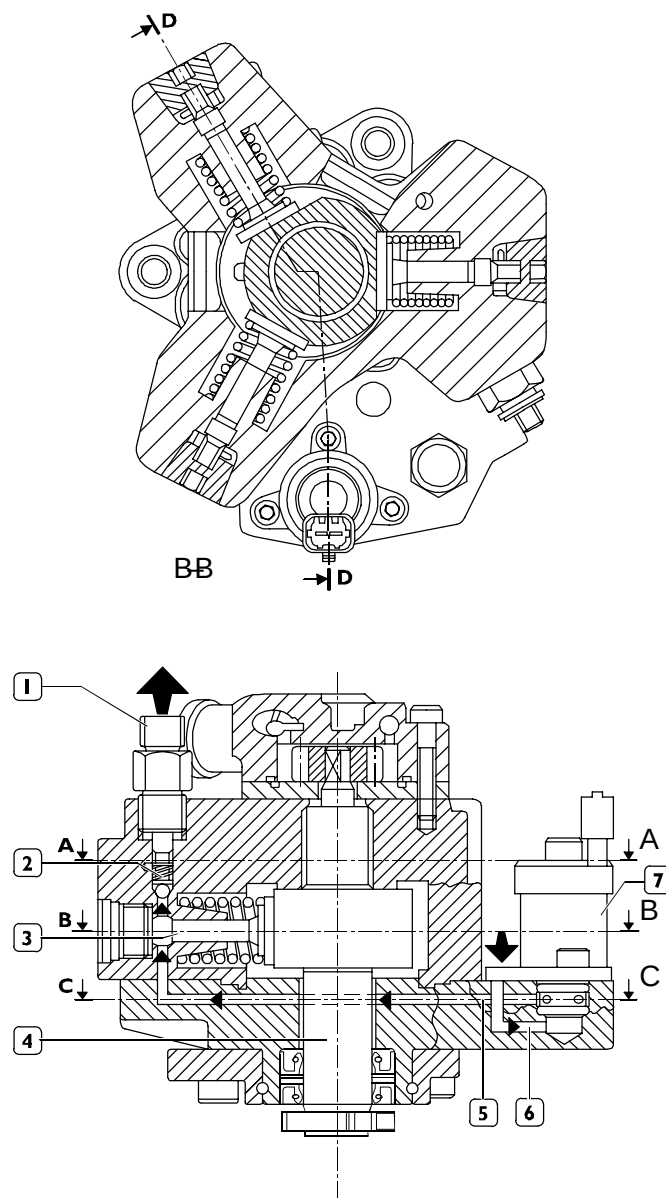
During aspiration the pump is powered via supply conduit (5).

The amount of fuel to be sent to the pump is set by pressure regulator (7).

The pressure regulator stops fuel inflow to the pump based on PWM input from the centre.

During pump compression, fuel reaches a pressure level sufficient to open the pressure valve to common rail (2), and supplies it via output (1).

Figure 208



1. Output for rail pressure – 2. Rail pressure valve – 3. Pump – 4. Pump shaft – 5. Pump supply conduit – 6. Pressure regulator supply conduit – 7. Pressure regulator – 8. High pressure pump – 9. Fuel input connection from filter – 10. Fuel output connection to filter support – 11. Fuel input connection from centre heat exchanger – 12. Fuel output connection from mechanical pump to filter – 13. Supply mechanical pump

88072

Pressure regulator

When the engine control centre pilots the pressure regulator via the PWM signal, solenoid (1) is activated, which in its turn generates movement of magnetic core (2).

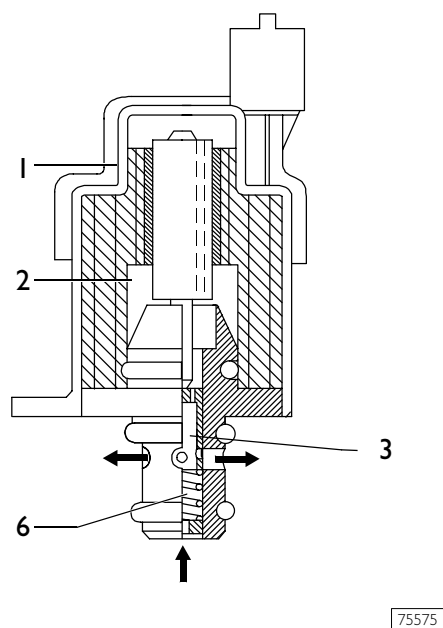
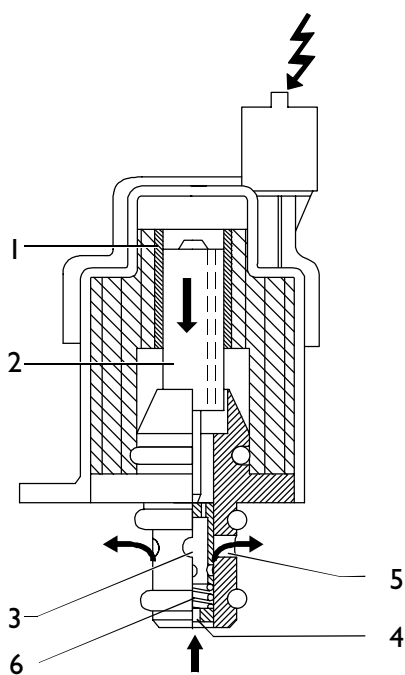
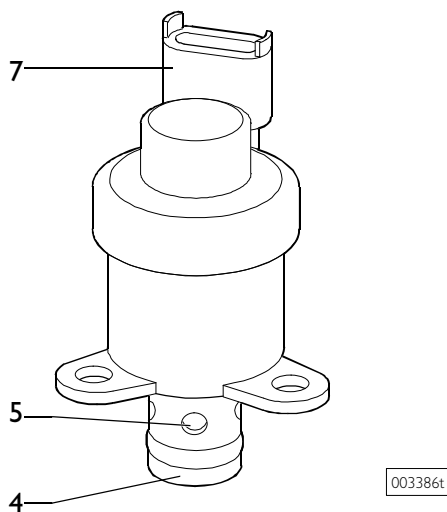
Core movement causes cylinder (3) axial displacement by fuel delivery partialization.

When solenoid (1) is not activated, the magnetic core is moved to its rest position by preload spring (6).

In these conditions, cylinder (3) is in a position to offer maximum fuel passage cross-section.

Control electro valve 78013 is connected to centre connector A pins 19 and 49.

Figure 209



1. Solenoid – 2. Magnetic core – 3. Cylinder – 4. Fuel input – 5. Fuel output – 6. Preload spring – 7. Connector

Rail (pressure accumulator) FIC

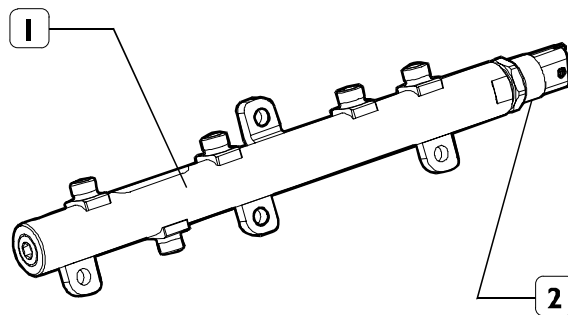
The hydraulic accumulator is mounted in the cylinder head on the side opposite aspiration.

By its volume, it damps fuel pressure oscillations owing to:

- ☐ high-pressure pump operation
- ☐ electro injector opening.

On hydraulic accumulator there is located the fuel pressure sensor.

Figure 210



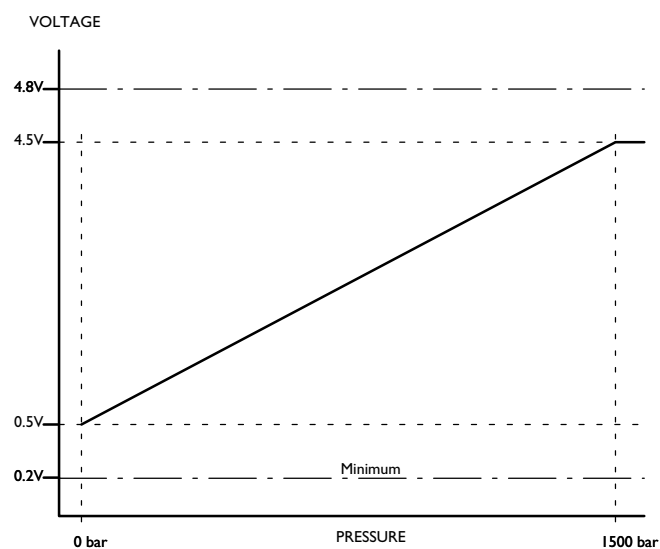
88418

1. Rail – 2. Pressure sensor

Pressure sensor

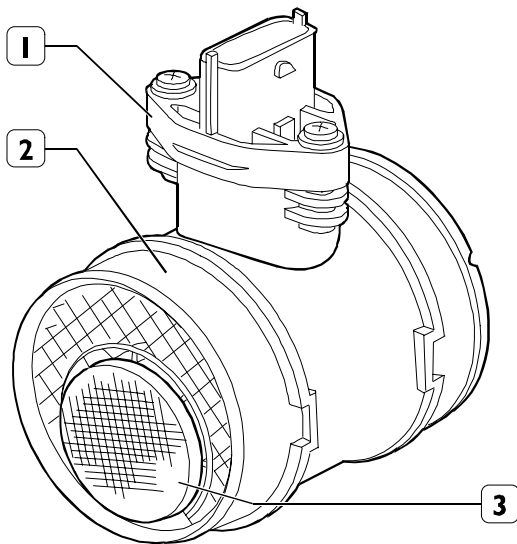
Fitted to a rail end, it measures fuel pressure present to the purpose of determining existing fuel pressure. Pressure value is used to control pressure and determine injection electric control duration (85157). It is connected to the central unit at pins A 8/43/28. It is fed at 5 V.

Figure 211



PRESSURE LIMITER OPERATING GRAPH

Figure 212

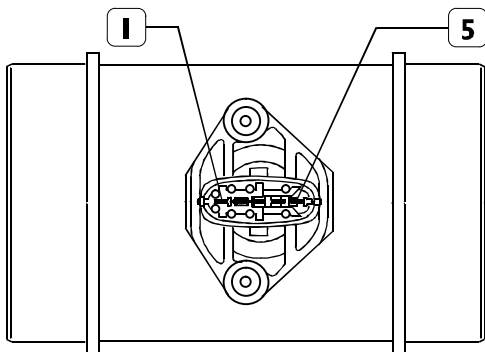


003333t

GAUGE

1. Connector – 2. Gauge body – 3. Air input grid

Figure 213



003334t

TECHNICAL VIEW OF GAUGE CONNECTOR

Air delivery gauge with EGR

Used in the EGR version to replace the one mounted on the engine aspiration manifold.

The gauge is of the heated film type and is located on their aspiration conduit between the turbine and the air filter.

The gauge contains the aspired air temperature sensor.

It is connected to the central unit on connector "A".

Pin 1 sensor – Pin A37 – temperature signal

Pin 2 sensor – Power supply (+15)

Pin 3 sensor – Pin A44 – mass

Pin 4 sensor – Pin A29 – reference voltage

Pin 5 sensor – Pin A42 – pressure signal

* **If present**

The operating principle is based on a heated membrane inserted in a measurement canal through which air to the engine flows.

The hot film membrane is kept at a constant temperature some 120 °C above incoming air level by the heating resistor.

The air mass traversing the measurement canal tends to subtract heat from the membrane so current must cross the resistor to maintain constant film temperature.

Current absorbed is proportional to the air mass flowing to the engine and is measured with a Wheatstone bridge and the signal is forwarded to the electronic centre.

Air temperature/pressure sensor (without EGR)

This component incorporates a temperature sensor and a pressure sensor (85156).

It is fitted on the engine intake manifold and measures the maximum flow rate of the intake air which is used to accurately calculate the amount of fuel to be injected at each cycle.

It is connected to the central unit on connector "A".

Pin 1 sensor – Pin A23 – earth –

Pin 2 sensor – Pin A53 – temperature signal

Pin 3 sensor – Pin A13 – 5V – supply –

Pin 4 sensor – Pin A40 – 0 ÷ 5V
pressure signal

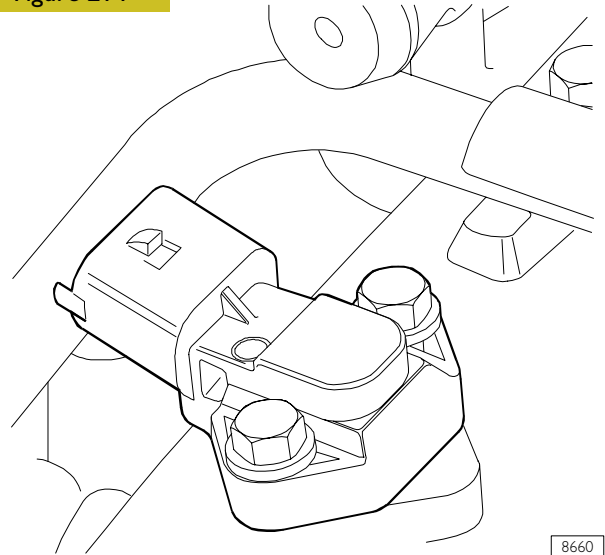
Course of sensor in relation to the temperature

Temperature	Resistance
– 40 °C	48.50 kOhm
– 20 °C	15.67 kOhm
0 °C	5.86 kOhm
20 °C	2.50 kOhm
40 °C	1.17 kOhm
60 °C	0.59 kOhm
80 °C	0.32 kOhm
100 °C	0.18 kOhm
120 °C	0.11 kOhm

Course of sensor in relation to the pressure:

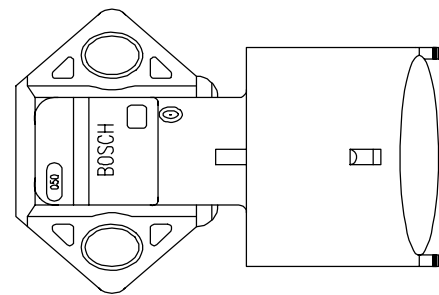
See graph opposite.

Figure 214



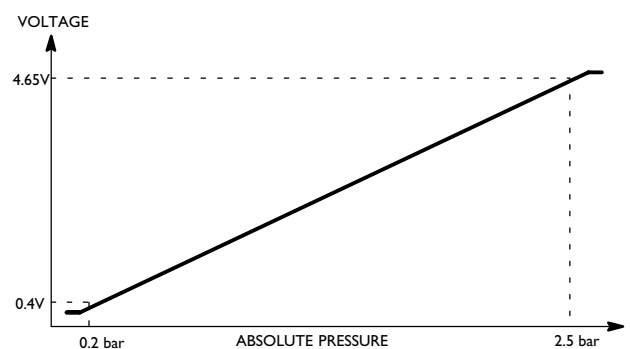
AIR FLOW METER

Figure 215



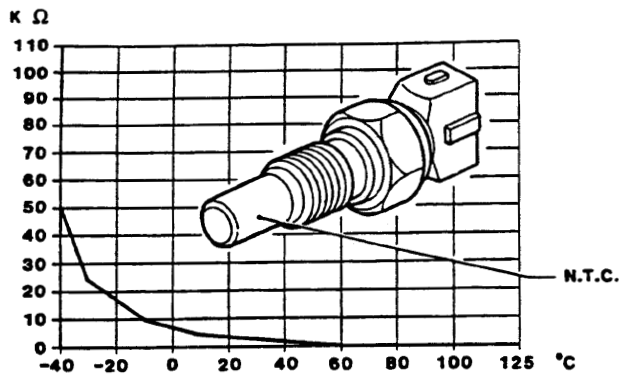
AIR FLOW METER CONNECTION

Figure 216



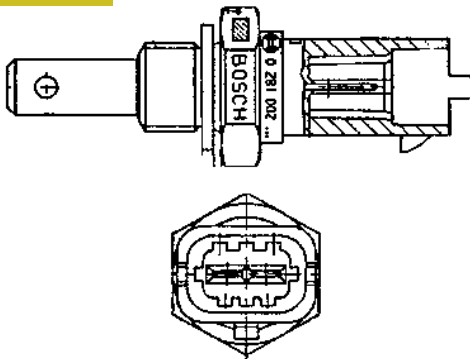
AIR FLOW METER OPERATING GRAPH

Figure 217



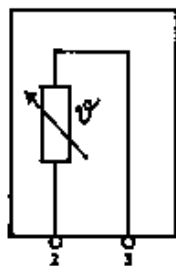
COURSE OF SENSOR RESISTANCE IN RELATION TO TEMPERATURE

Figure 218



TECHNICAL VIEW OF ENGINE COOLANT TEMPERATURE SENSOR

Figure 219



NTC

WIRING DIAGRAM

Atmospheric pressure sensor

This is integrated inside the control unit.

It measures the atmospheric pressure to correct the flow rate in relation to the altitude.

Engine coolant temperature sensor

This is an NTC sensor located on the thermostat box.

It detects the temperature of the coolant fluid to give the control unit information about the engine temperature conditions.

It is connected to pins 58 and 41 of connector A of the control unit.

Course of the sensor in relation to the temperature:

Temperature	Resistance
-40°C	48.30 kOhm
-20°C	15.46 kOhm
0°C	5.89 kOhm
20°C	2.50 kOhm
40°C	1.17 kOhm
60°C	0.59 kOhm
80°C	0.32 kOhm
100°C	0.19 kOhm
120°C	0.11 kOhm

Fuel temperature sensor

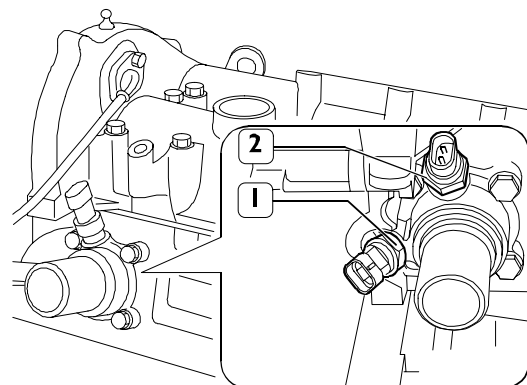
This is an NTC sensor located on the fuel filter.

It detects the temperature of the fuel to give the control unit information about the fuel oil temperature conditions.

It is connected to pins 52 and 51 of connector A of the control unit.

It is exactly the same as the engine coolant temperature sensor.

Figure 220



003324t

LOCATION OF ENGINE COOLANT TEMPERATURE SENSOR

1. EDC signal – 2. Signal instrument panel

Fuel filter

Cartridge degree of filtering: 5 micron

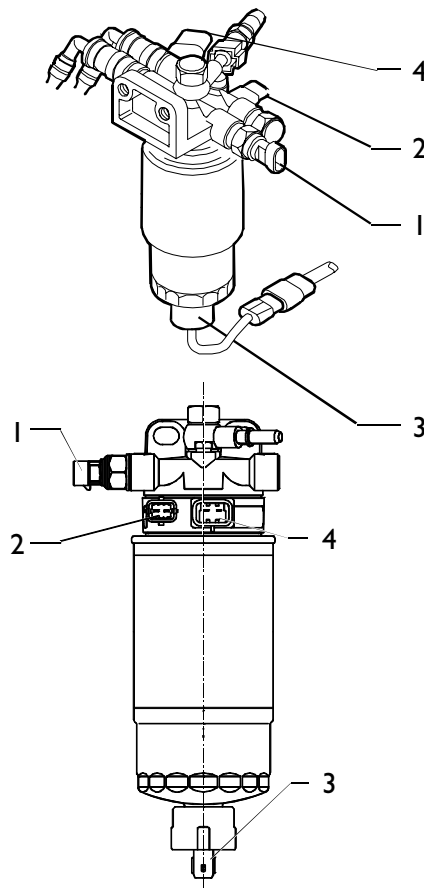
Differential operating pressure (obstruction indicator): 0.6 bar

The fuel temperature sent by the sensor to the electronic injection control unit allows very accurate calculation of the flow rate of the fuel to be injected in the cylinders.

It is located in a fairly accessible position in the left front part of the engine compartment.

1. Actuated by ECU via relay with fuel temperature below 3°C.
2. Filtering element
3. Shows presence of water through a warning lamp on instrument panel.
4. Differential pressure sensor calibrated at 0,6 bar : any clogging is shown by warning lamp
5. It is a NTC sensor connected to EDC for fuel temperature reading enabling electronic control to calculate the amount of diesel oil to be injected into the cylinders.
6. Installed on filter support for excess fuel return to tank.

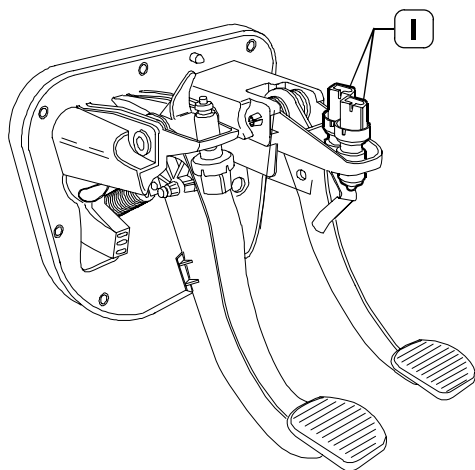
Figure 221



003312t

1. Clogged filter sensor - 2. Fuel temperature sensor - 3. Water sensor - 4. Heater

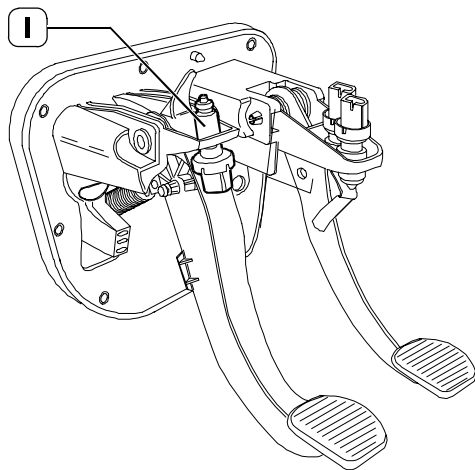
Figure 222



003326t

I. Brake pedal switches

Figure 223



003327t

I. Clutch pedal switch

Brake pedal switches

On brake pedal there are two switches; an N.C. (53565) is directly connected to the central unit at pins K17–K80. It provides signals (voltages) in order to warn the central unit about occurred braking.

Second switch 53501 N.A. provides a voltage for turning on stop lights.

Clutch pedal switch

An N.C. switch connected to electronic centre pin K58 is mounted on the clutch pedal (42374).

The “clutch pedal actuated signal” is used by the centre to identify gear condition selected and gear shifts.

In absence of the pedal pressed switch signal, the centre disengages the Cruise Control function.

Injectors

The solenoid valve controls the lift of the atomiser needle.

On the fuel inlet union a filter protects the injector for impurities. The injector is constructively the same as conventional ones, except that there is no needle return spring.

Access to the injectors is gained by releasing the side soundproof cover from the cylinder head. The fuel recovery pipe has a quick coupling.

The injector comprises two parts:

- ☐ actuator – atomiser composed of pressure rod (1), pin (2) and nozzle (3)
- ☐ control solenoid valve comprising a coil (4) and drive valve (5).

1st phase: rest position

The coil (4) is not activated and the shutter (6) is in the closed position.

The same fuel pressure acts in both the control area (7) and in the pressure chamber (8), but as the shutter (6) is closed, the needle (2) cannot be raised.

2nd phase: start of injection

The coil (4) is energised and causes the shutter (6) to move upwards.

The fuel of the control volume (9) flows towards the backflow duct (10) causing a drop in the pressure in the control area (7).

At the same time, the pressure of the fuel in the pressure chamber (8) causes the needle (2) to rise, resulting in fuel injection to the cylinder.

3rd phase: end of injection

The coil (4) is not activated and makes the shutter (6) return to the closed position, which re-creates a balance of forces that makes the needle (2) return to the closed position and consequently end injection.

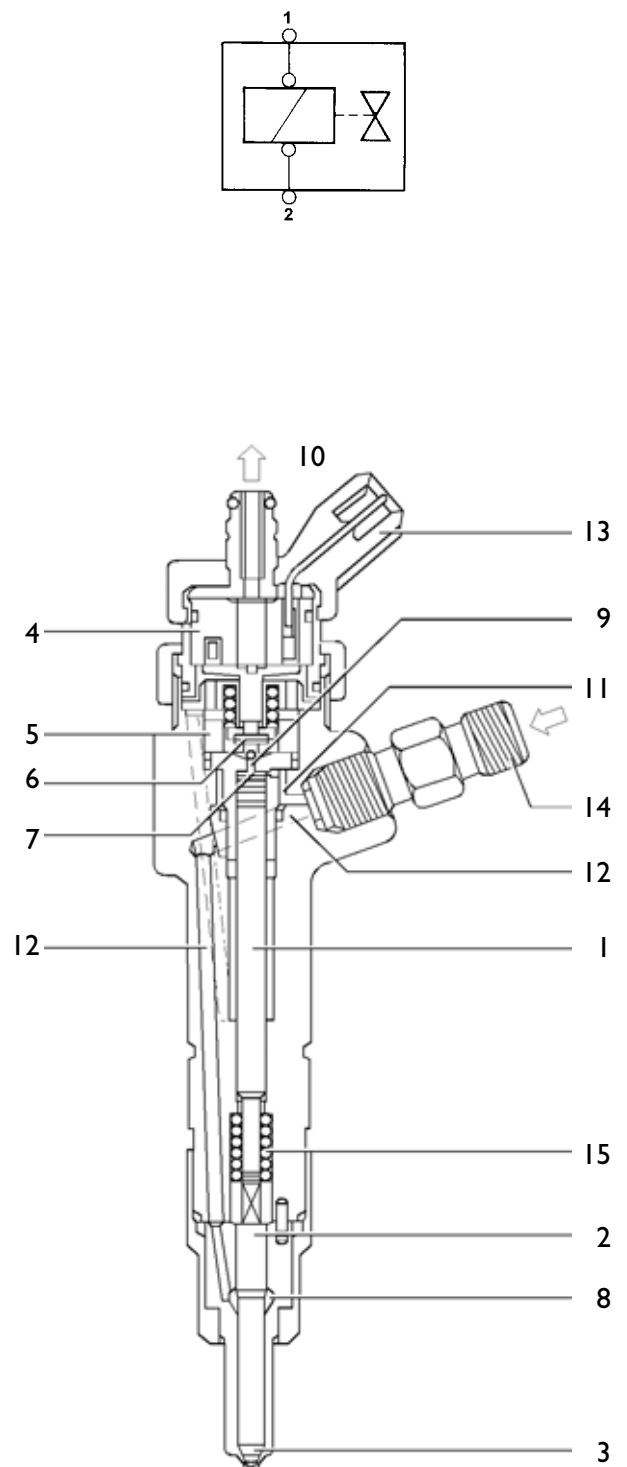
Injectors (78247)

The solenoid valve is of the N.C. type.

The injectors are connected individually to the control unit at the following pins:

- ☐ A16 / A47 cylinder 1 injector
- ☐ A2 / A31 cylinder 2 injector
- ☐ A1 / A46 cylinder 3 injector
- ☐ A17 / A33 cylinder 4 injector

Figure 224



INJECTOR WIRING DIAGRAM AND CROSS SECTION
 1. Pressure rod – 2. Needle – 3. Nozzle – 4. Coil – 5. Pilot valve – 6. ball shutter – 7. control area – 8. pressure chamber – 9. Control volume – 10. Backflow duct – 11. Control duct – 12. Supply duct – 13. Electrical connection – 14. High pressure fuel inlet – 15. Spring

Electromagnetic junction fan

The fan is provided with an electromagnetic junction monitored by the electronic centre pin A39 that activated the junction remote control switch, to optimise water cooling.

The electrical fan remote control switch is activated or deactivated by the centre according to the temperature of:

- ☐ the coolant
- ☐ over supply air
- ☐ the fuel

Engine coolant temperature

Activated at over 96 °C and deactivated at under 84 °C

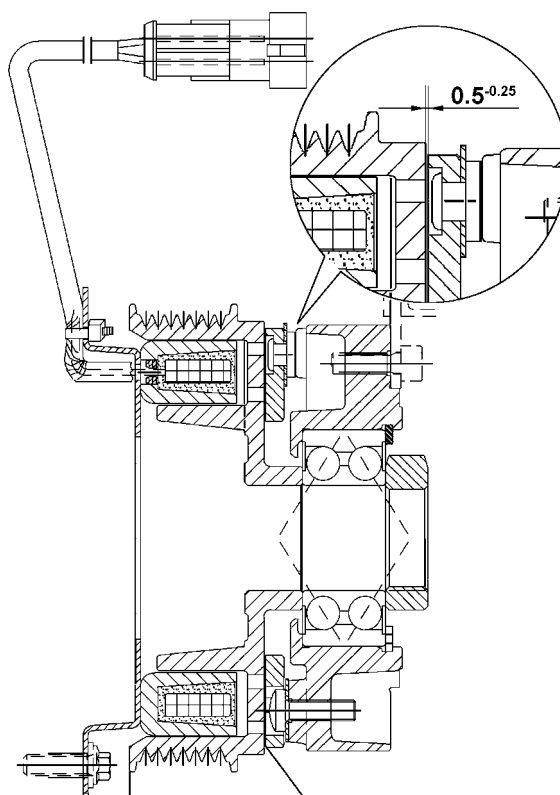
Over supply air temperature

Activated at over 75 °C and deactivated at under 65 °C

Fuel temperature

Activated at over 20 °C and deactivated at under 10 °C

Figure 225



88064

ELECTROMAGNETIC JUNCTION TECHNICAL VIEW
1. Coil – 2. Connector

CLIMATE CONTROL

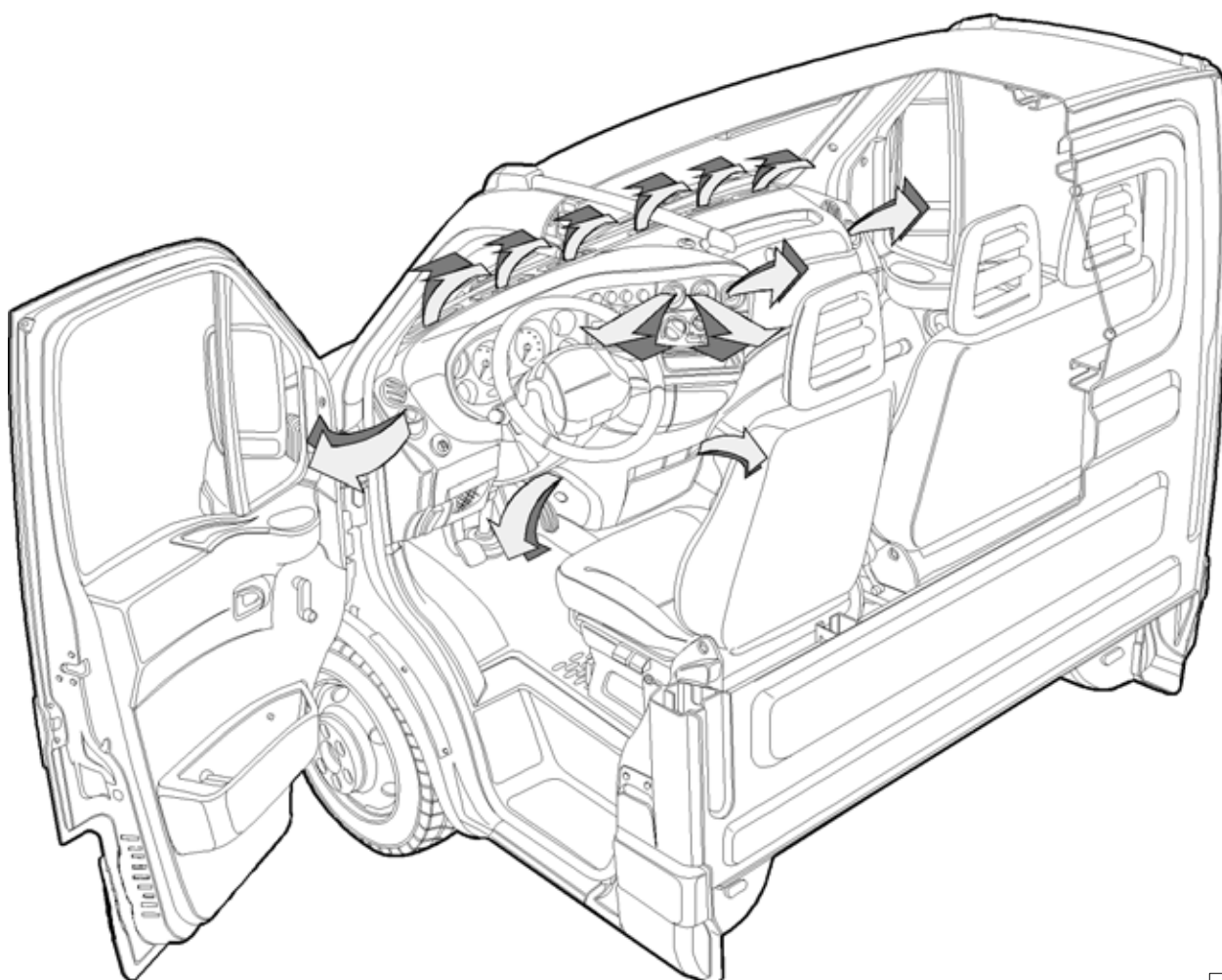
General

The function of the system is to automatically or manually control the temperature required in the cab.

According to the set required temperature the system controls:

- ☐ automatically:
 - air vent temperature
 - fan speed (continuous changing)
- ☐ manually:
 - fan speed
 - compressor on/off
 - outside air inlet/re-circulation
 - MAX DEF function (defrosting);
- ☐ totally manually (through bowden cables):
 - distribution of air flows to the air vents.

Figure 226



000250t

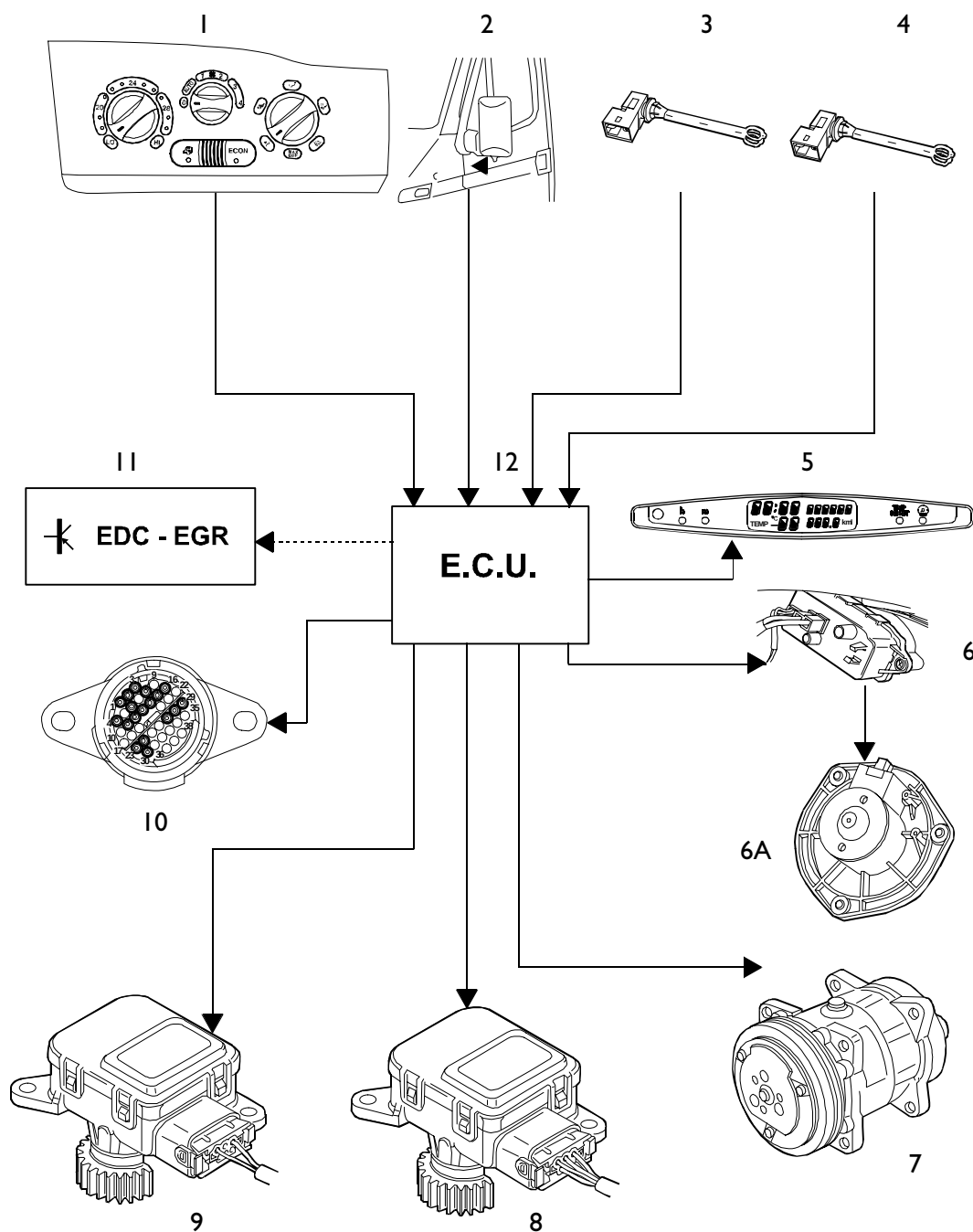
AIR FLOW DISTRIBUTION INSIDE THE CAB

Operating logic

A special electronic control unit controls automatic system operation checking the various parameters in order to provide the comfort required in the cab.

The control unit carries out and sets certain functions to obtain adaptation of the temperature of the cab to the one required depending on the requests received and the temperature conditions detected.

Figure 227



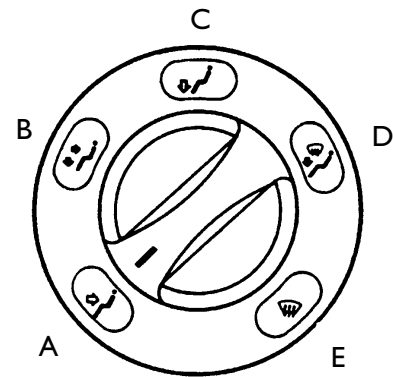
000251t

CLIMATE CONTROL ELECTRONIC CONTROL UNIT CONNECTIONS

1. Controls on dashboard: inside temperature sensor, re-circulation switch, "eco" switch, fan control potentiometer –
2. Outside temperature sensor – 3. Evaporator sensor – 4. Blown air sensor – 5. Digital display – 6. Fan electronic control module – 6A. Fan – 7. Compressor – 8. Gear motor for re-circulation function – 9. Air mixing gear motor –
10. Diagnostic connector – 11. EDC-EGR control unit – 12. Climate control electronic control unit

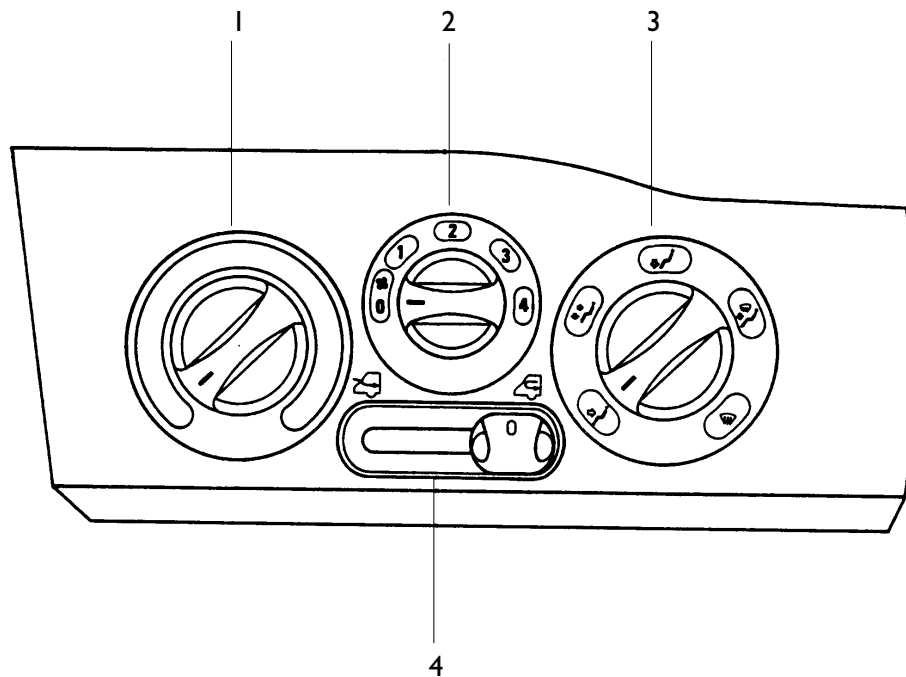
Air distribution knob

- A. Face area vents
- B. Face and floor area vents
- C. Floor area vents
- D. Floor and windscreen area vents
- E. Windscreen area vents

Figure 228

000252t

DETAIL OF AIR DISTRIBUTION KNOB

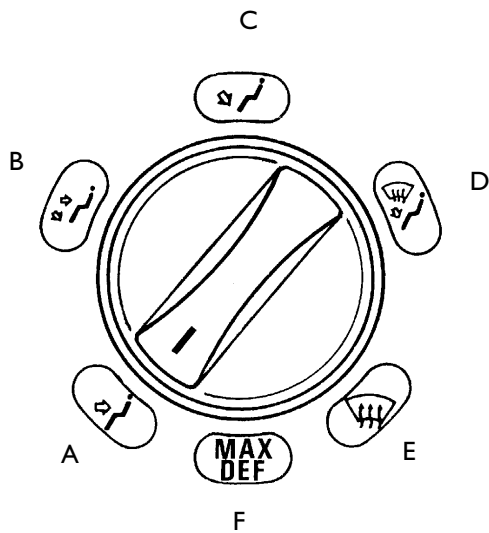
Figure 229

000241t

VENTILATION AND HEATING CONTROLS ASSEMBLY

1. Temperature adjustment knob – 2. Fan speed control knob – 3. Air distribution knob – 4. Outside air inlet or re-circulation

Figure 230



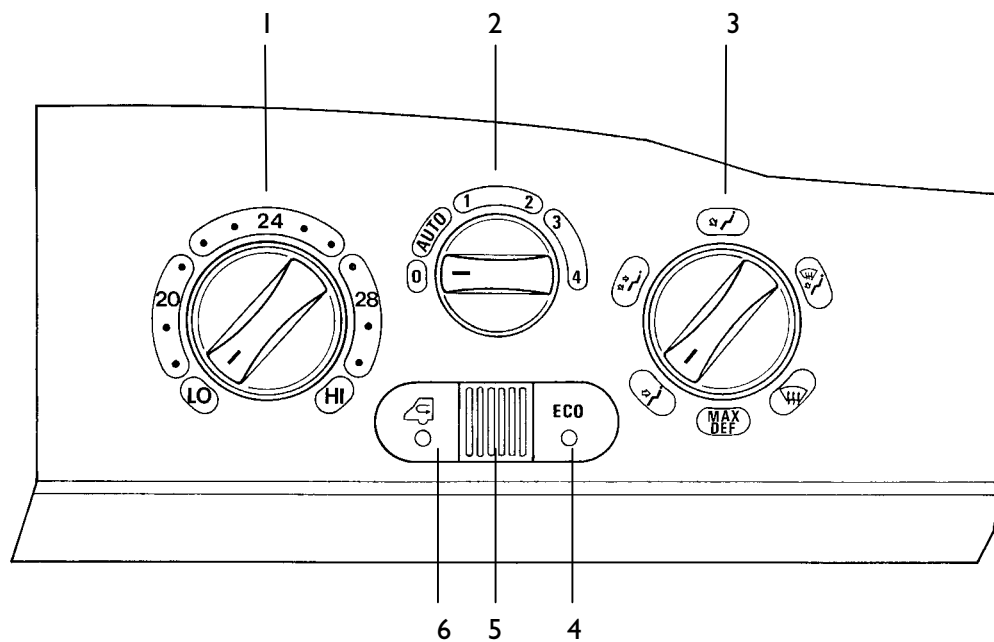
000254t

DETAIL OF AIR DISTRIBUTION KNOB

Air distribution knob with climate control system

- A. Face area vents
- B. Face and floor area vents
- C. Floor area vents
- D. Floor and windscreen area vents
- E. Windscreen area vents
- F. "Max def" function

Figure 231



00025et

CLIMATE CONTROL SYSTEM CONTROLS ASSEMBLY

1. Knob for setting required temperature – 2. Fan speed control knob – 3. Air distribution knob – 4. "Eco" function switch – 5. inside temperature sensor – 6. Re-circulation knob

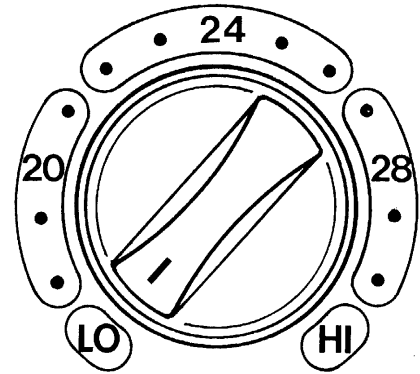
Setting the required temperature

To set the required temperature, turn the corresponding knob (Figure 232) to the 18 °C – 30 °C range.

This knob is connected to a potentiometer which detects the different angular positions and transmits a signal to the electronic control unit.

There are fifteen different positions (one per degree centigrade) with two extremes "LO" and "HI" which correspond respectively to the coldest and warmest positions.

Figure 232



000255t

KNOB FOR SETTING REQUIRED TEMPERATURE

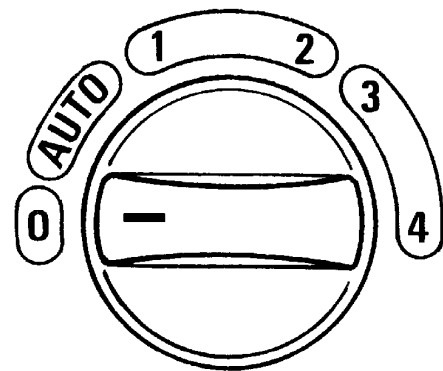
Setting the fan speed

To set the required fan speed, turn the corresponding knob (Figure 233) to the 0 to 4 range.

This knob is connected to a potentiometer which detects the different angular positions and transmits a signal to the electronic control unit.

There are six different positions, four of which are set manually (1, 2, 3, and 4), one for no air flow (0) and an automatic one controlled by the electronic control unit where the system itself selects the most suitable air flow to reach and/or maintain the required temperature (AUTO).

Figure 233



000256t

KNOB FOR SETTING FAN SPEED

“Auto” function

With the ventilation control knob in the “AUTO” position, the control unit will be able to control:

- ☐ The air temperature at the vents.
- ☐ Fan speed.

Function “0” (OFF)

With the ventilation control knob in the “0” position, the control unit acts as follows:

- ☐ No air flow (fan off).
- ☐ Mixing stopped in the previous position.
- ☐ Re-circulation on.
- ☐ Compressor off.
- ☐ “MAX DEF” function cut off if in progress.

“ECO” function

The engagement of this function is completely manual and is obtained through the special switch which makes it possible to disengage the compressor electromagnetic clutch.

Compressor control is however linked to other parameters, namely:

- ☐ Outside temperature (> 5 °C “ON”)
- ☐ Evaporator temperature (> 4 °C “ON”, < 3.5 °C “OFF”)
- ☐ Coolant fluid system pressure (min. 3.5 bar, max. 28 bar)
- ☐ EDC or EGR control unit (in versions where applicable).

“Re-circulation” function

The engagement of this function is completely manual and is obtained through the special switch which makes it possible to shut off the outside air inlet.

In the “MAX DEF” mode this function is cut off regardless of the position of the switch.

“MAX DEF” function

The engagement of this function is completely manual and is obtained turning the air distribution clockwise as far as the “MAX DEF” position.

The control unit will be informed of this request by the closing of a special N.O. microswitch in the rear part of the knob.

The engagement of this function involves:

- ☐ Maximum air flow, regardless of the position of the knob.
- ☐ Air distribution towards the windscreen through bowden cables.
- ☐ Mixing with the highest heat available (memorising the last position of the motor before leaving the adjustment state).
- ☐ Outside air inlet, regardless of the position of the re-circulation button.
- ☐ Compressor on, regardless of the position of the ECO button.

Switching this function off the gear motor moves back to the position memorised previously if the temperature adjustment knob is not at “LO” or “HI”.

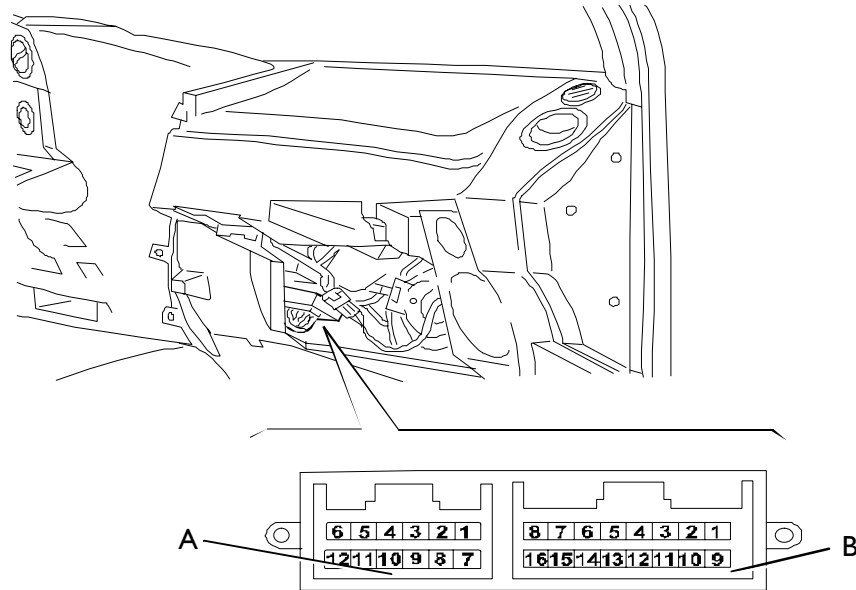
If this function is already selected when the ignition key is turned on, no position will be memorised, and switching it off, the adjustment procedure will be started with the mixing gear motor in the “HI” condition.

Electronic control unit

This automatically controls the system monitoring the thermodynamic parameters in order to provide the suitable climate (temperature and humidity) required by the driver.

It is located on the lower part of the passenger's side duct unit.

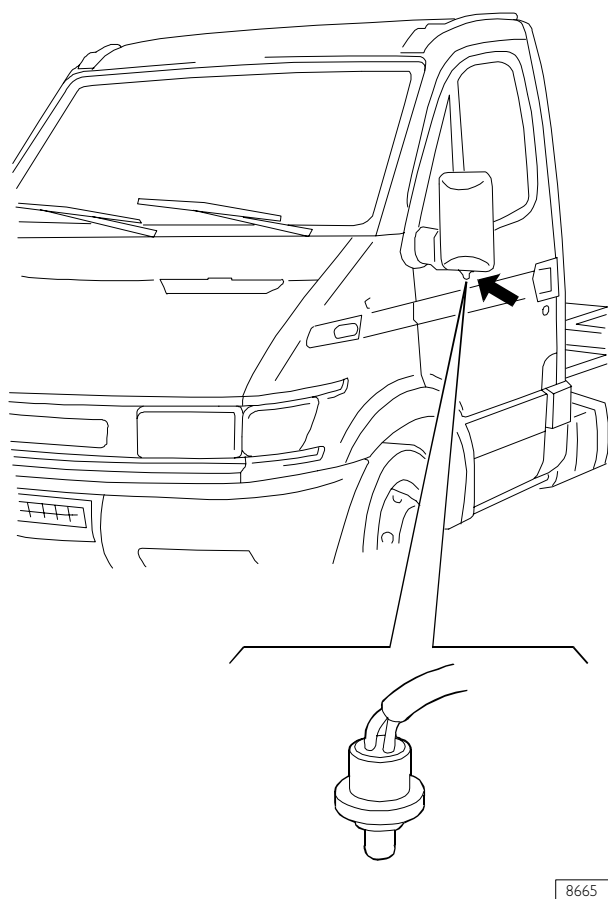
Figure 234



000257t

Connector	Pin	Function
A	1	Signal from treated air temperature sensor
	2	Signal from required temperature potentiometer
	3	Signal from ventilation control potentiometer
	4	Earth
	5	Positive from "ECO" switch
	6	Positive from microswitch for "MAX DEF" function
	7	Signal earth
	8	Supply (+15/A)
	9	5 V supply for ventilation, temperature and air mixing potentiometers
	10	Line L for 38-pin diagnostic connector pin 13
	11	Line K for 38-pin diagnostic connector pin 14
	12	Positive from switch for re-circulation
B	1	Positive (0 to 8 V) for fan electronic control module (0 to 6 V)
	2	Serial line for digital display (instrument cluster pin "B 11")
	3	Signal from evaporator temperature sensor
	4	Reference voltage from air mixing motor (0 to 5 V)
	6	Signal from outside temperature sensor
	7	Signal from inside temperature sensor
	8	Negative for fan electronic control module
	10	Supply (0 to 12 V) for re-circulation gear motor (+12 V for switching on re-circulation)
	11	Supply (0 to 12 V) for re-circulation gear motor (+12 V for switching on re-circulation)
	12	Key-operated positive supply
	13	Positive for compressor control relay
	14	Supply (0 to 12V) for air mixing gear motor (+12V with "HI" function)
	15	Supply (0 to 12V) for air mixing gear motor (+12V with "LO" function)
	16	Reference voltage from electronic fan control module (0 to 12 V)

Figure 235



OUTSIDE TEMPERATURE SENSOR

Outside air temperature sensor

This is an NTC sensor on the left wing mirror (Figure 235) for both types of drive.

The resistance at 25 °C is 10 kOhm.

Operating range between – 30 °C and + 50 °C.

It is connected to pin 6 connector B of the control unit.

Inside temperature sensor

This is an NTC sensor located on the climate control module (Figure 237) between the two “Re-circulation” and “ECO” switches.

It is ventilated and contains a motor/fan so that the temperature reading is not affected by stagnating air inside the dashboard which is not realistic.

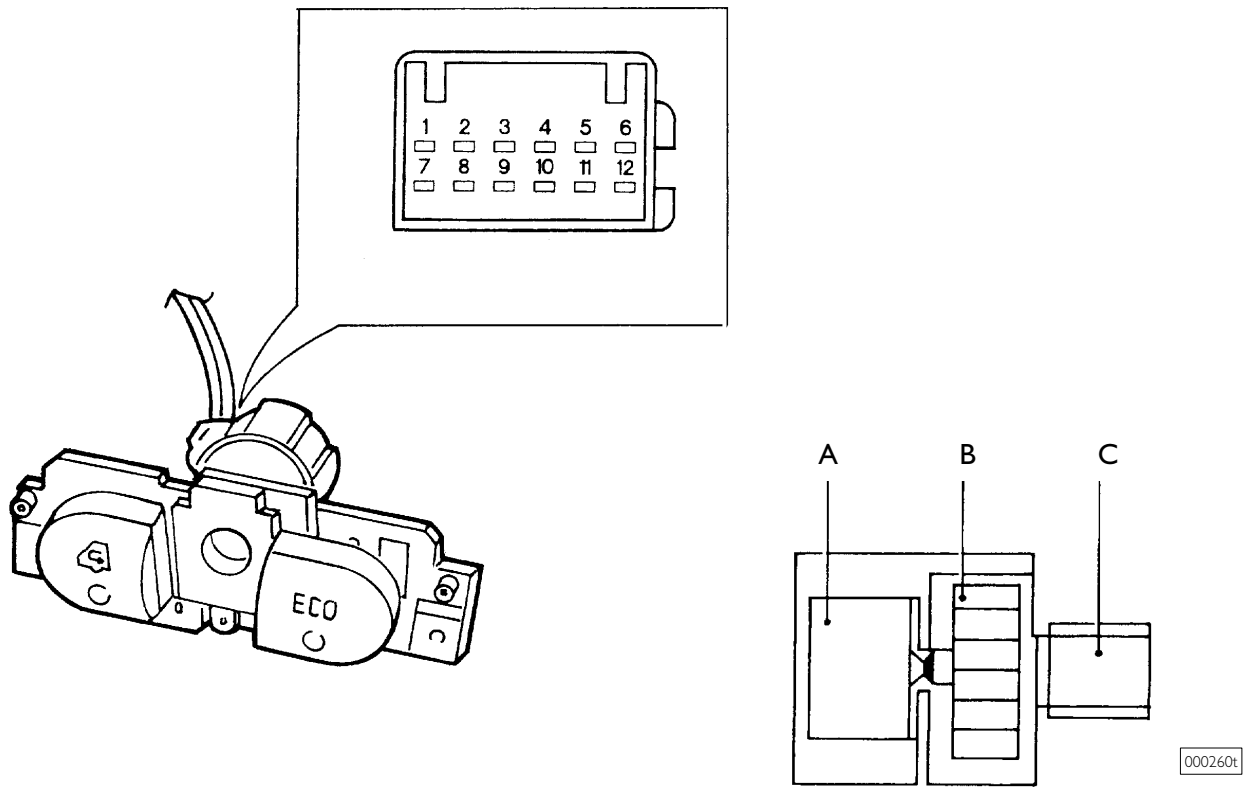
The resistance is 2.2 kOhm at 25 °C

Operating range between + 5 °C and + 45 °C

It is connected to pin 7 connector B of the control unit.

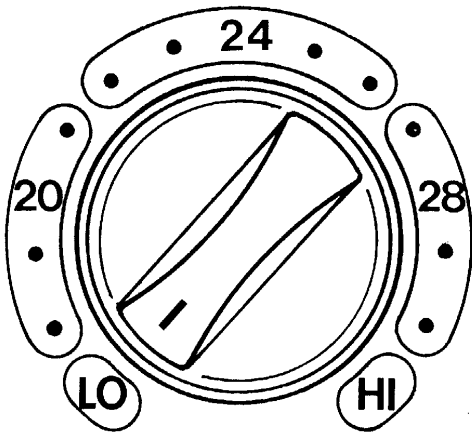
The motor **A**, fan **B** and sensor **C** are a single part and are connected to the wiring by a single connector at pin 12.

Figure 236



Pin	Function
1	Positive for motor
2	Negative for motor
3	Signal earth
4	Sensor signal for control unit (pin 7 connector B)
5	Positive from microswitch for “MAX DEF” function to control unit (pin 6 connector A)
6	—
7	Positive from “ECO” switch to control unit (pin 5 connector A)
8	Earth
9	+15/A supply
10	+15/A supply
11	Negative for lighting pictograms
12	Positive from re-circulation switch to control unit (pin 12 connector A)

Figure 237



000255t

REQUIRED TEMPERATURE SETTING KNOB

Blown air temperature sensor

This is an NTC sensor positioned inside the distributor unit upstream of the air distribution ports to the various vents.

The resistance at 25 °C is 10 kOhm.

Operating range between 0 °C and – 80 °C

It is connected to pin 1 connector A of the control unit.

Evaporator temperature sensor

This is an NTC sensor positioned inside the distributor unit and it is fitted directly on the evaporator. It causes switching on (>4 °C) and switching off (>3.5 °C) of the compressor electromagnetic clutch. The resistance at 25 °C is 10 kOhm.

Operating range between 0 °C and +25 °C.

It is connected to pin 3 connector B of the control unit.

Required temperature potentiometer

This is a potentiometer connected to the required temperature setting knob on the climate control panel.

It detects the various angular positions (Figure 238) and transmits a signal to the electronic control unit for a total of fifteen different positions (one per degree centigrade) two of which are the extremes "HI" and "LO" which correspond respectively to a request for the coldest and warmest positions.

It is supplied by the control unit at 5V.

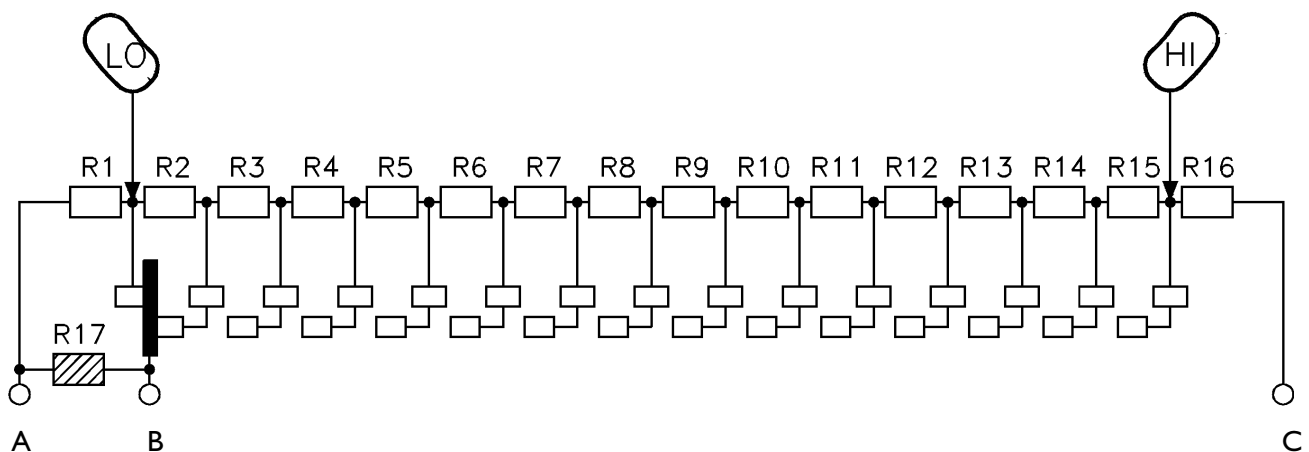
Resistances R1 to R16 are approx. 700 Ohm.

Resistance R17 is approx. 1.1 Mohm

The total resistance R is approx. 11 kOhm.

It is connected to pin 2 and 9 connector A of the control unit.

Figure 238



000265t

POTENTIOMETER POSITION DETECTION LAYOUT

A. Signal earth (ms7) – B. Signal to control unit pin 2 connector A – C. positive from control unit pin 9 connector A

Ventilation control potentiometer

This is a potentiometer connected to the ventilation control knob positioned on the climate control panel.

It detects the various angular positions (Figure 240) and transmits a signal to the electronic control unit for a total of six different positions, four of which are manual flows (1, 2, 3, and 4), one no air flow (0) and one automatic (AUTO).

It is supplied by the control unit at 5 V.

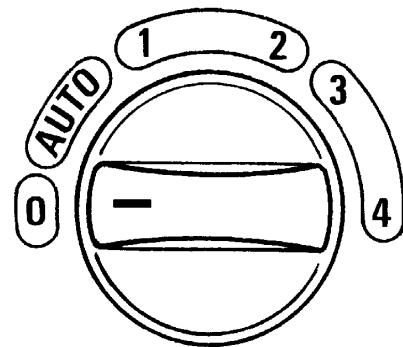
The resistance R1 to R7 are approx. 1.5 kOhm.

Resistance R8 is approx. 1.1 Mohm.

The total resistance R is approx. 10.5 kOhm.

It is connected to pins 3 and 9 connector A of the control unit.

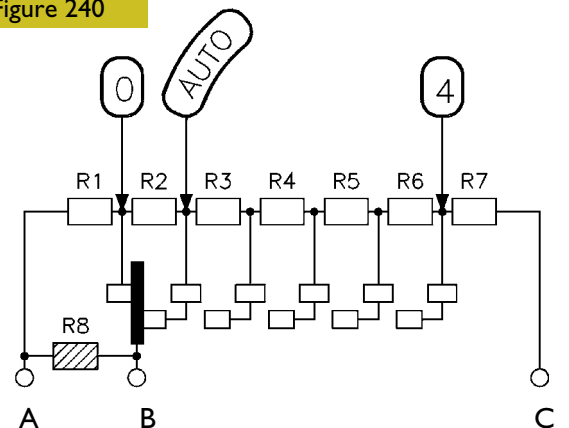
Figure 239



000256t

VENTILATION CONTROL KNOB

Figure 240

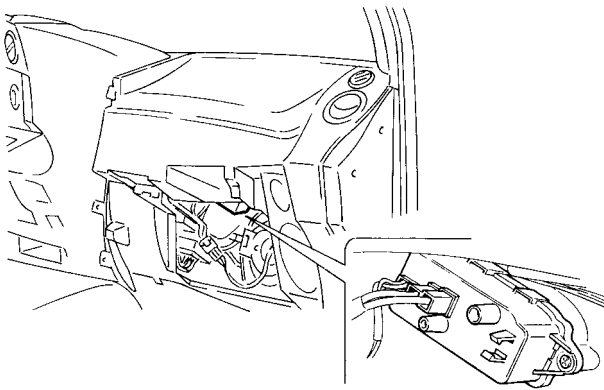


000266t

POTENTIOMETER POSITION DETECTION LAYOUT

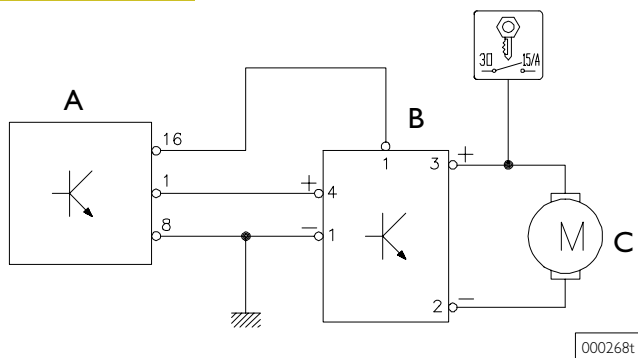
A. Signal earth (ms7) – B. signal to control unit pin 3 connector A – C. Positive from control unit pin 9 connector A

Figure 241



LOCATION OF FAN ELECTRONIC CONTROL MODULE

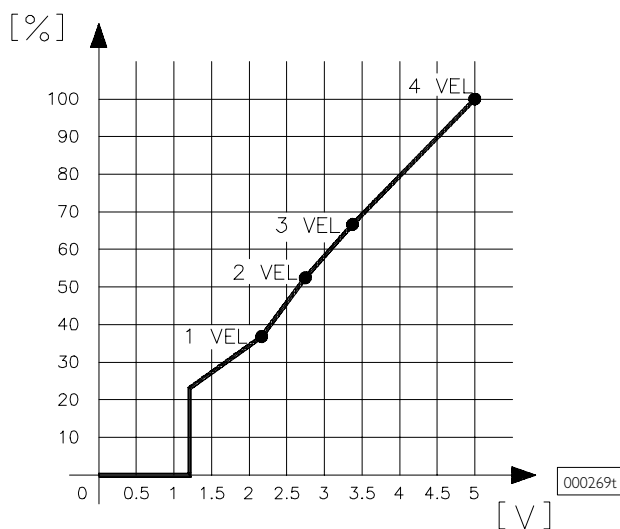
Figure 242



ELECTRONIC FAN CONTROL MODULE CONNECTIONS

A. Electronic fan control module – B. Electronic climate control unit – C. Fan motor

Figure 243



GRAPH OF AIR FLOW RATE IN RELATION TO VOLTAGE

Electronic fan control module

This is an electronic circuit inside the distributor unit near the fan (Figure 241) and it adjusts the fan speed.

It is driven by the control unit with a voltage of between 0 and 5 Volt.

It drives the fans with a voltage of 0 to 12 Volt.

It is connected to pins 1 and 16 connector B of the control unit.

The graph (Figure 243) represents the air flow rate in relation to the control voltage from the control unit.